

Next-Gen Muscarinic Could Produce (+)
Schizophrenia Data in 2H26 - Initiate BUY

In H2:26, we think oral ML-007C-MA (novel M1/M4 muscarinic) has a 30-40% PoS to generate superior Phase II schizophrenia data (\$1-2B+ oppty), driving pot'l +100% stock move and enabling the Street to assign more value to '007's second program in Alzheimer's psychosis (\$1-2B+ oppty), which has Phase II data in H2:27 (another pot'l +100%). Also in H2:26 is Phase II data for ML-004 (oral 5-HT1B/1D) in autism disorder (\$500M-\$1B call option). Initiate BUY & \$32 PT.

Big picture: MPLT has an intriguing Phase II pipeline that could ultimately disrupt the CNS/neuro landscape, starting with ML-007C-MA (oral M1/M4 agonist) in schizophrenia (SCZ) and Alzheimer's disease psychosis (ADP). Second asset ML-004 is an oral 5-HT1B/1D for autism spectrum disorder (ASD). Each program represents a \$1B+ blockbuster opportunity, and all of them will have pbo-controlled Phase II data in 2026-27 (3 shots on goal). As such, we like the risk/reward at \$660M cap, since \$1B+ in peak sales (conservative) x low 2-3x multiple = \$2-3B of value per program (unadjusted), which could be quickly realized over the next 1-2 years. Also, ML-007's program success could attract Big Pharma attention. Muscarinics in SCZ alone have been a high-value target, as evidenced by two major M&A deals with \$14B+ for BMJ/KRTX (Cobefny) and \$9B for ABBV/ CERE (emracladine).

In H2:26, ML-007C-MA's pbo-controlled Phase II ZEPHYR trial (N=300) could produce highly differentiated SCZ data, such as competitive efficacy, superior safety/tolerability and dosing convenience. (1) Having undergone careful iterations in Phase I, ML-007C-MA is a fixed-dose combo of ML-007 (a strong M1/M4 muscarinic agonist) + fesoterodine (a peripherally-acting anticholinergic). The former component drives efficacy, while the latter blunts any pro-cholinergic side effects caused by M1/M4 agonism (e.g. nausea, vomiting), (2) In SCZ, oral antipsychotics are the mainstay of treatment, but KRTX's oral Cobefny (M1/M4 approved in Sep 2024) is disrupting the SCZ paradigm, given its novel MOA and favorable efficacy/safety. Its sales trajectory (\$172M run rate) looks stronger than other antipsychotic SCZ launches (e.g. Latuda, Caplyta). (3) We think ML-007C-MA could have superior attributes to Cobefny, including competitive efficacy (and the pot'l to benefit all SCZ domains), cleaner safety/tolerability (e.g. less vomiting and GI/urinary side effects) and thus fewer discontinuations, and once-daily dosing (vs twice-daily Cobefny). MPLT could have strong IP protection as well.

Net-net, a DCF-based SOTP supports our \$32 PT (or a \$1.7B+ valuation), including: (1) \$17/sh to SCZ on \$1-2B+ sales (after a conservatively high dropout rate) and 30-40% PoS, (2) \$10/sh to ADP on \$1-2B+ sales and 10-20% PoS, and (3) \$4/sh to ASD on \$500M-1B+ sales a 10-20% PoS. **Risks:** (1) Phase II/III pbo risk, (2) Competition, but MPLT could be superior, (3) Phase I to Phase II translation (e.g. AE rates in actual pts are unknown), but the M1/M4 approach is partially de-risked.

FY (Dec)	2024E	2025E	2026E
EPS	(6.52)	(5.23)	(4.08)
Rev. (MM)	0.0	0.0	0.0
FY P/E	NM	NM	NM

INITIATING COVERAGE

RATING	BUY
PRICE	\$13.29*
PRICE TARGET % TO PT	\$32.00 +141%
52W HIGH-LOW	\$20.86 - \$13.28
FLOAT (%) ADV MM (USD)	33.2% 5.62
MARKET CAP	\$659.2M
TICKER	MPLT

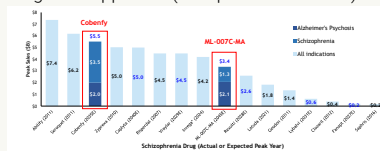
*Prior trading day's closing price unless otherwise noted.

FY (Dec)	CHANGE TO JEF		JEF vs CONS	
	2024	2025	2024	2025
REV	NA	NA	NA	NA
EPS	NA	NA	NA	NA

2024 (\$)	Q1A	Q2A	Q3	Q4	FY
EPS	(1.68)	(1.54)	(1.68)	(1.63)	(6.52)
PREV					

Exhibit 3 - We Think The Sales Precedents (Either Historical or Projected) of Recent Antipsychotics/Muscarinics in CNS Indications Support a Multiple \$2-4B Peak Sales Opportunity for MPLT's ML-007 (Unadjusted).

MPLT Is Currently in Phase II Studies in Schizophrenia (Data in H2:26) and Alzheimer's Psychosis (Data in H2:27). All Other Listed Drugs Are Approved (Except KRTX for ADP)



Source: Company reports, Visible Alpha, Jefferies Estimates
Company reports, Visible Alpha, Jefferies Estimates

Andrew Tsai * | Equity Analyst
(415) 229-1566 | atsai@jefferies.com

John Cox, Ph.D. * | Equity Associate
+1 (415) 229-8708 | jcox1@jefferies.com

Brian Balchin, ACA * | Equity Analyst
(415) 229-1522 | bbalchin@jefferies.com

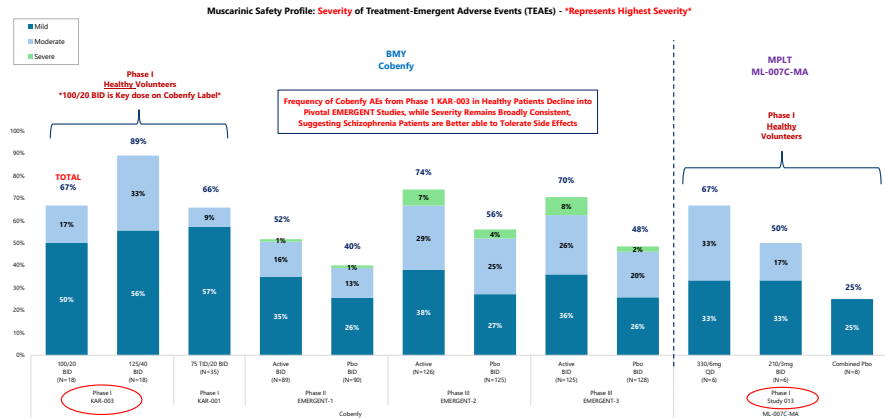
Matthew Barcus, Ph.D. * | Equity Associate
(415) 229-8703 | mbarcus@jefferies.com

Exhibit 1 - MPLT Newsflow

Timing	Expected Newsflow	Drug	Indication
H2:26	Phase II data in schizophrenia (Study 211; ZEPHYR)	ML-007C-MA	Schizophrenia
H2:26	Phase II data in autism spectrum disorder (IRIS)	ML-004	Autism Spectrum Disorder (ASD)
H2:26	Potentially initiate second study in ADP (Alzheimer's psychosis)	ML-007C-MA	Alzheimer's Disease Psychosis (ADP)
H2:26	Complete IND-enabling studies	ML-021	Parkinson's Disease
2026	Initiate IND-enabling studies	ML-009	Hyperactivity/Impulsivity
H2:27	Phase II data in ADP Alzheimer's psychosis (Study 221; VISTA)	ML-007C-MA	Alzheimer's Disease Psychosis (ADP)

Source: Jefferies estimates, Company reports

Exhibit 2 - MPLT's ML-007C Phase I Showed Lower % TEAEs (50-67%) vs KRTX Cobenfy's Phase I Healthy Volunteer studies (67%-89%), with the Severity of TEAEs Being Mild in 50%+ Cases for Both Drugs



Doses Shown for MPLT are the Phase II Target Doses; We present MPLT's Data Based on the Highest Severity AE Experienced by Patients For Comparability Purposes

Table of Contents

Executive Summary	6
Bull/Bear Case Arguments	12
Bull Case	12
Bear Case	13
Pipeline and Newsflow	15
MPLT Sum-of-the-Parts (SOTP) Analysis	16
Company Overview	16
Schizophrenia Background and Treatment Landscape	19
Diagnosis	19
PANSS Is Considered the "Gold Standard" for Assessing Antipsychotic Efficacy	19
Schizophrenia Affects ~3M Patients in the US	20
Side Effects Are Associated With Current Standard-of-Care (SoC) in Schizophrenia	20
Muscarinic Agonists (and Various Approaches) - Background and History	24
Cobenfy Launch Progress	26
Efficacy Comparisons of Current Schizophrenia Drugs	28
ML-007C-MA Data is Supportive of a Differentiated Profile in Schizophrenia	37
ML-007C-MA Phase II Expectations in Schizophrenia	37
ML-007C-MA Has Been Strategically Optimized Through Four Phase I Clinical Trials	38
Comparing the Phase I Safety Profile of ML-007C-MA to Muscarinics	43
Comparing Safety of ML-007 to Muscarinic KarXT Phase I Healthy Volunteer Data	45
Translatability into Phase 2: Cobenfy AE Rates Generally Decline From Phase I into Phase II	46
MPLT's ML-007C-MA Could be Highly Competitive Compared to Other Muscarinic Programs	57
Unpacking CERE Emraclidine Failure in Schizophrenia - Little/No Readacross to MPLT	57
Pivotal Phase III Data for NBIX's Direclidine M4 Agonist in 2027-28	58
Preclinical and in vitro Data Supports a Potential Best in Class M1/M4 Muscarinic Agonist in ML-007	59
ML-007C-MA in Alzheimer's Disease Psychosis	62
Rationale for Muscarinic Agonists in Alzheimer's Disease Psychosis (ADP)	62
Phase II Expectations for ML-007C-MA in ADP - Topline Data in H2:27	62
Cobenfy Data in ADP Expected by YE:25 Could Have Technical Readacross to MPLT Stock, But We Remain Constructive	64
Earlier Stage Pipeline Products Could Represent Sources of Valuation Upside	66
(1) ML-004 for Autism Spectrum Disorder (ASD)	66
(2) ML-021 in Parkinson's Disease	69
(3) ML-009 in Impulsivity, Agitation, and Irritability	70
Intellectual Property	71
ML-007	71

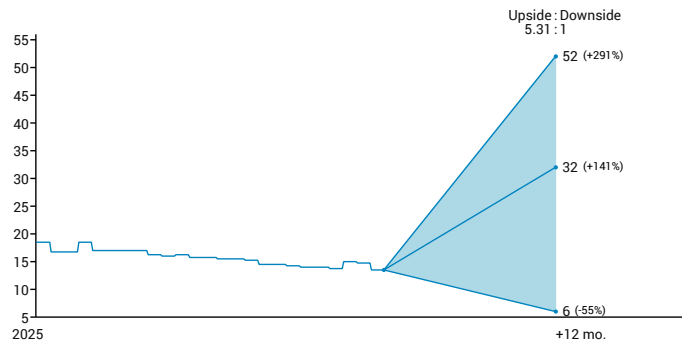
ML-004	71
ML-009	71
Licensing and Other Collaboration Agreements	72
Key Risk Factors	73
Management Biographies	74
Financials	75
DCF Valuations	75
Market Models	78
Income Statement	81
Balance Sheet	82
Cash Flow Statement	83

The Long View: Maplight Therapeutics

Investment Thesis / Where We Differ

Maplight Therapeutics (MPLT) is a clinical-stage biotech company focused on improving outcomes for patients with central nervous system (CNS) disorders, including schizophrenia. Their lead compound, ML-007C-MA (M1/M4) could offer a differentiated approach to treatment in both schizophrenia and Alzheimer's disease psychosis (ADP) by potentially improving dosing convenience, as well as safety/efficacy. Schizophrenia and ADP are both \$1B + opportunities in theory, making the risk/reward favorable at this valuation. MPLT also has a pipeline of other CNS/Neuro drugs for \$1B+ markets that are in Phase II trials (e.g., ML-004 in ASD) or could enter the clinic as early as 2026-27 to offer additional valuation support/upside.

Risk/Reward - 12 Month View



Base Case,
\$32, +141%

- Our base case is a SOTP analysis of ML-007C-MA in schizophrenia (\$1-2B peak sales and 30-40% PoS) and ADP (\$1-2B peak sales and 10-20% PoS), as well as \$500M-1B+ peak sales for ML-004 in ASD and 10-20% PoS.
- No credit is given for ML-021 in Parkinson's or ML-009 in hyperactivity/impulsivity
- For each program, we build the revenue and cash flow streams out to patent term expirations before probability-adjusting our DCF.
- Some credit given for cash

Upside Scenario,
\$52, +291%

- Our upside case includes a higher PoS for lead program ML-007C-MA in schizophrenia (55-65% PoS) and ADP (25-35% PoS), as well as ASD (15-25%) and earlier-stage pipeline programs (0-10%)
- No credit given for cash

Downside Scenario,
\$6, -55%

- Our downside case considers a lower PoS for lead program ML-007C-MA in schizophrenia (0-10%) and no credit for ADP. No credit for earlier-stage pipeline programs.
- Some credit given for cash

Sustainability Matters

Top Material Issue(s): 1) Product Quality and Safety: Lead drug ML-007C-MA could have a differentiated therapeutic profile than other epilepsy drugs, including no titration, a differing AE profile, and no food effect. MPLT should continue to demonstrate ML-007C-MA is safer and more convenient than its competitors, as this is a major point of differentiation. **2) Employee Engagement, Diversity and Inclusion:** Smaller biotech companies must ensure that they have sufficient talent to develop drugs (whereas large pharma/biotech may provide more stability). This is especially critical as companies look to supplement their commercial portfolio with newer drugs and or BD opportunities.

Company Target(s): None

Qs to Mgmt: 1) What gives you confidence in the safety profile of your schizophrenia and ADP drug vs competitors, including Cobenfy? **2)** What steps have you taken to prepare for 3+ Phase II studies in CNS/Neuro indications in the form of relevant employee and/or Board hires?

Catalysts

- **H2:26:** Phase II **ZEPHYR** data for ML-007C-MA in schizophrenia
- **H2:26:** Phase II **IRIS** data for ML-004 in autism spectrum disorder (ASD)
- **2026:** IND updates for ML-021 in Parkinson's disease and ML-009 in hyperactivity/impulsivity
- **H2:27:** Phase II **VISTA** data for ML-007C-MA in Alzheimer's disease psychosis (ADP)

Executive Summary

In H2:26, we think oral ML-007C-MA (novel M1/M4 muscarinic) has a 30-40% PoS to generate superior Phase II **schizophrenia** data (\$1-2B+ oppty), driving a potential +100% stock move and enabling the Street to assign more value to ML-007's second program in **Alzheimer's psychosis** (\$1-2B+ oppty), which has Phase II data in H2:27 (another potential +100% move). Also in H2:26 is Phase II data for ML-004 (oral 5-HT1B/1D) in **autism disorder** (an estimated \$500M-\$1B call option). Initiate with BUY rating and \$32 PT.

Big picture: MPLT has an intriguing Phase II pipeline that ultimately could disrupt the CNS/neuro landscape, starting with **ML-007C-MA (M1/M4 agonist)** in schizophrenia (SCZ) and Alzheimer's disease psychosis (ADP) and ML-004 (5-HT1B/1D) in autism spectrum disorder (ASD). Each program offers a \$1B+ blockbuster opportunity, and all of them will share pbo-controlled Phase II data in 2026-27 (3 shots on goal), potentially driving significant stock upside. As such, we like the risk/reward at \$660M cap, since we estimate \$1B+ in peak sales (conservative) x low 2-3x multiple = \$2-3B of value per program (unadjusted) that could be quickly realized within the next 1-2 years.

- We think **ML-007C-MA** is elegantly designed to not only produce competitive efficacy in SCZ/ADP, but also offer potentially superior safety/tolerability and dosing convenience (both attributes can drive efficacy higher).
- Behind ML-007C-MA is a pipeline of molecules for other large indications, including **ML-004** (oral 5-HT1B/1D) for autism spectrum disorder (ASD), which will have Phase II IRIS data in H2:26 (\$1B+ oppty). Earlier stage **ML-021** (M4 agonist) for Parkinson's and **ML-009** (GPR52 PAM) for hyperactivity/impulsivity could also begin delivering value over 2026-27 when Phase I/II studies officially start/ramp up.

Our assertion is supported by historical precedents: (1) BMY spent \$14B+ for KRTX following (+) Phase III SCZ data with **Cobefy/KarXT (M1/M4 agonist)**, which is now approved and guided to have multi-billion dollar peak sales potential (cons is at \$5B+), (2) ABBV acquired CERE for \$9B following (+) Phase Ib SCZ results with **emracladine (M4 PAM)**. However, emracladine's Phase II setback in Nov 2024 allows MPLT to now leapfrog CERE, and we'd argue that both M1/M4 are necessary for meaningful efficacy. (3) Other notable recent CNS transactions include JNJ's \$14B+ acquisition of ITCI in Jan 2025 for Caplyta (an approved antipsychotic in SCZ/BPD/MDD) and ABBV's \$900M+ upfront to Gilgamesh for a novel psychedelic (GM-2505) in MDD. Bottom line, we think Big Pharma's appetite for potentially disruptive CNS drugs is rising, and we think ML-007C-MA for SCZ/ADP fits that narrative.

In H2:26, ML-007C-MA's Phase II ZEPHYR trial (N=300) could produce highly differentiated efficacy/safety data in schizophrenia (SCZ). (1) ML-007C-MA is a fixed-dose combination (FDC) of ML-007 (a strong M1/M4 muscarinic agonist) and fesoterodine (a peripherally-acting anticholinergic, or PAC). The former component is meant to drive efficacy, while the latter is meant to blunt any pro-cholinergic side effects caused by M1/M4 agonism (e.g. nausea, vomiting), (2) In SCZ, oral antipsychotics are the mainstay of treatment, though KRTX's oral Cobefy (approved) is a novel approach (M1/M4) whose (+) Phase II/III efficacy data looks quite competitive to antipsychotics (e.g. Cobefy's PANSS score reductions are towards the high-end of what antipsychotics can achieve). M1/M4 muscarinics also avoid traditional antipsychotic side effects like weight gain, EPS symptoms (rigidity/stiffness), and anxiety. (3) We think MPLT's '007 could have superior properties to Cobefy (and antipsychotics by extension), including competitive-to-stronger efficacy (within SCZ and in other multiple indications), cleaner safety/tolerability (less vomiting, less GI/urinary side effects) resulting in fewer discontinuations and higher drug compliance, and a once-daily dosing option in SCZ (vs twice-daily with Cobefy).

By elegantly synchronizing the agonist (M1/M4) and antagonist (PAC) activity through several iterations in Phase I, ML-007 could ultimately address the positive, negative and cognitive symptoms of SCZ (whereas all SCZ drugs only benefit only positive symptoms). By carefully optimizing ML-007+PAC in Phase I, MPLT could offer:

- (1) a safer and more tolerable approach, including **fewer instances of pro-cholinergic AEs** caused by M1/M4 (e.g. nausea, vomiting) and **anti-cholinergic AEs** caused by the peripheral-acting anticholinergic (e.g. GI constipation, dry mouth, urinary pain, urinary retention, eye disorders). MPLT could also have fewer severe AEs as well. Indeed, we are encouraged to see 0% vomiting in Phase I (vs 6-28% for Cobenfy).
- (2) a more **convenient dosing profile** (e.g. once-daily QD vs twice-daily BID Cobenfy in SCZ, and BID vs 3x daily TID Cobenfy in ADP), which are valuable and potentially underappreciated attributes (per our expert/physicians discussions). In addition, ML-007C-MA's titration requirement is only 1 day in schizophrenia (vs 1-2 weeks for Cobenfy) and 1 week in ADP (vs 4-5 weeks for Cobenfy). ML-007C-MA also does not have a fasting requirement or need to be taken with food (although Phase II recommends patients to take with food). Alternatively, Cobenfy's drug exposure changes with food (anticholinergic trospium drops by ~80%), so it must be dosed 1 hour prior to meal time or 2 hours after (e.g. a fasting requirement).
- (3) Ultimately, better safety/convenience could result in a **stronger efficacy profile**, although we'd add ML-007's M1/M4 receptor activation (efficacy) also looks stronger than that of Cobenfy's M1/M4 with xanomeline.
- (4) MPLT could have **stronger IP protection** than Cobenfy, whose xanomeline+trospium are essentially two generic components.

Some additional thoughts on ML-007's Phase I safety/tolerability results in healthy volunteers (vs KRTX's Phase I healthy volunteer studies and Phase II/III SCZ studies). ML-007's Phase I studies showed favorable safety/tolerability results with once- and twice-daily dosing (QD/BID), minimal titration requirements and no need for fasting - all while achieving CSF exposures expected to be clinically relevant in Phase II/III studies. Note Cobenfy's ongoing 52-week OLE studies in schizophrenia reported **discontinuation rates of ~47-77%**.

- (1) **Vomiting is a driver of real-world discontinuations with Cobenfy.** ML-007 already shows signs of differentiation with a **0% vomiting rate** in Phase I healthy volunteers, including with 210/3mg BID (N=6) and 330 6mg QD (N=6). Maintaining the AE rate in Phase II ZEPHYR could set ML-007 apart from Cobenfy's **vomiting AEs of 9-16%** from the pivotal Phase II/III studies (EMERGENT). In Phase I, Cobenfy's vomiting rate was **6-28%** with 100/20mg BID (N=18) and 125/40mg BID (N=18).
- (2) **Nausea is another driver of real-world discontinuations with Cobenfy.** MPLT's Phase I nausea rates are **50%** for the Phase II doses (330-6mg QD, 210/3mg BID), optically above **22%** for Cobenfy (100/20mg BID and 125/40mg BID). (a) However, MPLT's finding is based on a small N=6 patient sample size (e.g. 1 patient can alter the overall % rate meaningfully, given a small denominator). (b) In addition, the study was evaluated in healthy volunteers, so AEs could attenuate in actual Phase II schizophrenia patients, who are taking multiple drugs at once (and can perhaps tolerate more AEs), (c) **By logic, vomiting is a worse side effect than nausea** (e.g. nausea leads to vomiting, but vomiting does not lead to nausea), (d) MPLT's Phase I AE rates are based on **7 days of dosing** (vs 5 days), whereas KRTX's Phase I is 5-day dosing - so not apples-to-apples comparison. (e) MPLT's Phase I did **not employ titration** (Phase II will have a short 1-day titration), whereas KRTX had already employed a 3-day titration, (f) MPLT's 50% nausea is **mostly mild** in severity (some moderate), whereas 1/3 of KRTX's nausea rate could be moderate severity, (g) MPLT's population includes **GLP-1 users** (which causes nausea)

- **(3) Some of KRTX Cobenfy's AE rates in Phase I healthy volunteers are undisclosed.** For Cobenfy, we have AE rates for relevant markers like nausea, vomiting, constipation, urinary retention, sweating, high blood pressure, and dizziness. However, we do not have AE rates for **dyspepsia (abdominal discomfort)**, which is a common AE in real-world practice.
- **(4) MPLT's constipation and urinary retention rates of 0% compare favorably to Cobenfy's 6% (constipation) and 0-11% (urinary retention).** Severe constipation could be a liability in the real-world. We also know urinary retention is listed as a warning in Cobenfy's label.

Separately, contrary to investor thinking, we think Cobenfy's launch in SCZ (approved Sep 2024) is trending favorably in a crowded SCZ market. Q3:25 sales of \$43M are up from \$35M in Q2 and \$27M in Q1. The trajectory looks stronger than Caplyta's initial SCZ launch (2020) and competitive to Latuda's initial SCZ launch (2011), before the two antipsychotics had secured additional label expansions. In any case, physicians will be increasingly familiar with the muscarinic class by the time ML-007C-MA potentially reaches the market anyway. That said, we do think MPLT's potential 2029-30 launch could look even more favorable should its efficacy/safety/tolerability attributes play out favorably.

Exhibit 4 - Comparing MPL-007 Phase I Healthy Adult Data vs KRTX's Phase I Healthy Adult with KarXT (Cobenfy) - and KRTX's Data Evolution Into Pivotal Phase II/ III SCZ Studies

** Only 100/20mg BID is approved and on Cobenfy Label

Drug Phase	ML-007C-MA Phase 1		KarXT Phase 1		KarXT Phase 2	KarXT Phase 3	KarXT Phase 3
Safety Population	Healthy Volunteers		Healthy Volunteers		Schizophrenia Patients	Schizophrenia Patients	Schizophrenia Patients
Dose	210/3 mg BID (N=6)	330/6 mg QD (N=6)	100/20 mg BID (N=18)	125/40mg BID (N=18)	EMERGENT-1 (N=89)	EMERGENT-2 (N=126)	EMERGENT-3 (N=125)
Subjects with any TEAE	3 (50%)	4 (67%)	12 (67%)	16 (89%)	52%	74%	70%
Mild	2 (33%)	2 (33%)	9 (50%)	10 (56%)	35%	38%	36%
Moderate	1 (17%)	2 (33%)	3 (17%)	6 (33%)	16%	29%	26%
Severe	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1%	7%	8%
Procholinergic TEAEs							
Nausea	3 (50%)	3 (50%)	4 (22%)	4 (22%)	17%	19%	19%
Vomiting	0%	0%	5 (28%)	1 (6%)	9%	14%	16%
Hyperhidrosis	1 (17%)	2 (33%)	1 (6%)	2 (11%)	2%	-	-
Salivary Hypersecretion	0%	-	1 (6%)	0%	-	-	-
Anticholinergic TEAEs							
Dry mouth	0%	-	2 (11%)	5 (28%)	9%	5%	0%
Dysuria	0%	-	0 (0%)	4 (22%)	-	-	-
Constipation	0%	-	1 (6%)	1 (6%)	17%	21%	13%
Urinary Retention	0%	-	0 (0%)	2 (11%)	-	-	-
Eye disorders	0%	-	0 (0%)	5 (28%)	-	5%	-
Other TEAEs							
Dizziness	3 (50%)	2 (33%)	4 (22%)	7 (39%)	3%	9%	0%
Postural dizziness	0%	-	2 (11%)	2 (11%)	0%	0%	0%
Headache	0%	-	3 (17%)	4 (22%)	7%	13%	11%

Source: Jefferies; Company Reports

(1) MPLT's Phase II Target Doses. We present MPLT's Data Based on the Highest Severity AE Experienced by Patients For Comparability Purposes.

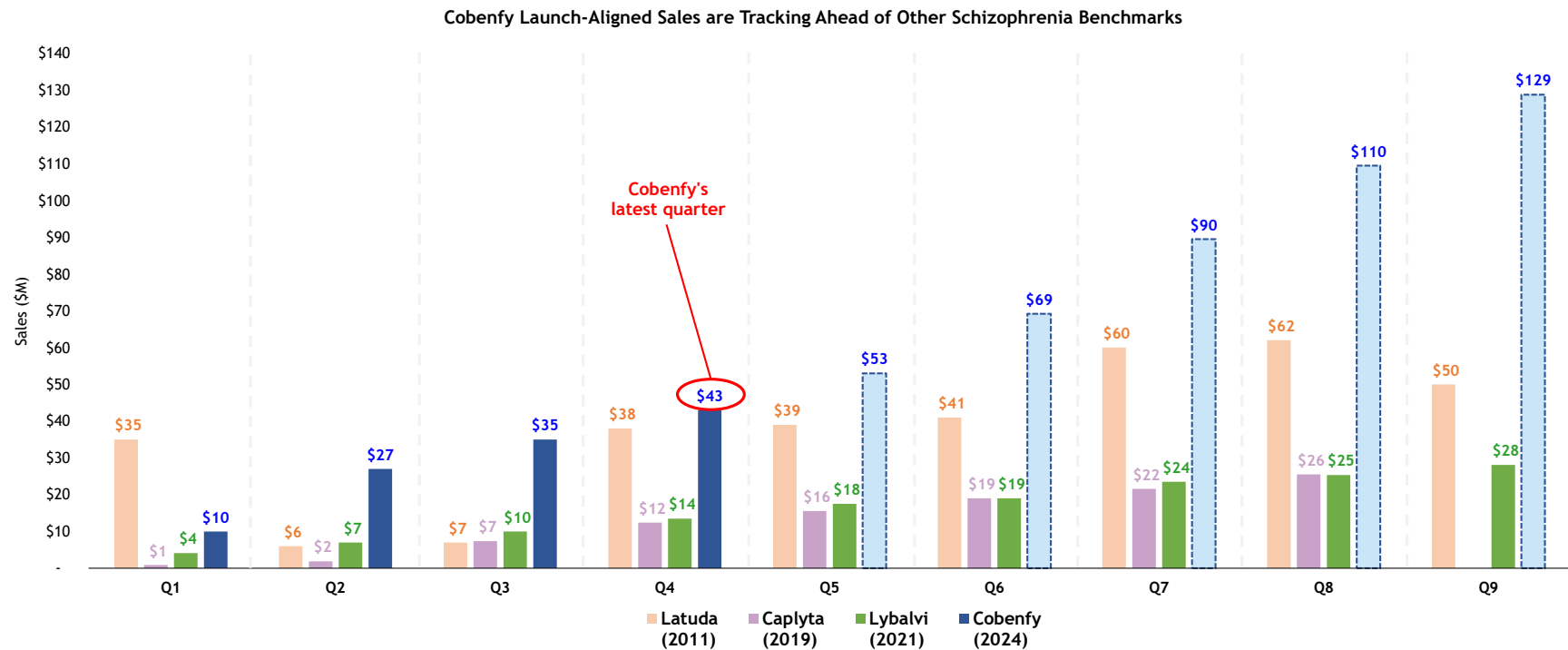
(2) KRTX's Phase II/III EMERGENT Trial Dosing: Xanomeline 50 mg/trospium 20 mg BID on days 1-2 followed by xanomeline 100mg/ trospium 20mg BID on days 3-7. The dose is increased to xanomeline 125mg/ trospium 30mg BID on days 8-34 unless the subject is experiencing adverse events from the xanomeline 100 mg/trospium 20 mg dose.

Exhibit 5 - Cross Trial Efficacy Comparisons of Muscarinic Agonists

Drug/Study Information							4-6 Week PANSS*			
Company	Name	Molecule	MOA	Status	Study	Study Phase	BL	Drug	Pbo	Pbo-adj
KRTX, BMY	Cobenfy	xanomeline + trospium Cl	M1-M4 + anti-cholinergic	Approved 2024	KAR-004	II	-	-17.4	-5.9	-11.5
					KAR-007	III	98.2	-21.2	-11.6	-9.6
					KAR-009	III	96.5	-20.6	-12.2	-8.4
NBIX	NBI-1117568	direclidine	Selective M4 orthosteric agonist	Phase III	SCZ2028	II	97.0	-18.2	-10.8	-7.4
CERE, ABBV	CVL-231	emraclidine	Selective M4 receptor PAM	Phase II	SCH-001	Ib	93.0	-19.5	-6.8	-12.7
					EMPOWER-1	II	98.3	-16.5	-13.5	-3.0
					EMPOWER-2	II	97.2	-14.2	-16.1	1.9
AVERAGE:							96.7	-18.2	-11.0	-7.2
MEDIAN:							97.1	-18.2	-11.6	-8.4

Source: Jefferies; Company reports, FDA review documents

Exhibit 6 - Cobenfy Sales are Tracking and Expected to Outpace Other Approved Schizophrenia Benchmarks When Launch Aligned



Source: Company reports, Jefferies, Visible Alpha

Latuda first quarter sales were impacted by drug stocking. Caplyta bipolar depression approval granted Dec 20, 2021 (late in Q8 of launch). Latuda bipolar approval granted June 2013, during 10th quarter of launch. Lybalvi approved in BP1 and SCZ initially in June 2021, but 50% of sales inferred to be attributed to SCZ based on company commentary. Dashed Cobenfy bars represent cons estimates from Visible Alpha

Ultimately, **ML-007C-MA is leapfrogging muscarinic competitors as its Phase II data in H2:26 could be registrational-quality**, meaning that MPLT could need just one add'l (+) Phase III to file (e.g., pot'l 2029-30 approval). **CERE/ABBV's emraclidine (M4 PAM)** may remain in Phase II, following two pivotal Phase II study failures in Nov 2024. **NMRA's NMRA-861 (M4 PAM)** is in Phase I SAD/MAD with data in Q1:26, and NBIX's **NBI-568 (direclidine, M4 PAM)** has Phase III data from at least one of three Phase III's over 2027-28. **(4)** As such, MPLT could be a prime M&A candidate post Phase II data, but not necessarily prior to Phase II data (in our view), given ABBV's layout of \$9B for emraclidine (post Phase Ib data) ultimately resulted in two Phase II failures.

Other 2025-27 catalysts that could enable MPLT stock to move higher: **(1)** By YE:25, BMY will have **Phase III data (ADEPT-2) in ADP** for Cobenfy. Positive data could de-risk MPLT's own Phase II VISTA results in ADP in H2:27. If data is negative, we'd argue ML-007C-MA could be a stronger M1/M4 agonist offering a better tolerability profile, and that trial execution/conduct can differ across programs. In addition, BMY noted in Q3:25 EPS that it is assessing trial conduct across studies, scrubbing data to avoid any unusual trends (mgmt remains blinded to ADEPT-2 results) and looking for both protocol deviations (e.g. treatment protocol breaches, improper monitoring, incorrect dose administration) and questionable trends at certain sites (e.g. excessive pbo effect, higher discontinuations, etc). In our view, BMY could technically elect to expand enrollment/upscale the study to ensure statistical significance. Although Cobenfy's pot'l ADP failure could spark headline fears to MPLT by raising a question on the role of muscarinics in indication beyond SCZ, we'd argue one cannot take away the fact Cobenfy is approved (and unequivocally is efficacious) in schizophrenia, an indication that certainly serves as a value driver for MPLT. **(2)** Phase II data for **ML-004 in ASD (IRIS)** slotted for H2:26. **(3)** Again, ML-00C-MA's own **Phase II VISTA results in ADP** in H2:27, **(4)** Pipeline progression for **ML-021 in Parkinson's disease** and **ML-009 in hyperactivity/impulsivity** over 2026-27.

Net-net, a DCF-based SOTP supports our \$32 PT (or a \$1.7B+ valuation), including: **(1)** \$17/sh to SCZ on \$1-2B+ peak sales and 30-40% PoS, **(2)** \$10/sh to ADP on \$1-2B+ sales and 10-20% PoS, and **(3)** \$4/sh to ASD on \$500M-1B+ sales a 10-20% PoS. **Risks:** **(1)** Phase II trial execution (e.g. high pbo risk), **(2)** Competition, but MPLT could be superior (and second-to-market), **(3)** Phase I to Phase II translation, but M1/M4 approach is partially de-risked.

Bull/Bear Case Arguments

Bull Case

- History consistently confirms that oral pills have become \$1B+ blockbusters for CNS/neuro indications with millions of US patients.
- There are actually 3 shots on goal (all in 2026) relative to the current valuation. All three Phase II studies are placebo-controlled and should be stock-moving, with the autism program serving a free call option, in our view. Minimum value baked into ML-007 for Alzheimer's ADP as well.
- Ultimately, assigning \$1-2B peak sales in SCZ and another \$2B+ in ADP suggest significant valuation potential (on a conservative 2-3x multiple). Not all indications need to succeed for significant valuation upside.
- In fact, atypical antipsychotics with inferior safety profiles have surpassed \$1B+ in sales, while other antipsychotics with similar efficacy but a simply cleaner safety profile have reached (or are expected to reach) \$1B+ in peak sales.
- As a novel muscarinic agonist (a different mechanism from an atypical antipsychotic), BMJ guides Cobenfy to reach multi-billion-dollar potential. Consensus estimates approach \$5B+ across all its potential indications (beyond SCZ).
- Even though the schizophrenia market is crowded, there is clearly room/space for a potentially superior product, especially across multiple CNS indications (beyond SCZ). If ML-007 is superior in efficacy/safety (over BMJ/KRTX's Cobenfy) as a convenient once-daily pill, then \$1-2B in schizophrenia alone is easily conservative. ML-007 could then command pricing power as it would become the go-to 1L/2L schizophrenia pill. In addition, platform value could be created, leading the Street to place greater value in other indications like Alzheimer's psychosis (ADP) - irrespective of how Cobenfy's own ADP outcome (since one can argue ML-007 is different in SCZ)
- Cobenfy's clinical/regulatory success following (+) Phase II/III schizophrenia data helps de-risk ML-007, which could turn out to be a superior product over Cobenfy.
- Cobenfy's commercial performance thus far is due to structural barriers of adoption, not anything fundamental to muscarinics. Additionally, ML-007C-MA is likely to have an improved profile compared to Cobenfy (e.g. cleaner tolerability and thus fewer discontinuations, leading to much higher real-world compliance). Also, physicians will be increasingly familiar with muscarinics by the time ML-007C-MA potentially reaches the market anyway.
- Dosing convenience (once-daily) is of particular importance in schizophrenia and the elderly. Should ML-007C-MA be eligible for QD dosing in schizophrenia and BID dosing in the elderly (ADP), this would afford a real competitive advantage over Cobenfy, which is BID (2x daily) in schizophrenia patients and TID (3x daily) in ADP
- ML-007 has an optimized PK synchronization of its agonist (M1/M4) and antagonist components (PAC), without sacrificing strong activation of both M₁ and M₄ receptors in the central nervous system. As such, ML-007 could address the positive, negative and cognitive symptoms of schizophrenia.
- ML-007C-MA (stronger M1/M4 agonism) could have an improved tolerability profile vs Cobenfy (M1/M4), due to an improved PK profile whereby MPLT has successfully matched plasma exposures of ML-007 (M1/M4) and a peripherally acting anti-cholinergic (PAC) to offset cholinergic side effects. Procholinergic side effects include nausea, vomiting, while anticholinergic side effects include urinary retention, constipation, eye disorders, dysuria (painful urination) - all of which are important to uptake in SCZ.
- In Phase I, ML-007 shows 50% nausea and 0% vomiting compared to Cobenfy's 22% nausea and 28% vomiting. Vomiting is likely to be the most detrimental side effect to patients/physicians (vs nausea). Plus, safety side effects could be lower in schizophrenia patients (vs healthy volunteers), enabling ML-007C-MA's safety data to simply improve over time.
- ML-007's ongoing Phase II studies in SCZ and ADP could be considered supporting pivotal studies, which would expedite timelines and probability of success (if those trial are successful) as MPLT would only need one add'l (+) Phase III study for each indication
- No approved treatment options in Alzheimer's psychosis, representing a \$3-5B+ oppty (at a minimum).
- BMJ is also exploring a whole host of programs with Cobenfy, including a long-acting injectable (LAI) in SCZ, Alzheimer's psychosis (ADP), adjunctive bipolar 1 mania (BP1), Alzheimer's cognition, and Alzheimer's agitation. Any program success could readthrough positively to MPLT, enabling the Street to assign more value to MPLT.

- MPLT could be a prime M&A target, following two blockbuster acquisitions of companies with muscarinics. ABBV completed its acquisition for CERE for \$8.7B in Aug 2024 and BMY completed its acquisition of KRTX for \$14B in Mar 2024.
- The failure of CERE's emraclidine (M4) in two Phase II trials is likely more indicative of the M4 sole agonist class of muscarinics (vs M1/M4 dual agonism like ML-007C-MA). NBIX's Phase II trial with M4 validates this assertion. If anything, the Phase III success of Cobenfy (M1/M4) should be supportive of ML-007C-MA's potential profile. Besides, the emraclidine dose may not have been well-optimized before Phase III study. ML-007C-MA's profile has been optimized across four Phase I studies.
- ML-007C differentiation of **0% vomiting rates** in Phase I healthy volunteers persists into the pivotal Phase II ZEPHYR study, which would set it apart from Cobenfy's vomiting AEs of 9-16% from the EMERGENT pivotal studies. Also, nausea rates stepping down meaningfully from 50% in Phase I healthy volunteers would further differentiate; if 0% vomiting signal is shown, then we see nausea rates coming in-line with Cobenfy's 17-19% as sufficient, so a super bull case could present itself should nausea rates also degrade to below Cobenfy's 17-19% e.g. 12%.
- Although Phase I nausea rates are 50% for the Phase II doses (330-6mg QD, 210/3mg BID), this is based on a small N=6 patient sample size (1 patient can alter the overall % rate meaningfully). In addition, the study was evaluated in healthy volunteers, so AEs should attenuate in actual Phase II schizophrenia patients who are typically taking multiple drugs (and can thus tolerate more AEs). MPLT's nausea is mostly mild and transient too.
- By logic, vomiting is a more severe side effect than nausea (e.g. nausea leads to vomiting, not vice versa).
- ML-007 has patent protection, including a coveted composition of matter (CoM) patent expiring in 2031-32 (without extension). In addition, ML-007 could have polymorph patents extending protection until 2045, suggesting a long tail value

Bear Case

- Relatively high expectations with current valuation. One must have conviction that ML-007 can easily become a \$1B+ product in schizophrenia alone and/or have high confidence ML-007's Phase II SCZ study can succeed and offer superior/differentiated properties.
- BMY/KRTX will actually share Cobenfy Phase II ADP data in Q4:25, which could cause headline fears to MPLT stock, which is also pursuing ADP as a secondary indication for ML-007
- Not all antipsychotics or other traditional oral pills (e.g. antidepressants) have become blockbuster pills. The devil will be in the details.
- A relatively high risk/reward story, despite Cobenfy's regulatory success. If ML-007 fails Phase II, then one cannot ascribe value to ADP and investors will need to have confidence in autism or MPLT's earlier-stage programs - which have limited data.
- The story is ultimately centered around ML-007, which must succeed in Phase II. In addition, it may be challenging to develop the conviction that MPLT's ML-007C-MA will work in indications beyond schizophrenia. Within schizophrenia, we have relatively small amounts of healthy volunteer safety/tolerability data.
- MPLT still needs to succeed in several Phase II/III studies before ML-007 is approved. In CNS/psychiatry, high placebo response rates in clinical studies are notoriously challenging to circumvent. Arguably, controlling for placebo responses can be even more challenging in other indications, like bipolar mania (should MPLT choose to pursue those indications). As such, management quality and study execution are important.
- Schizophrenia drug launches have routinely disappointed the Street. Rather, it is more so the indication expansions after SCZ that drive most of the sales for a CNS drug (e.g. MDD major depression, BPD bipolar depression, GAD general anxiety, BP bipolar mania).
- Cobenfy's launch trajectory is looking more modest than Street expectations, raising questions if \$1B+ schizophrenia alone can be a blockbuster opportunity (which places more reliance on Cobenfy's indication expansion opportunities in ADP Alzheimer's psychosis)
- Within schizophrenia, we've seen positive Phase II/III data from other muscarinics like KRTX Cobenfy, NBIX's direclidine, and CERE's emraclidine. However, emraclidine failed two Phase II trials. The Street may still remain unsure about MPL-007 (even if it succeeds in Phase II) until the drug generates results in pivotal Phase III studies.
- The muscarinic space is becoming crowded with other potential fast followers.

- ML-007C-MA may not actually have that much of an improved tolerability profile in actual schizophrenia patients. Phase I data in healthy volunteers showed 50% nausea compared to Cobenfy's 22% nausea and 28% vomiting.
- Dosing convenience for ML-007C-MA (once-daily in schizophrenia) may not be as important a differentiator in schizophrenia patients, who are taking an average of 7 pills per day anyway.
- Cobenfy hasn't been successful commercially yet (commercial sales in SCZ yet to be fully de-risked). It is unclear how the muscarinics class will be able to alter physician habits away from prescribing antipsychotics. At this juncture, most physicians are likely familiar with managing antipsychotics and their tolerability profiles in their schizophrenia patients quite well.
- Unclear whether Big Pharma remains interested in another muscarinic for schizophrenia. After ABBV's \$8.7B acquisition of CERE, emraclidine ultimately failed two Phase II trials. In addition, after BMY's \$14B acquisition of KRTX, the Cobenfy launch has been slower than expected.
- Nausea and vomiting are the two key AEs that drove most of the discontinuations for Cobenfy in KRTX's pivotal EMERGENT schizophrenia studies and open-label extensions. While ML-007C showed zero cases of vomiting in the Phase I healthy volunteers study forms the basis of the asset's safety differentiation (vs 6%-22% vomiting for Cobenfy's Phase I healthy volunteers study KAR-003), the bear case is if vomiting emerges as a safety signal in the pivotal ZEPHYR study.
- Phase I nausea rates for ML-007C are relatively high at 50% (vs 22% for Cobenfy), so another bear argument would be if these rates did not degrade to levels that match those for Cobenfy's pivotal studies i.e. 17-19%. Even if ML-007C continues to show zero vomiting cases in pivotal studies, Bears could still argue that nausea is sufficient to drive discontinuations i.e. it does not necessarily have to be vomiting.

Pipeline and Newsflow

Exhibit 7 - MPLT Newsflow

Timing	Expected Newsflow	Drug	Indication
H2:26	Phase II data in schizophrenia (Study 211; ZEPHYR)	ML-007C-MA	Schizophrenia
H2:26	Phase II data in autism spectrum disorder (IRIS)	ML-004	Autism Spectrum Disorder (ASD)
H2:26	Potentially initiate second study in ADP (Alzheimer's psychosis)	ML-007C-MA	Alzheimer's Disease Psychosis (ADP)
H2:26	Complete IND-enabling studies	ML-021	Parkinson's Disease
2026	Initiate IND-enabling studies	ML-009	Hyperactivity/Impulsivity
H2:27	Phase II data in ADP Alzheimer's psychosis (Study 221; VISTA)	ML-007C-MA	Alzheimer's Disease Psychosis (ADP)

Source: Jefferies estimates, Company reports

Exhibit 8 - MPLT Pipeline

Program	Target	Modality	Indication	Designation	Stage	Milestone
ML-007C-MA	M1/M4 Agonist	Oral small molecule	Schizophrenia	N/A	Phase II (ZEPHYR)	- Phase II data expected H2:26
			Alzheimer's Disease Psychosis (ADP)	N/A	Phase II (VISTA)	- Phase II data expected H2:27
ML-004	5-HT _{1B/1D} Agonist	Oral small molecule	Autism Spectrum Disorder (ASD)	N/A	Phase II (IRIS)	- Phase II data expected H2:26
ML-021	M4 Antagonist	Oral small molecule	Parkinson's Disease	N/A	Preclinical	- Complete IND-enabling studies in 2026
ML-009	GPR52 Positive Allosteric Modulator	Oral small molecule	Hyperactivity/Impulsivity	N/A	Preclinical	- Initiate IND enabling studies in 2026

Source: Jefferies, Company reports

MPLT Sum-of-the-Parts (SOTP) Analysis

Exhibit 9 - MPLT SOTP Worksheet

NPV (\$M)	Bear Case	%	Base Case	%	Bull Case	%
ML-007C-MA - Schizophrenia	\$2	37%	\$17	53%	\$25	49%
Probability	5%		40%		60%	
NPV	\$110		\$878		\$1,318	
ML-007C-MA - Alzheimer's Disease Psychosis	\$0	0%	\$10	31%	\$19	37%
Probability	0%		15%		30%	
NPV	\$0		\$508		\$1,016	
ML-004 - Autism Spectrum Disorder	\$0	0%	\$4	14%	\$6	11%
Probability	0%		20%		25%	
NPV	\$0		\$230		\$288	
ML-007C-MA - Alzheimer's Disease Cognition	\$0	0%	\$0	0%	\$1	1%
Probability	0%		0%		5%	
NPV	\$0		\$0		\$29	
ML-021 - Parkinson's Disease	\$0	0%	\$0	0%	\$1	1%
Probability	0%		0%		5%	
NPV	\$0		\$0		\$29	
ML-009 - Hyperactivity/Impulsivity	\$0	0%	\$0	0%	\$1	1%
Probability	0%		0%		5%	
NPV	\$0		\$0		\$32	
Cash	\$4	63%	\$1	3%	\$0	0%
Weight	40%		10%		0%	
\$/Share	\$6		\$32		\$52	
Implied Market Cap (\$M)	\$298		\$1,664		\$2,711	
Upside/downside	-57%		140%		291%	

Stock Price	\$13.29
Share Count (in M) - YE:26	52.2
Cash (Q4:26)	\$471
Cash/Share	\$9.02
Debt (Q4:26)	\$0

Source: Jefferies estimates, Company reports

eharrison@maplightr.com - Maplight Therapeutics - 27/03/2026 06:21 AM UTC

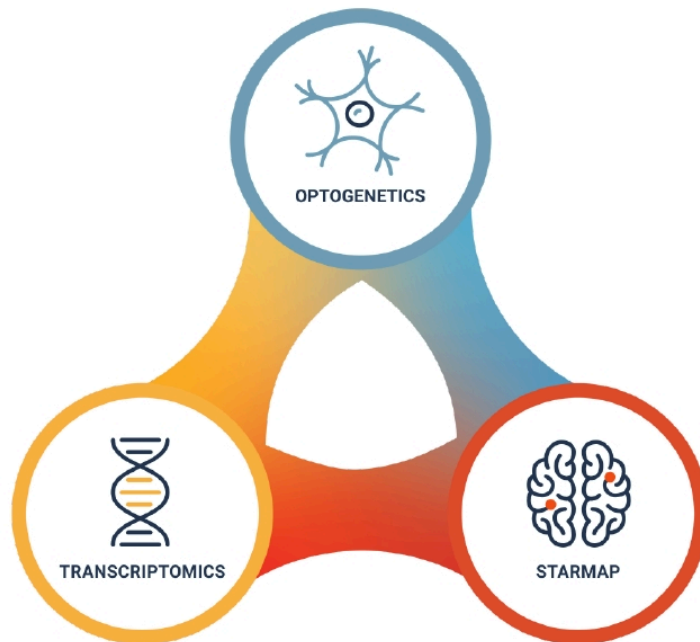
Company Overview

Maplight Therapeutics (MPLT) is a clinical-stage biopharmaceutical company focused on improving outcomes for patients with debilitating central nervous system (CNS) disorders. Founded by globally recognized leaders in psychiatry and neuroscience, MPLT addresses a critical gap in circuit-specific pharmacotherapies through its proprietary discovery platform. This platform identifies neural circuits causally linked to disease and targets them for therapeutic modulation, enabling the development of treatments with potential advantages in efficacy, safety, tolerability, and ease of use.

- Lead Program - ML-007C-MA:** MPLT's lead candidate, **ML-007C-MA**, is a fixed-dose combination of an M1/M4 muscarinic agonist (ML-007) and a peripherally acting anticholinergic (PAC) fesoterodine. It is designed to activate muscarinic receptors in the CNS while mitigating peripheral cholinergic side effects. ML-007C-MA is initially being developed for **schizophrenia** and **Alzheimer's disease psychosis (ADP)** - two areas with significant unmet need. Schizophrenia affects over 20 million people globally, including ~2-3 million in the U.S., and current dopaminergic antipsychotics only treat positive symptoms while causing serious side effects such as extrapyramidal, metabolic abnormalities, and QTc prolongation. ADP affects approximately 40% of the 7 million U.S. Alzheimer's patients, yet no FDA-approved therapies exist. ML-007C-MA has been evaluated in four Phase I trials involving N=270 participants and over 1,500 doses, demonstrating favorable safety, tolerability, and exposure without fasting requirements. The ongoing **ZEPHYR Phase II trial** for schizophrenia and **VISTA Phase II trial** for ADP are expected to deliver topline results in H2:26 and H2:27, respectively. We think ML-007C-MA could fetch \$1-2B + peak sales in schizophrenia (30-40% PoS) and \$1-2B+ peak sales in ADP (10-20% PoS).
- Additional Pipeline:** Beyond ML-007C-MA, MPLT has a pipeline of other compounds to provide sources of valuation upside. **(1) ML-004**, a 5-HT1B/1D agonist in Phase II (IRIS trial) for social communication deficits and irritability in autism spectrum disorder (ASD), with results anticipated in H2:26. **(2) ML-021**, an M4 antagonist in preclinical development for motor deficits in Parkinson's disease, targeting IND-enabling study completion in H2:26. **ML-009** is a GPR52 positive allosteric modulator (PAM) for hyperactivity and agitation-related disorders, with IND-enabling studies set to begin in 2026.

Strategic Positioning: The recent FDA approval of KRTX/BMY's **Cobenfy** (xanomeline and trospium chloride), a M1/M4 muscarinic agonist + mAChR antagonist, validates the therapeutic potential of muscarinic receptor modulation. However, Cobenfy has undesirable side effects (e.g., nausea/vomiting) and inconvenient dosing regimens (BID, or 2x daily for schizophrenia and T1D, or 3x daily for ADP) that involve slow titration. As such, we think MPLT could offer a differentiated therapy in ML-007C-MA with best in class potential by addressing limitations of current antipsychotics and muscarinics. Cobenfy is currently guided for multi-billion dollar potential across its many approved and pipeline indications (e.g., schizophrenia, ADP, etc.), with cons estimates reaching \$5B+. As such, our combined \$2-4B+ peak sales for ML-007C-MA (between schizophrenia and ADP) could be conservative.

MPLT's Discovery Platform for Addressing Neural Circuit-Specific Pharmacotherapies



Source: Company reports

Explore Causal Link Between Neural Circuit Activity and Disease Symptoms

Identify Potential Druggable Targets Within These Disease-Relevant Neural Circuits

Validate Expression, Selectivity and Location of Potential Targets in Brain Tissue

Schizophrenia Background and Treatment Landscape

Diagnosis

Schizophrenia is a common psychiatric disorder characterized by recurrent psychosis in that patients have a disconnected view of reality. Diagnosis requires patients to present two of five following symptoms (DSM-5 Criteria): Delusions, hallucinations, disorganized speech, disorganized or catatonic behavior, and/or negative symptoms. In addition, patients must experience social and/or occupational dysfunction for at least six months. Schizophrenic symptoms can be grouped into three categories:

- **Positive symptoms are psychotic behaviors** that rarely occur in healthy people, and people with positive symptoms could "lose touch" with reality. Common symptoms include (1) visual hallucinations (false perceptions, occurs in 40-80% of patients), (2) delusions (false beliefs, up to 80% of patients), (3) thought disorders (unusual ways of thinking) and (4) movement disorders (agitated movements).
- **Negative symptoms disrupt normal emotions and behaviors.** Symptoms include (1) "flat affect" (reduced expression of emotions), (2) diminished feelings of pleasure, (3) reduced speaking, (4) diminished motivation, and (5) impaired work and social functioning.
- **Cognitive symptoms can severely impede memory and thinking.** Some patients experience challenges of distilling information, paying attention or remembering things.

PANSS Is Considered the "Gold Standard" for Assessing Antipsychotic Efficacy

Introduced in a 1987 publication, the positive and negative symptoms scale (PANSS) utilizes a 45-minute interview to evaluate the severity of schizophrenia. Physicians score patients from 1 to 7 on 30 different symptoms, including positive symptoms, negative symptoms, and general psychopathology symptoms. As such, scores can range from 30 (min) to 210 points (max). The original 1987 publication tested the PANSS scale on over 100 schizophrenia patients, and determined that their mean scores were ~77 points, including 18.2 on the positive scale, 21.0 on the negative scale, and or 37.7 on the general psychopathology scale. *In comparison, we estimate the baseline PANSS scores across historical studies are around the 94-97 range on average.*

Exhibit 10 - Positive and Negative Syndrome Scale (PANSS)

Positive (p)	Score	Negative (n)	Score	Cognitive/ General (g)	Score
P1. Delusions	1-7	N1. Blunted Affect	1-7	N1. Somatic Concern	1-7
P2. Conceptual Disorganization	1-7	N2. Emotional Withdrawal	1-7	N2. Anxiety	1-7
P3. Hallucinatory Behavior	1-7	N3. Poor Rapport	1-7	N3. Guilt Feelings	1-7
P4. Excitement	1-7	N4. Passive/ Apathetic Social Withdrawal	1-7	N4. Tension	1-7
P5. Grandiosity	1-7	N5. Difficult in Abstract Thinking	1-7	N5. Mannerism/Posturing	1-7
P6. Suspiciousness/ Persecution	1-7	N6. Lack of Spontaneity and Convo Flow	1-7	N6. Depression	1-7
P7. Hostility	1-7	N7. Stereotyped Thinking	1-7	N7. Motor Retardation	1-7
Total	7-49	Total	7-49	N8. Uncooperativeness	1-7
				N9. Unusual Thought Content	1-7
				N10. Disorientation	1-7
				N11. Poor Attention	1-7
				N12. Lack of Judgement and Insight	1-7
				N13. Disturbing of Volition	1-7
				N14. Poor Impulse Control	1-7
				N15. Preoccupation	1-7
				N16. Active Social Avoidance	1-7
				Total	16-112

Each item is scored on presence/ severity: (1 = absent, 2 = minimal, 3 = mild, 4 = moderate, 5 = moderate severe, 6 = severe, 7 = extreme)
PANSS total score minimum = 30, maximum = 210
Source: Jefferies research, SciELO

Schizophrenia Affects ~3M Patients in the US

Schizophrenia occurs in ~1% of the general population (e.g., 3M US patients) and <1% among persons age 65 and older. Interestingly, the peak age of onset is 18-25 years in men and 26-45 years in women, although an earlier or later onset is still common. The specific cause is unknown though schizophrenia has a biologic basis. Also, schizophrenics have a lower life expectancy than the general population; a patient's lifespan is about 10 years less than the general population.

Side Effects Are Associated With Current Standard-of-Care (SoC) in Schizophrenia

Antipsychotic medications are used as first-line (1L) therapies, but they are known to cause significant side effects. In fact, literature suggests that 74% of US patients discontinue medication within 18 months. The current medications can be broken out into **typical** (first-gen) and **atypical** (second-gen) antipsychotics. Both classes of drugs are designed to block abnormally high dopamine signals, and atypical antipsychotics are also known to influence another chemical messenger called serotonin.

- **Typical antipsychotics (first-gen):** First introduced in the 1950s, the first-generation antipsychotics are known to cause Parkinson-like (extrapyramidal) side effects, impairing motor control and causing tremors, spasms, and muscle rigidity. In some cases, the symptoms could become permanent even if patients were to stop treatment. *In comparison, ML-007C-MA has not shown evidence of extrapyramidal side effects.*
- **Atypical antipsychotics (second-gen):** First introduced in the ~1990s, second-generation drugs offer a different side effect profile than the older agents, namely by reducing the instances of Parkinson-like effects. While a majority of schizophrenia patients in the US are treated with atypical antipsychotics, the newer drug class tends to cause weight gain, metabolic issues, non-insulin dependent (Type II) diabetes, cardiovascular side effects, sexual side effects, sleep disturbances, and motor disturbances. *In comparison, has not shown signs of weight gain, metabolic issues, or CV side effects at go-forward dose regimens.*
- **Muscarinic agonists:** Recently re-emerging as a novel therapeutic approach (e.g., Cobenfy), muscarinic receptor agonists aim to modulate cholinergic signaling in the brain, which plays a role in cognition and mood regulation. Unlike dopamine-targeting antipsychotics, these agents act on muscarinic receptors (particularly M1 and M4 subtypes) to potentially improve both psychotic and non-psychotic symptoms without the dopaminergic side effects seen in traditional therapies. Early clinical data suggest that muscarinic agonists may reduce psychosis while offering benefits in cognition and negative symptoms, though tolerability remains a concern due to risks of gastrointestinal disturbances (e.g., nausea/vomiting), and excessive salivation. These pot'l side effects result in gentle real-world titration for KRTX's Cobenfy. Fasting is also required, with the label indicating it must be taken an hour before a meal, or two hours afterward. *In comparison, ML-007C-MA may not cause as frequent vomiting, allows for a simpler titration regimen (single-dose in schizophrenia and one-week in ADP), and has no food effect.*
- **Most antipsychotics demonstrate only modest effectiveness (if any) in treating non-psychotic symptoms:** While antipsychotic drugs help reduce the psychotic or **positive symptoms** (e.g., hallucinations, delusions, suspiciousness) in about 70% of schizophrenia patients, they exhibit minimal efficacy on **negative symptoms** (e.g., diminished emotional expression, lack of motivation) and much of the **cognitive disturbances** attributed to schizophrenia. *In comparison, ML-007C-MA may be able to benefit negative symptoms and cognition akin to other muscarinic agonists (albeit this needs to be confirmed in large ongoing pbo-controlled studies).*
- **Discontinuations are frequent:** A 2005 NEJM publication discovered that 74% of patients taking typical or atypical antipsychotics discontinued treatment within 18 months, due to side effects

or a lack of efficacy, suggesting a large unmet need for new therapies such as lumateperone. Cobenfy showed discontinuation rates of 2-6% in Phase I and EMERGENT-1/2/3 studies. *In comparison, ML-007C-MA showed 0% pbo-adj discontinuation rates.*

Atypical Antipsychotic Background and History

Atypical antipsychotics, also known as second-generation antipsychotics, emerged in the late 20th century as a response to the limitations of first-generation (typical) antipsychotics, particularly their extrapyramidal side effects and limited efficacy in treating negative and cognitive symptoms of schizophrenia. The prototypical atypical agent, clozapine (Clozaril), was first synthesized in 1958 and demonstrated superior efficacy in treatment-resistant schizophrenia, albeit with serious risks such as agranulocytosis that initially hindered its widespread use. Its reintroduction in the 1990s, under strict monitoring protocols, catalyzed the development of a new class of antipsychotics including risperidone (**Risperdal**), olanzapine (**Zyprexa**), quetiapine (**Seroquel**), and aripiprazole (**Abilify**),

Unlike typical antipsychotics, which primarily act as dopamine D2 receptor antagonists, atypicals exhibit broader receptor binding profiles, targeting serotonergic (5-HT2A), dopaminergic (D1-D4), adrenergic, histaminergic, and even muscarinic receptors. This polypharmacology was initially thought to confer improved efficacy and tolerability, particularly in reducing extrapyramidal symptoms and enhancing treatment of negative symptoms. However, accumulating evidence has challenged the notion of a class-wide superiority, suggesting that individual drug profiles, rather than generational labels, better predict clinical outcomes.

Treatment Resistance and Positive Symptom Limitations: Approximately ~30% of patients with schizophrenia exhibit treatment resistance to atypical antipsychotics (e.g., treatment-resistant schizophrenia or TRS), defined as inadequate symptom control despite trials of at least two antipsychotic agents. While clozapine remains the gold standard for treatment-resistant cases, its efficacy is limited to 25-30%+ in this population, and its side-effect burden of metabolic syndrome and hematologic risks remains substantial. Atypicals also **mostly address positive symptoms** (e.g., hallucinations, delusions), with **minimal impact on negative symptoms** (e.g., anhedonia, avolition) and **cognitive domains** (e.g., memory, executive function). Cognitive impairment, affecting up to 80% of patients, is increasingly recognized as a core feature of schizophrenia, yet remains largely unresponsive to current pharmacotherapies.

Exhibit 11 - Comparison of Selected Atypical Antipsychotics (Second-generation antipsychotics)

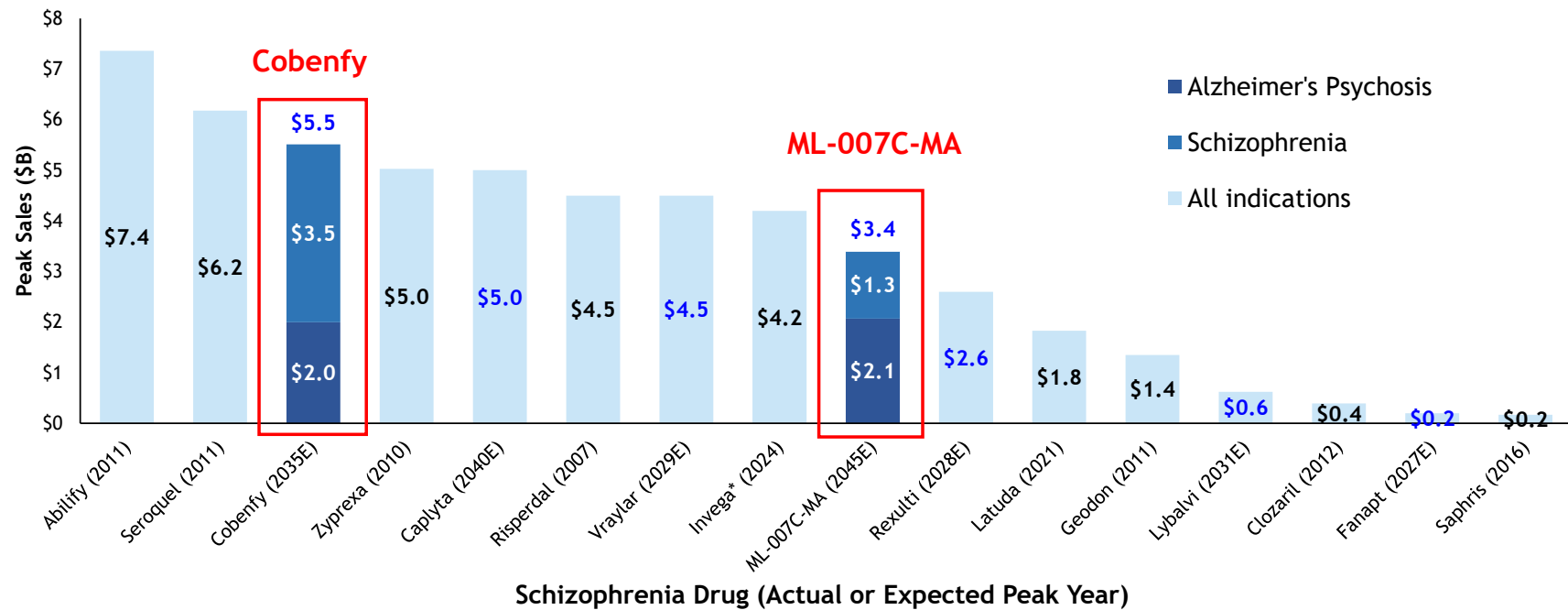
Company	Branded	Generic	Mode	Indication**	Approved	Peak Sales	Peak Year	Blackbox?	Safety Notes
Novartis (NVS)	Clozaril	clozapine	Oral	TR-SCZ, Reducing Suicide Risk	1989	\$0.4B	~2012	Neutropenia, Hypotension, Bradycardia, Syncope, Mortality, Seizure, Myocarditis, Cardiomyopathy	White blood cell decrease, heart inflammation
Janssen (JNJ)	Risperdal	risperidone	Oral	SCZ, BP1 (Manic/Mixed)*, ASD (Irritability)	1993	\$4.5B	2007	Mortality	Less sedating but more extrapyramidal side effects
Eli Lilly (LLY)	Zyprexa	olanzapine	Oral, IM	SCZ, BP1 (Manic/Mixed), BP1 depression*, TRD*	1996	\$5.0B	2010	Mortality	Less extrapyramidal side effects but increases weight and blood sugar
AstraZeneca (AZN)	Seroquel	quetiapine	Oral	SCZ, BP1 (Manic), Bipolar depression	1997	\$6.2B	2011	Mortality, Suicidality	Less motor side effects but more weight gain and hypertension
Pfizer (PFE)	Geodon	ziprasidone	Oral , IM	SCZ, BP1 (Manic/Mixed)*	2001	\$1.4B	2011	Mortality	Heart arrhythmia, hypertension, dizziness
Otsuka/ Bristol Myers (BMJ)	Abilify	aripiprazole	Oral , IM	SCZ, BP1 (Manic/Mixed), MDD*, ASD (Irritability), TS	2002	\$7.4B	2011	Mortality, Suicidality	Weight gain, headache, agitation, anxiety, insomnia, nausea
Johnson & Johnson (JNJ)	Invega/ Invega Sustenna	paliperidone	IM	SCZ, SZA	2006	\$4.2B	2024	Mortality	Restlessness, weight gain, sedation
Allergan (AGN)	Saphris	asenapine	Oral	SCZ, BP1 (Manic/Mixed)	2009	\$0.2B	2016	Mortality	Stroke, ischemic attacks, QT prolongation, seizures, hypotension
Vanda (VND)	Fanapt/ Zomaril	lisperidone	Oral	SCZ, BP1 (Manic/Mixed)	2009	\$0.2B	2027	Mortality	Weight gain, seizures, hypotension, suicide, cognitive impairment
Sunovion	Latuda	Lurasidone	Oral	SCZ, BP1 depression	2010	\$1.8B	2021	Mortality, Suicidality	Stroke, ischemic attack, neuroleptic malignant syndrome, weight gain
Otsuka/ Lundbeck	Rexulti	brexpiprazole	Oral	SCZ, MDD, ADA*	2015	\$2.6B	2028	Mortality, Suicidality	Stroke, ischemic attacks, weight gain, hypotension, seizures
Allergan (AGN)/ AbbVie (ABBV)	Vraylar	cariprazine	Oral	SCZ, BP1 (Manic/Mixed), BP1 depression, MDD	2015	\$4.5B	2029	Mortality, Suicidality	Stroke, ischemic attacks, weight gain, hypotension, seizures
Intracellular (ITC)/ Johnson & Johnson (JNJ)	Caplyta	lumateperone	Oral	SCZ, Bipolar depression, MDD	2019	\$5.0B	2040	Mortality	Stroke, ischemic attacks, weight gain, metabolic changes, hypotension, seizures
Alkermes (ALKS)	Lybalvi	olanzapine and samidorphan	Oral	SCZ, BP1 (Manic/Mixed)	2021	\$0.6B	2031	Mortality	Stroke, ischemic attacks, weight gain, metabolic changes, hypotension, seizures

* Efficacy Comparison of Approved Second-Generation Antipsychotics (Positive, Placebo-Controlled Datasets Mentioned in Their Respective FDA Labels)

** Schizophrenia (SCZ); Treatment-resistant schizophrenia (TR-SCZ); Schizoaffective disorder (SZA); Bipolar 1 disorder (BP1); Major depressive disorder (MDD); Tourettes syndrome (TS); Autism spectrum disorder (ASD); Treatment-Resistant Depression (TRD); Alzheimer's Disease Agitation (ADA)

Source: Jefferies research, Factset, Visible Alpha, Company reports

Exhibit 12 - We Think The Sales Precedents (Either Historical or Projected) of Recent Antipsychotics/Muscarinics in CNS Indications Support a Multiple \$2-4B Peak Sales Opportunity for MPLT's ML-007 (Unadjusted).
MPLT Is Currently in Phase II Studies in Schizophrenia (Data in H2:26) and Alzheimer's Psychosis (Data in H2:27). All Other Listed Drugs Are Approved (Except KRTX for ADP)



Source: Company reports, Visible Alpha, Jefferies Estimates
Company reports, Visible Alpha, Jefferies Estimates

Muscarinic Agonists (and Various Approaches) - Background and History

Muscarinic acetylcholine receptors (mAChRs) have been recognized for centuries as therapeutic targets, with ancient uses of muscarinic antagonists for respiratory ailments and more recent applications in conditions like COPD, overactive bladder, and glaucoma. The mAChR family consists of five subtypes (M1-M5), with M1 and M4 emerging as primary targets for neurological diseases such as schizophrenia and Alzheimer's disease psychosis/cognition.

Despite early promise, the field faced significant early setbacks. First-generation muscarinic agonists and antagonists, while effective in preclinical models, were plagued by dose-limiting cholinergic side effects, including salivation, gastrointestinal distress, cardiovascular effects, and seizures due to poor receptor subtype selectivity. Notably, clinical trials of xanomeline (an M1/M4-preferring agonist) in the 1990s showed symptomatic benefit in AD and schizophrenia but were halted due to intolerable peripheral side effects. Similar challenges led to the discontinuation of other programs by other biopharmaceutical companies.

That said, there has been a rejuvenation of advancements in the muscarinic drug class lately, largely driven by the following factors:

- **Structure-Based Drug Design:** High-resolution crystal structures of mAChRs have enabled rational design of ligands with improved subtype and functional selectivity.
- **Allosteric Modulation:** Positive allosteric modulators (**PAMs**) bind to sites distinct from the orthosteric acetylcholine binding pocket, offering greater receptor subtype selectivity and a ceiling effect that reduces overdose risk. **M₁-** and **M₄-**selective PAMs have shown cognitive and antipsychotic benefits in preclinical models, with improved tolerability profiles compared to orthosteric agonists. However, achieving the right balance between intrinsic agonist activity and co-operativity with endogenous acetylcholine remains a challenge.
- **Pharmacological Tailoring:** Recent insights into receptor biology, such as the concepts of receptor reserve and signaling bias, allow for the design of ligands that preferentially activate beneficial signaling pathways (e.g., arrestin-dependent pathways for cognition) while avoiding those that mediate adverse effects. This has led to the development of partial agonists and biased ligands that can selectively target CNS circuits without triggering peripheral toxicity.
- **Peripheral Antagonist Pairing:** A major advancement in muscarinic agonist development has been the use of peripherally acting muscarinic antagonists (PACs) co-formulated with central muscarinic agonists. This strategy aims to preserve CNS efficacy while reducing cholinergic side effects, particularly gastrointestinal (GI) symptoms such as nausea, vomiting, and diarrhea, which have historically limited the clinical utility of muscarinic agonists. The recent FDA approval of **Cobenfy** validated this approach. Similarly, **MPLT's ML-007C-MA** employs this co-formulation strategy, enabling pot'l convenient dosing without fasting requirements and improved tolerability.

Roles of M1 and M4 agonism in schizophrenia treatment: M1 agonism might have a larger role in cognitive/negative symptoms of schizophrenia, while M4 primarily modulates dopaminergic tone, reducing hyperdopaminergic signaling linked to positive symptoms (e.g., hallucinations/delusions). As such, dual M1/M4 agonist might have a greater capability of encompassing the entire spectrum of schizophrenia symptoms on PANSS.

Exhibit 13 - Muscarinic Agonist Comparison Table

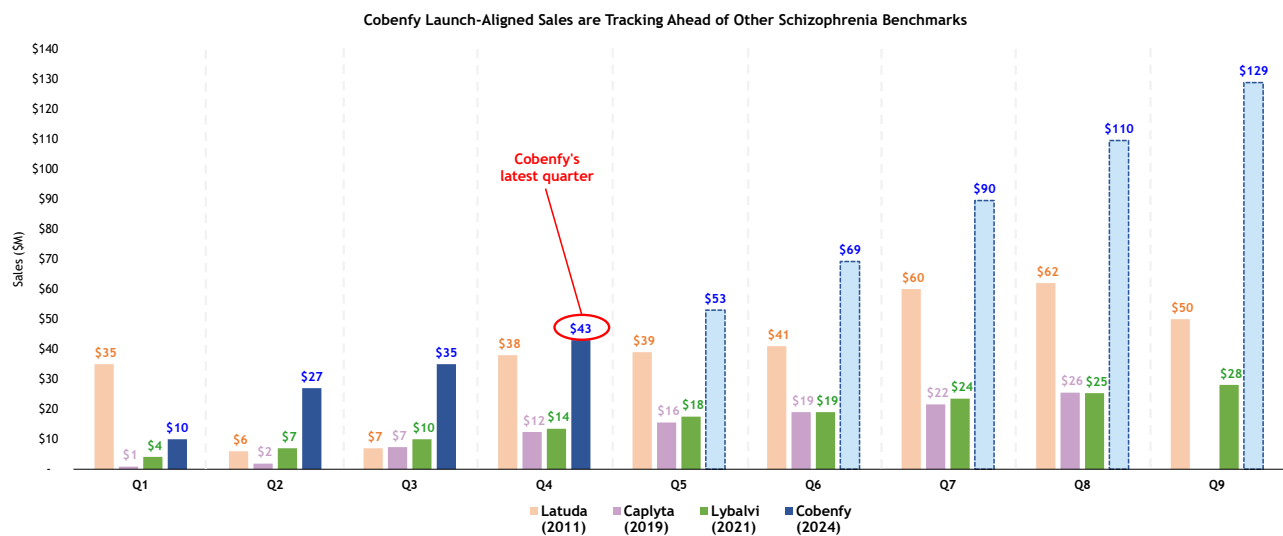
Company	Drug	Target	Lead Indication(s)	Status
Karuna, Bristol Myers Squibb (KRTX/BMY)	Cobenfy (xanomeline and trospium)	M1/M4 agonist + mAChR antagonist	Schizophrenia, ADP, ADC, ADA, BP-I	Approved (SCZ) 2024
Maplight (MPLT)	ML-007C-MA	M1/M4 agonist + mAChR antagonist	Schizophrenia, ADP	Phase II
Cerevel, AbbVie (CERE/ABBV)	Emraclidine	M4 PAM	Schizophrenia, ADP	Phase II
Neumora (NMRA)	NMRA-861	M4 modulator	Schizophrenia	Phase I
	NMRA-898	M4 modulator	Schizophrenia	Early discovery
	NMRA-266	M4 PAM	Schizophrenia	Phase I (clinical hold)
Neurocrine, Sosei (NBIX)	NBI-567	M1 agonist	CNS indications	Phase I
	NBI-568 (direclidine)	M4 agonist	Schizophrenia	Phase III
	NBI-569	M4 agonist	CNS indications	Phase I
	NBI-570	M1/M4 agonist	Schizophrenia	Phase I
	NBI-986	M1/M4 antagonist	Movement disorders	Phase I
Terran Biosciences	TerXT	M1/M4 agonist + mAChR antagonist pro-drugs	Schizophrenia	TBD
Gilgamesh	Undisclosed	M1/M4 agonist	Schizophrenia	Early discovery
Neurosterix	Undisclosed	M4 PAM	Schizophrenia	-
Takeda (TAK)	TAK-071	M1 PAM	Parkinson's	Discontinued
Merck (MRK)	MK-7622	M1 PAM	Alzheimer's disease	Discontinued

Source: Jefferies, Company reports

Cobenfy Launch Progress

BMJ has guided Cobenfy to be a pot'l multi-billion dollar product across several indications, including schizophrenia, bipolar-I mania, and alzheimer's disease psychosis (ADP), cognition, and agitation (ADA). For reference, BMJ acquired KRTX for \$14B over 2023-24. Despite some investor skepticism around the commercial success so far, Cobenfy's trajectory since its Q4:24 launch is arguably to be expected for a rollout in a tough indication like schizophrenia. While uptake has indeed been gradual, this is consistent with the entrenched prescribing behaviors in schizophrenia, which is a market that has been dominated by antipsychotic generics for decades. Importantly, BMJ mgmt has repeatedly emphasized that the launch is tracking ahead of all branded schizophrenia benchmarks. Indeed, this point holds up against Latuda, Caplyta, and Lybalvi launches - three branded schizophrenia launches that were initially in launched in schizophrenia only, or whose sales split in schizophrenia can be inferred from management commentary (e.g., Lybalvi).

Exhibit 14 - Cobenfy Sales are Tracking and Expected to Outpace Other Approved Schizophrenia Benchmarks When Launch Aligned



Source: Company reports, Jefferies, Visible Alpha
Latuda first quarter sales were impacted by drug stocking. Caplyta bipolar depression approval granted Dec 20, 2021 (late in Q8 of launch). Latuda bipolar approval granted June 2013, during 10th quarter of launch. Lybalvi approved in BP1 and SCZ initially in June 2021, but 50% of sales inferred to be attributed to SCZ based on company commentary. Dashed Cobenfy bars represent cons estimates from Visible Alpha

Key Launch Metrics:

- Prescriptions:** Weekly TRx volumes have climbed from ~1,000 in Q4:24 to over 2,000 by Q2:25 and surpass 2,400 in Q3:25. This steady growth aligns with expectations for a paradigm-shifting therapy in a more conservative, generic/antipsychotic-dominated treatment landscape.
- Revenue:** Net sales progressed from \$10M in Q4:24 (partial quarter) to \$27M in Q1:25, \$35M in Q2:25, and \$43M in Q3:25 totaling \$105M year-to-date. Management anticipates stronger H2 performance as access and physician familiarity deepen.
- Access:** Virtually 100% coverage across Medicaid and Medicare was secured within months - a gating factor for uptake given >80% of schizophrenia patients are in these channels.
- Physician Feedback:** Feedback remains highly positive, citing robust efficacy across positive and negative symptoms and cognition. Intent-to-prescribe metrics are strong, and trialist growth continues weekly.

Key Levers to Growth Expansion:

- Expansion into hospital settings. BMJ has cited that 20-25% of new schizophrenia starts originate here, then transition into community settings.

- Peer-to-peer education and Phase IV switch study (readout early 2026) to accelerate conversion from entrenched D2 therapies.
- Broader lifecycle management potential expansions into bipolar-I mania, alzheimer's disease psychosis (ADP), alzheimer's disease cognition, and alzheimer's disease agitation (ADA).

Nevertheless, we think key features of ML-007C-MA could make it differentiated from KRTX's Cobenfy, which could translate to improved commercial success:

(1) **ML-007C-MA may have an improved tolerability profile.** The rates of vomiting in ML-007C-MA's Phase I study (0%) on N=6 are below that of Cobenfy in a similar-stage KAR-003 Phase I trial (e.g., 28% on N=18). We acknowledge that nausea is technically not improved (e.g., 50% for ML-007C-MA and 22% for Cobenfy), but the lack of a clear vomiting signal could suggest that nausea signals are less severe than Cobenfy. Additionally, ML-007C-MA's signal on a small N=6 could improve on a larger sample size.

(2) **ML-007C-MA may be eligible for less frequent and more convenient dosing than Cobenfy's BID/TID (2x/3x daily) dosing** and fasting requirements (1 hour before or 2 hours after meals). The Phase II ZEPHYR study is evaluating both QD and BID dosing regimens for ML-007C-MA and the Phase II VISTA program is evaluating BID dosing in ADP. The Phase I Study 013 for ML-007C-MA also revealed no food effect, so in the ongoing Phase II trials, ML-007C-MA will be dosed close to a meal. Fasting requirements and BID/TID dosing for Cobenfy could represent significant limitations for both elderly and schizophrenia patients.

(3) **We expect ML-007C-MA to have an advantage on drug titration as well, alleviating patient/physician burdens and potentially improving real-world discontinuations.** Real-world Cobenfy patients (per BMY) are slowly titrated from 50-100mg/20mg over 1-2 weeks compared to the 8-day titration from clinical trials. The Cobenfy label recommends dose titration from 50-100/20mg for at least two days, and a final increase to 125mg/20mg based on patient tolerability and response. Within the Phase III ADEPT-2/3/4 studies for ADP, Cobenfy requires a 4-5-week titration period. ML-007C-MA is using a short one-dose titration in the ZEPHYR Phase II schizophrenia and a 1-3 week titration for the VISTA ADP program.

(4) **Per BMY commentary, the apparent physician enthusiasm does validate muscarinics as a class, which is arguably (+) for MPLT.** Most notable is feedback on cognition - a key differentiating feature compared to antipsychotics. Therefore, the bar for "success" for ML-007C-MA remains favorable, with muscarinics ushering in a differentiated approach to schizophrenia treatment.

(5) **Physician familiarity with muscarinics may continue to grow over time, before ML-007C-MA would reach the market in 2029-30.** We'd argue that barriers to adoption (e.g., payor access, entrenched habits) for Cobenfy are structural, so we expect that MPLT could leverage these learnings to optimize their own pot'l launch sequencing. Structural limitations to the Cobenfy launch may also be refined by the time MPLT makes it to market, enabling a "second-mover" advantage.

Net-net: ML-007C-MA offers a cleaner tolerability profile, simpler QD/BID dosing without food restrictions, and faster titration versus Cobenfy's regimen. With growing muscarinic adoption and Cobenfy paving the way, MPLT is well positioned for a meaningful second-mover advantage.

Exhibit 15 - Cobenfy Dosing Instructions Recommend a One Week Titration and Fasting Requirement

-----INDICATIONS AND USAGE-----

COBENFY is a combination of xanomeline, a muscarinic agonist, and trospium chloride, a muscarinic antagonist, indicated for the treatment of schizophrenia in adults. (1)

-----DOSAGE AND ADMINISTRATION-----

- Assess liver enzymes and bilirubin prior to initiating treatment with COBENFY and as clinically indicated during treatment. (2.1)
- Assess heart rate at baseline and as clinically indicated during treatment with COBENFY. (2.1)
- Recommended starting dosage of COBENFY is 50 mg/20 mg orally twice daily for at least two days, then increase the dosage to 100 mg/20 mg twice daily for at least five days. (2.2)
- Dosage may be increased to 125 mg/30 mg orally twice daily based on patient tolerability and response. (2.2)
- See the full prescribing information for the recommended titration and maximum recommended dosage. (2.2)
- Take at least 1 hour before a meal or at least 2 hours after a meal. Do not open capsules. (2.2)
- Geriatric patients: Recommended starting dosage of COBENFY is 50 mg/20 mg orally twice daily. Consider a slower titration. The maximum recommended dosage is 100 mg/20 mg twice daily. (2.3)

-----DOSAGE FORMS AND STRENGTHS-----

Capsules (xanomeline/trospium chloride): 50 mg/20 mg, 100 mg/20 mg, 125 mg/30 mg (3)

Source: FDA label

Efficacy Comparisons of Current Schizophrenia Drugs

We analyzed 15+ different drugs from 50+ well-controlled Phase II/III trials (incl. N=1 Phase Ib) to ascertain typical PANSS reductions across successful and unsuccessful trials. Of the atypical antipsychotics, we see an average Week 4 PANSS reduction of **-16.8**, vs **-10.0** pbo, which corresponded to an average pbo-adj effect of **-6.7**. On Week 6 PANSS reductions, we see average reductions of **-18.6** vs **-10.7** pbo, corresponding to an average effect size of **-8.0**. In general, median reductions look in-line/similar to averages across both time points. For muscarinics, of N=7 schizophrenia trials, we see average Week 4-6 PANSS reductions of **-18.2** and **-11.0** for pbo, corresponding to an average pbo-adj effect size of **-7.2**. **Net-net:** Efficacy data for muscarinic trials (including N=2/7 trial failures) suggests the class can deliver at least comparable efficacy (albeit differentiated toward negative/cognitive symptoms) on PANSS compared to established antipsychotics.

We also analyzed placebo effects for schizophrenia drugs across 50+ placebo-controlled trials over the years. The average placebo effect at Week 4 was **-9.2**, **-9.9** at Week 5, and **-10.2** at Week 6. Interestingly, placebo effects did seem to increase in magnitude from trials for drugs approved in 2000-2009 at Week 4-6 (**-8.3-8.5**) to 2010-2019 (**-11.2-13.1**). However, in drugs approved from 2020+ (including those not yet approved), trials have shown generally lower average placebo responses of **-10.2-11.0** at Week 4-6. We think sponsors could be getting more disciplined about trial conduct and patient/site selection. We assume modern schizophrenia programs could be leaning into tighter baseline severity thresholds, and multi-step eligibility reviews to screen out early improvers and "professional" patients. Companies may also be carefully selecting sites, confining geographies, potentially contributing to blinded data quality. Finally, more centralized review, risk-based monitoring, and rater/participant training is also likely contributing to some extent. The net effect is less heterogeneity in placebo arms, intuitively consistent with our findings finding that Week 4-6 placebo changes have eased from the 2010s to 2020+. That said, data will likely continue to evolve through the remainder of the decade.

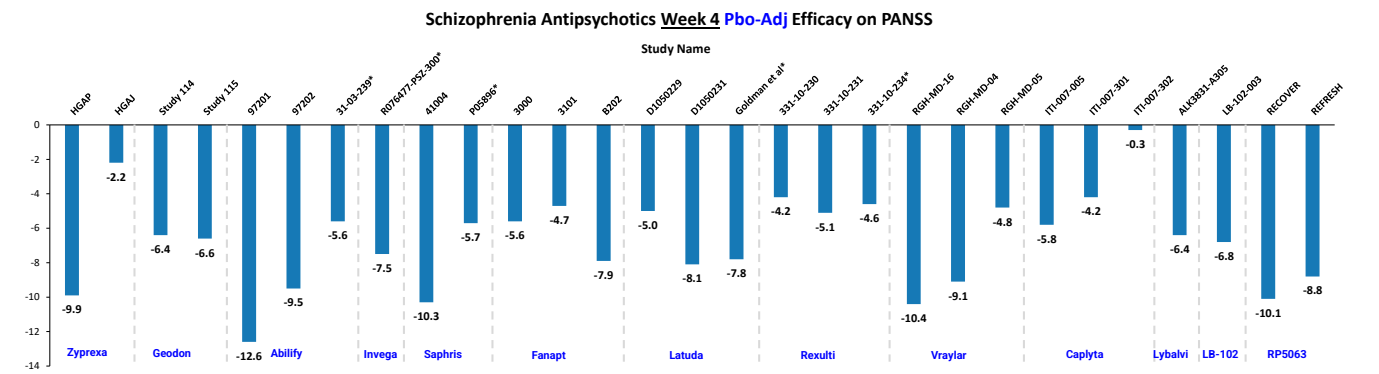
MPLT is minimizing placebo (pbo) response in its Phase II schizophrenia trial through tightly controlled operations and quality safeguards. The Phase II ZEPHYR study is running at 25 US sites, each extensively vetted using rater quality/concordance metrics from recent acute schizophrenia studies, recent enrollment in positive acute trials, 1:1 PI interviews, and external consultant input. Site-level enrollment is gated by early data quality to prevent rater drift. Participants must have moderate-to-severe symptoms at screening and baseline, with exclusion of >20% improvement from screening to baseline, and complete a 3-step eligibility review plus a duplicate-patient exclusion process. The N=300 enrollment numbers are conservative to avoid quality erosion. Because diminished quality control drives pbo response, MPLT has in-sourced site relationships and data-quality oversight, deploying its own cadre of CRAs, each covering a small number of sites on this single study for real-time monitoring and rapid intervention. Quality is reinforced by independent central review and a regular cadence of blinded data analytics focused on high rater quality and concordance, enabling early detection of outliers without unblinding. Human-factor risks are addressed via placebo-response training for all participant-facing staff and expectancy-bias education for participants. Finally, MPLT selected a CRO with a strong recent schizophrenia track record to align vendor execution to these controls. **Net-net:** Limited sites and geography, strict and multi-layered eligibility, gated and conservative enrollment, in-house CRA oversight with real-time monitoring, blinded central analytics, and targeted training create a focused framework to suppress pbo and preserve a pot'l pbo-drug separation.

Exhibit 16 - Cross Trial Efficacy Comparison of Atypical Antipsychotic Schizophrenia Drugs Sorted by Drug Approval Date

Drug/Study Information					4 Week PANSS				6 Week PANSS		
Company	Name	Molecule	Approval	Study	BL	Drug	Pbo	Pbo-adj	Drug	Pbo	Pbo-adj
LLY	Zyprexa	olanzapine	1996	HGAP	98.3	-11.9	-2.0	-9.9	-12.3	-1.9	-10.4
				HGAJ	90.1	-15.0	-12.8	-2.2	-17.7	-13.4	-4.3
AZN	Seroquel	quetiapine	1997	Study 4*	96.9	-	-	-	-28.4	-19.2	-9.2
PFE	Geodon	ziprasidone	2001	Study 114	95.8	-20.2	-13.8	-6.4	-23.5	-21.0	-2.5
				Study 115	89.5	-8.4	-1.8	-6.6	-8.3	-0.4	-7.9
				Study 303	85.2	-	-	-	-3.6	1.7	-5.3
BMY	Abilify	apiprazole	2002	97201	97.9	-15.5	-2.9	-12.6	-	-	-
				97202	92.6	-14.5	-5.0	-9.5	-	-	-
				138001	92.8	-	-	-	-15.0	-2.3	-12.7
				Study 4	90.0	-	-	-	-11.3	-5.3	-6.0
				31-03-239*	94.0	-24.6	-19.0	-5.6	-28.6	-21.2	-7.4
JNJ	Invega	paliperidone	2006	R076477-SCH-303	94.6	-	-	-	-23.3	-4.1	-19.2
				R076477-SCH-304	94.1	-	-	-	-17.5	-8	-9.5
				R076477-SCH-305	94.6	-	-	-	-23.3	-4.1	-19.2
				R076477-PSZ-300*	90.6	-15.6	-8.1	-7.5	-18.2	-8.1	-10.1
AGN	Saphris	asenapine	2009	041004	96.5	-16.9	-6.6	-10.3	-15.9	-5.3	-10.6
				041023	88.9	-	-	-	-16.2	-10.7	-5.5
				041021	90.8	-	-	-	-14.5	-11.1	-3.4
				041022	86.9	-	-	-	-9.4	-9.9	0.5
				P05896*	98.6	-18.4	-12.7	-5.7	-	-	-
VNDA	Fanapt	iloperidone	2009	3000	95.8	-8.7	-3.1	-5.6	-8.6	-4.1	-4.5
				3101	92.9	-11.5	-6.8	-4.7	-	-	-
				B202	-	-17.5	-9.6	-7.9	-26.8	-18.5	-8.3
Sunovion	Latuda	lurasidone	2010	D1050006	90.0	-	-	-	-17.0	-6.2	-10.8
				D1050196	94.4	-	-	-	-14.1	-5.5	-8.6
				D1050229	96.0	-19.1	-14.1	-5.0	-23.4	-17.0	-6.4
				D1050231	96.6	-21.2	-13.1	-8.1	-25.7	-16.0	-9.7
				D1050049	93.1	-	-	-	-13.6	-12.3	-1.3
				D1050233	97.5	-	-	-	-26.5	-10.3	-16.2
				Goldman et al*	94.5	-17.0	-9.2	-7.8	-18.6	-10.5	-8.1
Lundbeck, Otsuka	Rexulti	brexpiprazole	2015	331-10-230	95.1	-15.5	-11.3	-4.2	-20.0	-13.5	-6.5
				331-10-231	95.9	-14.1	-9.0	-5.1	-20.7	-12.0	-8.7
				331-07-203	96.2	-	-	-	-22.0	-17.3	-4.7
				331-10-234*	101.1	-17.8	-13.2	-4.6	-22.8	-17.4	-5.4
AGN	Vraylar	cariprazine	2015	RGH-MD-16	96.7	-18.3	-7.9	-10.4	-20.2	-9.5	-10.7
				RGH-MD-04	95.7	-18.1	-9.0	-9.1	-23.0	-14.3	-8.7
				RGH-MD-05	96.3	-21.1	-16.3	-4.8	-29.1	-19.2	-9.9
ITCI, JNJ	Caplyta	lumateperone	2019	ITI-007-005	88.1	-13.2	-7.4	-5.8	-	-	-
				ITI-007-301	90.1	-14.5	-10.3	-4.2	-	-	-
				ITI-007-302	90.4	-13.6	-13.3	-0.3	-14.6	-15.1	0.5
ALKS	Lybalvi	olanzapine + samidorphan	2021	ALK3831-A305	101.8	-23.9	-17.5	-6.4	-	-	-
LBRX	LB-102	N-methyl amisulnride	Phase III	LB-102-003	93.9	-16.1	-9.3	-6.8	-	-	-
RVPH	RP5063	brilaroxazine	Phase III	RECOVER	99.1	-23.9	-13.8	-10.1	-	-	-
				REFRESH	85.9	-20.2	-11.4	-8.8	-	-	-
AVERAGE:					93.9	-16.8	-10.0	-6.8	-18.6	-10.7	-8.0
MEDIAN:					94.5	-16.9	-9.6	-6.4	-18.4	-10.6	-8.2

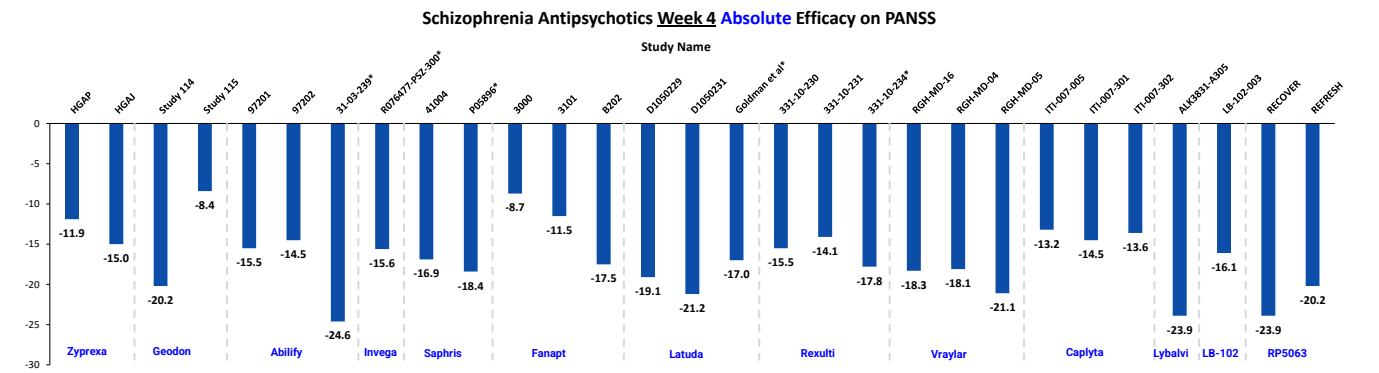
Source: Jefferies, Company reports, FDA review documents
*Pediatric/Adolescent study, BL is baseline PANSS score

Exhibit 17 - Schizophrenia Atypical Antipsychotics Week 4 Pbo-adj Efficacy on PANSS



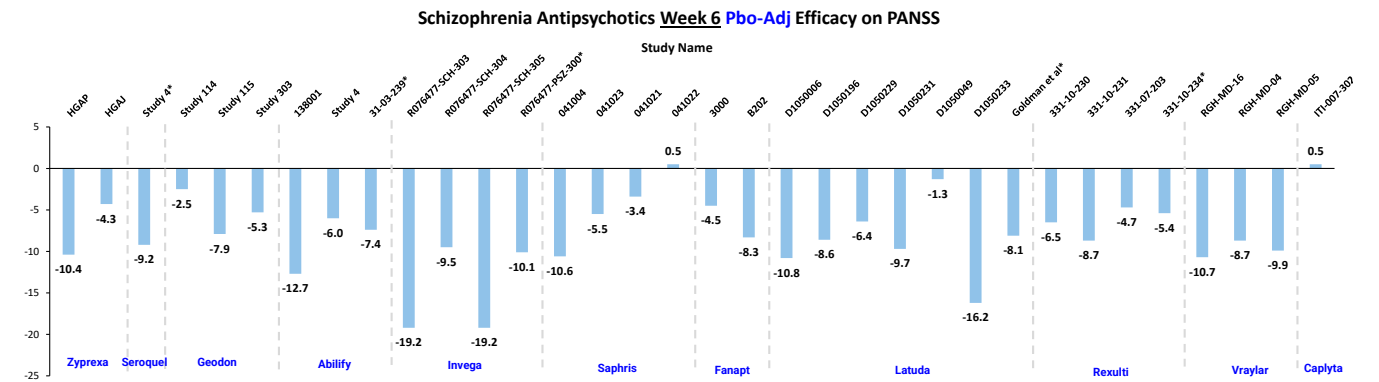
Source: FDA review documents, Company reports, Jefferies estimates
Some values may be estimated by Jefferies from company-reported charts/materials

Exhibit 18 - Schizophrenia Atypical Antipsychotics Week 4 Absolute Efficacy on PANSS



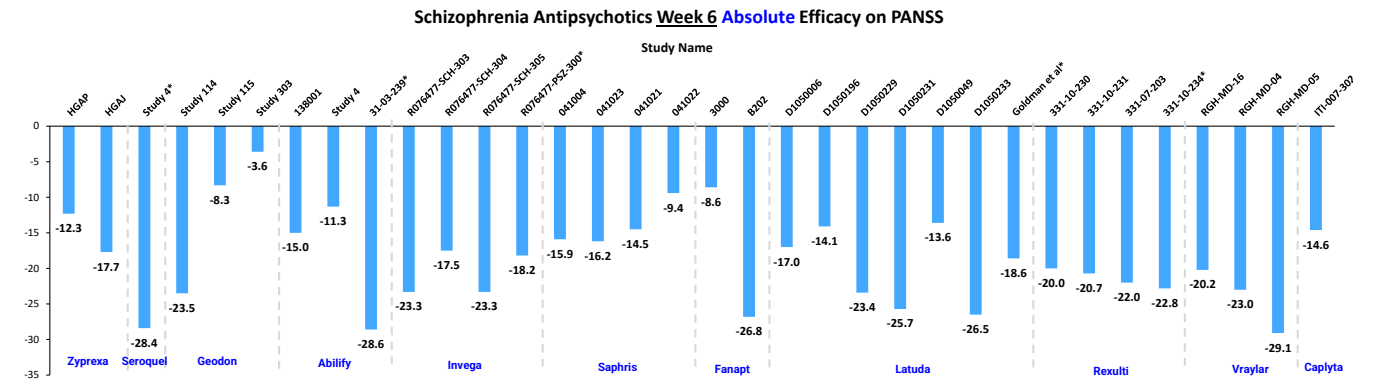
Source: FDA review documents, Company reports, Jefferies estimates
Some values may be estimated by Jefferies from company-reported charts/materials

Exhibit 19 - Schizophrenia Atypical Antipsychotics Week 6 Pbo-adj Efficacy on PANSS



Source: FDA review documents, Company reports, Jefferies estimates
Some values may be estimated by Jefferies from company-reported charts/materials

Exhibit 20 - Schizophrenia Atypical Antipsychotics Week 6 Absolute Efficacy on PANSS



Source: FDA review documents, Company reports, Jefferies estimates
Some values may be estimated by Jefferies from company-reported charts/materials

Exhibit 21 - Cross Trial Efficacy Comparisons of Muscarinic Agonists

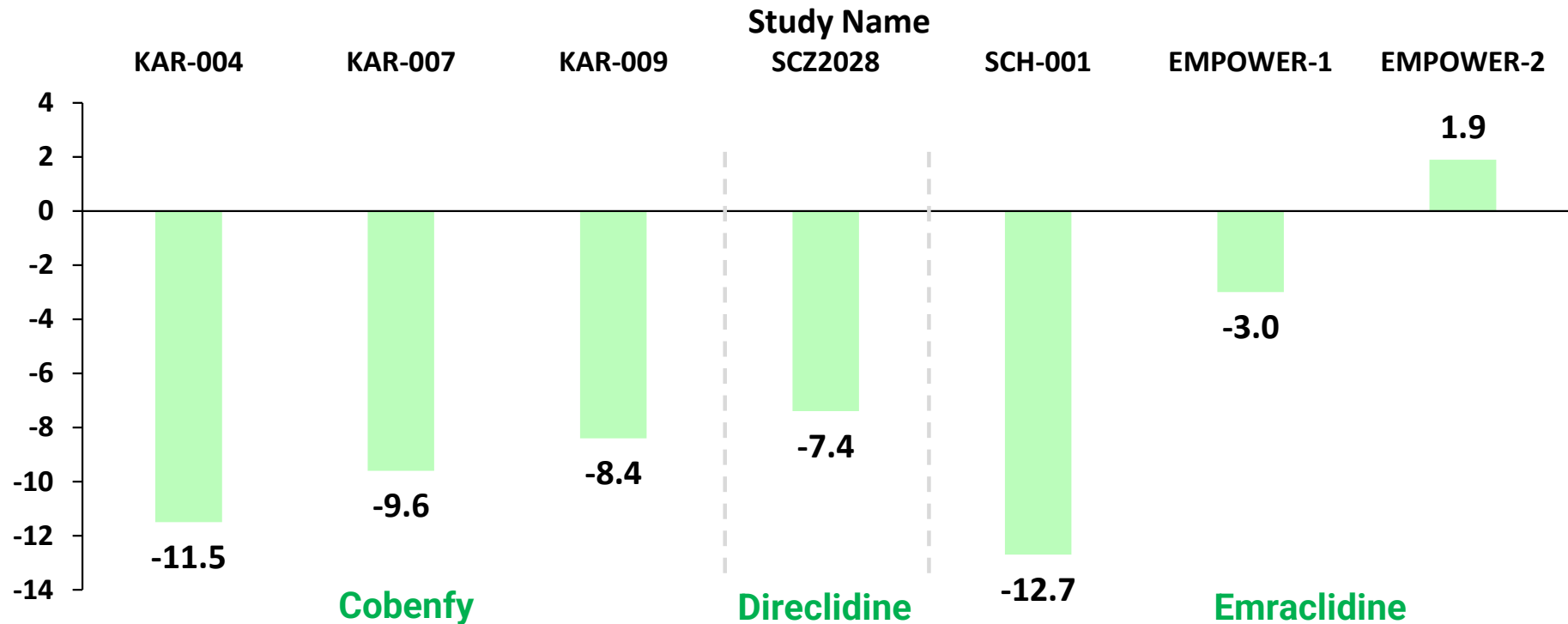
Drug/Study Information							4-6 Week PANSS*			
Company	Name	Molecule	MOA	Status	Study	Study Phase	BL	Drug	Pbo	Pbo-adj
KRTX, BMY	Cobenfy	xanomeline + trospium Cl	M1-M4 + anti-cholinergic	Approved 2024	KAR-004	II	-	-17.4	-5.9	-11.5
					KAR-007	III	98.2	-21.2	-11.6	-9.6
					KAR-009	III	96.5	-20.6	-12.2	-8.4
NBIX	NBI-1117568	direclidine	Selective M4 orthosteric agonist	Phase III	SCZ2028	II	97.0	-18.2	-10.8	-7.4
CERE, ABBV	CVL-231	emraclidine	Selective M4 receptor PAM	Phase II	SCH-001	Ib	93.0	-19.5	-6.8	-12.7
					EMPOWER-1	II	98.3	-16.5	-13.5	-3.0
					EMPOWER-2	II	97.2	-14.2	-16.1	1.9
AVERAGE:							96.7	-18.2	-11.0	-7.2
MEDIAN:							97.1	-18.2	-11.6	-8.4

Source: Jefferies, Company reports, FDA review documents

eharrison@maplightrx.com - Maplight Therapeutics - 27/03/2026 06:21 AM UTC

Exhibit 22 - Schizophrenia Muscarinics Week 4-6 Pbo-adj Efficacy on PANSS

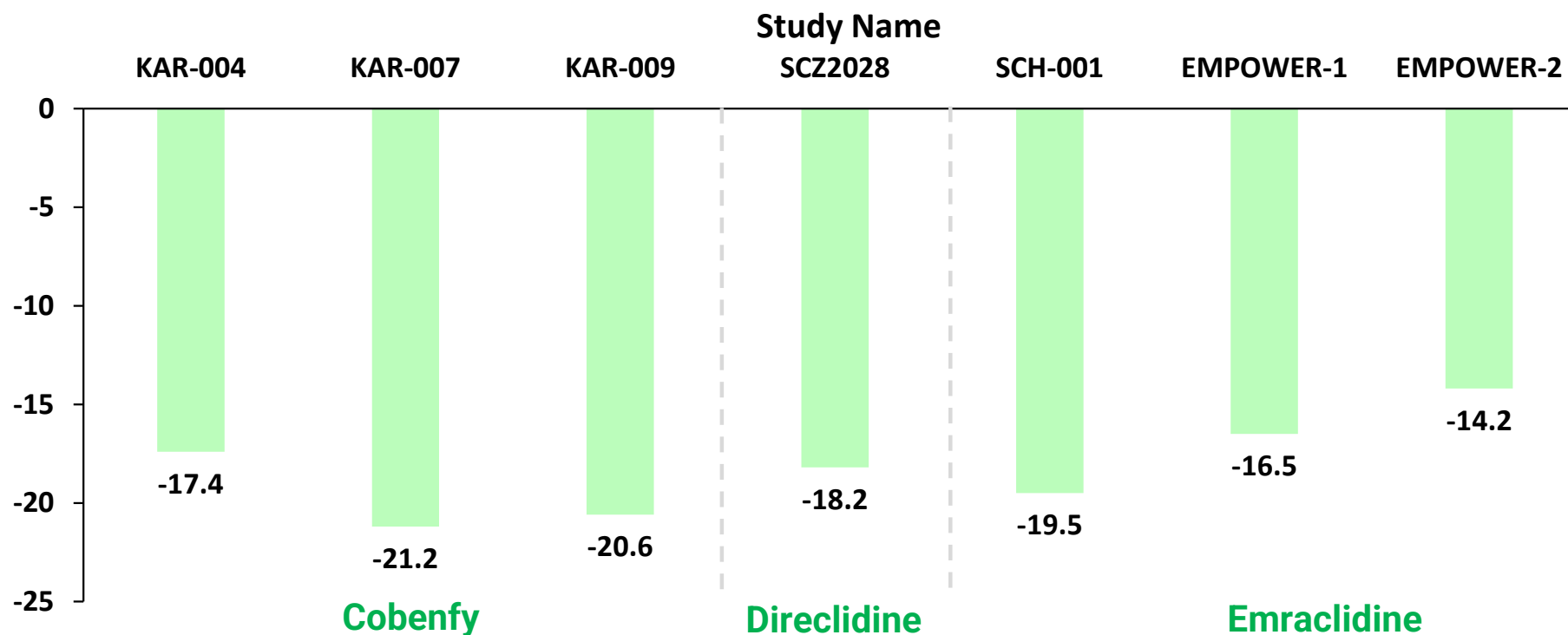
Schizophrenia Muscarinic Week 4-6 Pbo-Adj Efficacy on PANSS



Source: FDA review documents, Company reports, Jefferies estimates
Some values may be estimated by Jefferies from company-reported charts/materials

Exhibit 23 - Schizophrenia Muscarinics Week 4-6 Absolute Efficacy on PANSS

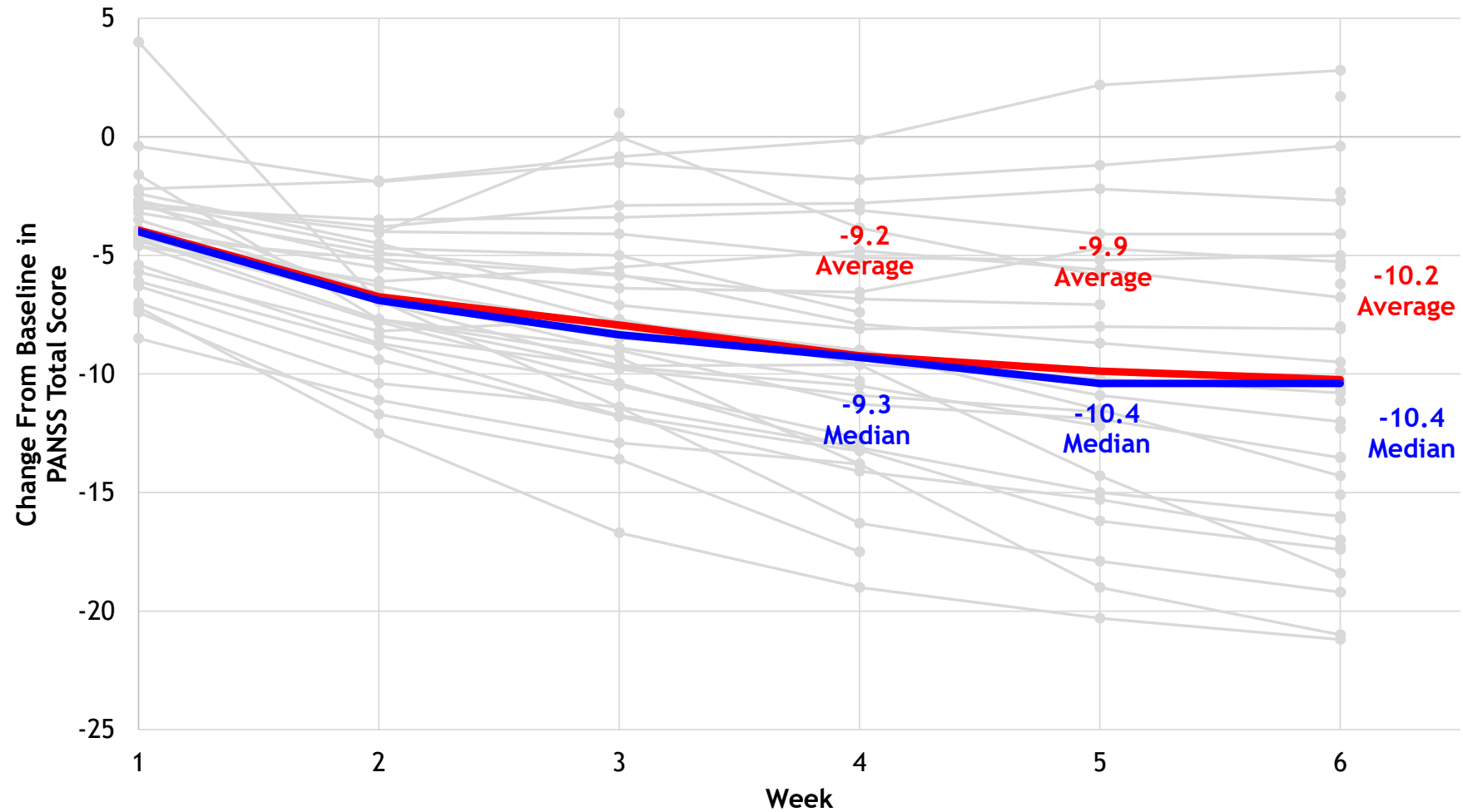
Schizophrenia Muscarinic Week 4-6 Absolute Efficacy on PANSS



Source: FDA review documents, Company reports, Jefferies estimates
Some values may be estimated by Jefferies from company-reported charts/materials

Exhibit 24 - Analyzed Placebo Effects Across 50+ Pbo-Controlled Schizophrenia Trials

Placebo Effect Across 50+ Pbo-Controlled Phase II/III SCZ Trials



Source: Jefferies research, FDA review documents, Company reports
Some values may be estimated by Jefferies from company-reported charts/materials

Exhibit 25 - Week 4 PANSS Placebo Responses Over Time
Week 4 PANSS Pbo Responses in Schizophrenia Trials by Drug Approval Date

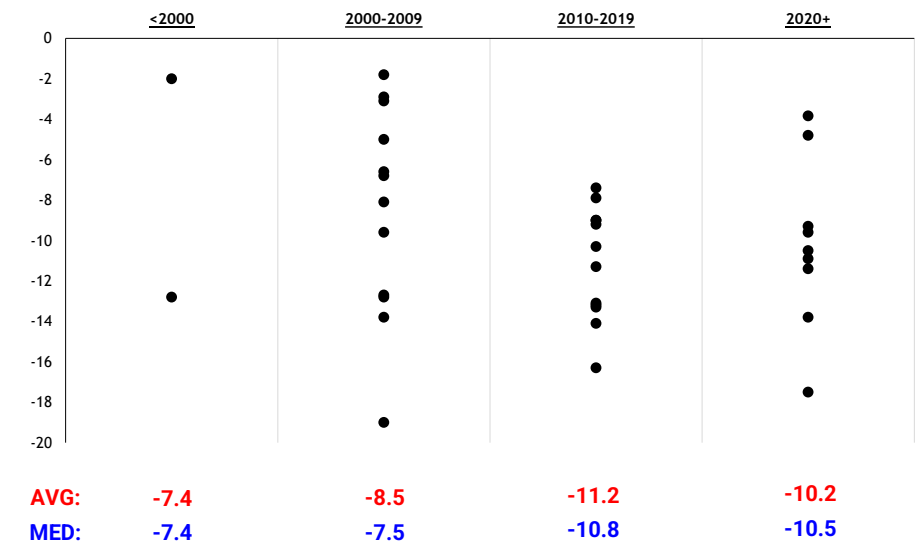
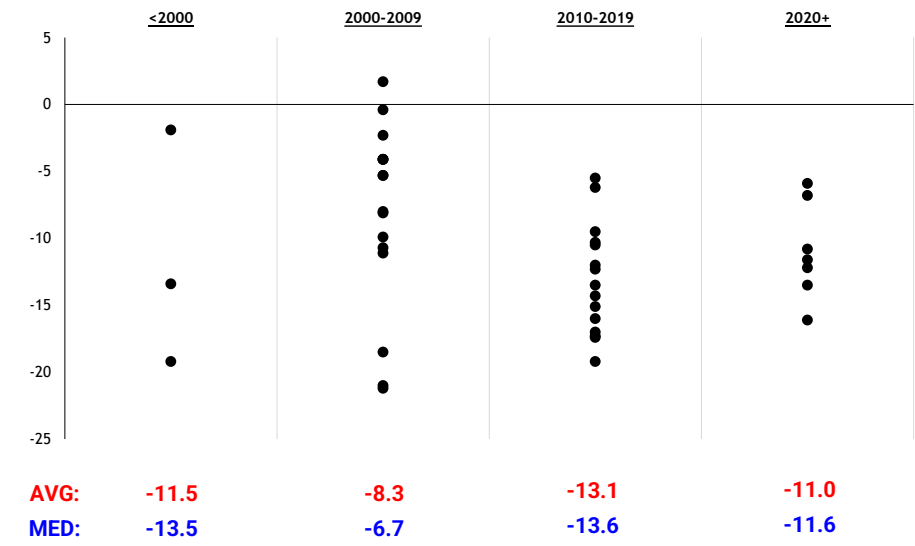


Exhibit 26 - Week 5-6 PANSS Placebo Responses Over Time
Week 5-6 PANSS Pbo Responses in Schizophrenia Trials by Drug Approval Date



Source: Jefferies research, FDA approval documents, Company reports
Drugs yet to be approved were included in the 2020+ bucket. Some values may be estimated by Jefferies from company-reported charts/materials

Source: Jefferies research, FDA approval documents, Company reports
Drugs yet to be approved were included in the 2020+ bucket. Some values may be estimated by Jefferies from company-reported charts/materials

acute-exacerbation-phase patients showed a **-17.4 absolute** Week 5 PANSS change for the 125/30mg BID dose compared to -5.9 pbo (e.g., **-11.6 pbo-adj**).

That said, we do acknowledge subtle nuances in potential mixed scenarios for the readout: If QD dosing is not viable relative to BID dosing, but safety/tolerability and efficacy is in-line, we could see perceived differentiation technically harder to justify for the Street. That said, we would argue a lack of a fasting requirement and a much shorter titration regiment (e.g., single dose for ML-007C-MA vs 1-2 weeks for Cobenfy) could still technically make ML-007C-MA a best-in-class muscarinic for schizophrenia. We'd also be careful to assign readacross to the Phase II VISTA program in ADP, which is exercising BID dosing and could remain intact. If safety liabilities emerge in Phase II or look optically worse than Cobenfy's Phase II KAR-009 study, it may be important to assess/consider levels of discontinuations due to AEs, or the temporal nature of AEs (e.g., if they are quickly resolved upon repeat dosing). For reference, Cobenfy showed 2-6% discontinuations due to AEs in the EMERGENT1-3 studies.

In a worst case scenario, if we see no efficacy on PANSS at all (which seems unlikely), the Street might need greater confidence that the pipeline (e.g., ML-004, ML-021, or ML-009) can deliver meaningful efficacy in other large CNS disorders (autism, Parkinson's, or hyperactivity/impulsivity). It may be difficult to the Street to assign value to ADP in this scenario as well.

Net-net: We like the set-up into Phase II schizophrenia data in H2:26, and think ML-007C-MA is well-positioned to deliver best-in-class efficacy/safety as a novel muscarinic M1/M4 agonist. Even in mixed scenarios, we'd still argue that titration/food advantages over Cobenfy could position ML-007C-MA as a compelling treatment option for physicians/patients.

ML-007C-MA Has Been Strategically Optimized Through Four Phase I Clinical Trials

ML-007 alone or combined/co-formulated with a peripheral acting anticholinergic (PAC) fesoterodine has been rigorously evaluated in four separate Phase I clinical trials to optimize the dosing regimen. Ultimately, MPLT established an optimal ML-007:PAC ratio of 100:1 to 600:1, balancing efficacy with mitigation of peripheral cholinergic and anticholinergic side effects. In total, N~270 patients healthy participants were enrolled and 1,500+ doses of ML-007 have been administered. Historically, the IND for ML-007C-MA was placed on clinical hold in May 2024 due to nonclinical findings associated with a single-agent animal study of ML-007. The hold was quickly removed in Jul 2024 following the FDA's review of add'l clinical/nonclinical data, signaling (+) mgmt execution.

(1) Study 001 (N=58): Demonstrated dose-dependent target engagement with ML-007 and achieved the maximum tolerated dose (MTD). Pairing ML-007 with a PAC blocked peripheral cholinergic effects.

(2) Study 011 (N=106): Immediate-release oral solution tested **22 dosing paradigms** across **13 cohorts**, using salivary volume and muscarinic activity as real-time PD markers. Ratios <100:1 resulted in more anticholinergic events (e.g., dry mouth, tachycardia) and ratios >600:1 resulted in more procholinergic events (e.g., nausea, diarrhea).

(3) Study 012 (N=24): Extended-release ML-007 and PAC demonstrated **close PK synchronization**, maintaining ratios within target range despite non-optimized release profiles.

(4) Study 013 (N=82): First use of bi-layer co-formulated tablet (ML-007C-MA), now in Phase II. Doses up to 210/3 mg BID and 330/6 mg QD achieved steady-state ratios within target range with low variability (CV ~30%).

Exhibit 28 - Summary of Four Phase I Clinical Programs for ML-007

Study 001



Immediate Release

N = 58

- Dose dependent target engagement with ML-007
- Single agent maximum tolerated dose (MTD) achieved
- Pairing with PAC blocked peripheral cholinergic effects

Study 011



Immediate Release

N = 106

- Wide range of ML-007 to PAC ratios evaluated across 13 cohorts
- Identified optimal ratios to balance pro- and anti-cholinergic peripheral effects

Study 012



Co-Administered Extended Release

N = 24

- Close matching of ML-007 and PAC peripheral exposures
- Low rates of TEAEs observed that were mild and transient in nature
- Exposure profile supported potential BID / QD dosing

Study 013



Co-Formulated Extended Release

N = 82

- First study that evaluated co-formulated bi-layer tablet
- Four cohort design, included two cohorts with multi-day dosing paradigm
- Evaluated food effects, titration and dosing in healthy adults and healthy elderly
- Finalized study design of Phase 2 trials in schizophrenia & ADP

Source: Company reports

Study 01

ML-007-001 (or Study 01) was a randomized, placebo-controlled Phase I first-in-human (FIH) trial assessing PK, safety, and tolerability of ML-007 oral solution in N=58 healthy adults.

Group 1 received single ascending (SAD) doses (0.8mg, 2.5mg, 8.2mg, 16mg, 32mg, and 49mg) to determine the maximum tolerated dose (MTD), which was 32 mg. At this dose, CSF concentrations reached ~50ng/mL, which was well above the estimated minimum-efficacious level of 20ng/mL from preclinical models. Group 2 explored co-administration with PAC extended release (ER) of 6mg prior to ML-007, demonstrating reduced procholinergic effects and improved tolerability, supporting PAC's role in mitigating peripheral side effects.

Study 011

ML-007-011 was a Phase I randomized, three-part SAD/MAD trial in N=106 healthy adults and elderly participants, evaluating ML-007 oral solution with and without PAC oral solution.

The study aimed to: **(1)** determine PAC dosing and timing to mitigate ML-007-associated cholinergic effects (e.g., Part 1), **(2)** assess multiple-dose safety/tolerability in healthy adults (e.g., Part 2), and **(3)** evaluate the same in elderly participants (e.g., Part 3). Single doses of ML-007 up to 164mg (~5x the single-agent MTD) were well tolerated when co-administered with PAC, and no intolerable dose was reached. Multiple doses up to 111 mg with PAC were tolerated over 5–7 days in both adult and elderly cohorts. Most TEAEs were mild and self-limited, with no SAEs reported. TEAE rates declined as PAC dose was reduced, achieving a more optimal balance of pro- and anticholinergic effects. Importantly, elderly participants exhibited similar safety and tolerability profiles to younger adults. Ultimately, MPLT determined ML-007 and PAC ratios <100:1 resulted in more anticholinergic events (e.g., dry mouth, tachycardia) and ratios >600:1 resulted in more procholinergic events (e.g., nausea, diarrhea).

Study 012

ML-007-012 (Study 012) was the first trial to evaluate the extended-release (ER) formulation of ML-007 in a two-part, single-center, open-label Phase I study with N=24 healthy adults.

The objectives were to assess safety, tolerability, and PK of escalating single doses of ML-007 ER (with and without PAC ER), compare ER vs oral solution under fed/fasted conditions (Part 1), and evaluate BID dosing strategies for future trials (Part 2).

- **Part 1:** Two cohorts (N=16) escalated QD doses up to 250 mg ML-007 ER with PAC ER.
- **Part 2:** One cohort (N=8) escalated BID doses up to 165 mg ML-007 ER with PAC ER.

ML-007 ER (with or without PAC ER) was generally well tolerated, with **lower AE rates than oral solution**. All TEAEs were mild; no moderate/severe AEs, SAEs, or deaths occurred. BID dosing did not increase TEAE rates versus QD, and no new TEAEs emerged compared to prior profiles. The study also assessed an orally dissolving tablet (ODT) formulation intended for as-needed use. PK and safety were consistent with oral solution. **Net-net:** MPLT determined that ML-007 administered with a PAC ER would be potentially compatible with QD dosing.

Study 013

MPLT's Phase I study ML-007C-MA-013 (Study 013) evaluated the bi-layer co-formulated tablet ML-007C-MA, now being advanced in Phase II ZEPHYR schizophrenia and VISTA ADP trials.

This single-center, four-cohort study assessed safety, tolerability, and pharmacokinetics (PK) under fasted and fed conditions, across single and multiple doses in healthy adults and multiple doses in healthy elderly participants, while exploring higher doses than previously tested and the impact of titration.

Key Findings:

- **Cohort 1:** Single 165/3 mg dose under fasted and fed conditions showed higher ML-007 PK exposure with food, no food effect on PAC PK, and improved tolerability when fed.

- **Cohort 2:** Escalating doses up to 210/3 mg BID and 330/6 mg QD were safe; tolerability improved with a titration step.
- **Cohorts 3 & 4:** Multi-dose regimens over 7 days confirmed maintenance doses of 165/3 mg BID, 210/3 mg BID, 270/6 mg QD, and 330/6 mg QD were well tolerated in adults, with no severe or serious TEAEs. Most adverse events were mild, cholinergic in nature, and stable over time. Common TEAEs included chills, dizziness, headache, nausea, hyperhidrosis, and salivary hypersecretion. At Phase II target doses, vomiting and constipation were absent, and no anticholinergic TEAEs occurred. In elderly participants, BID dosing up to 210/3 mg was well tolerated, while 330/6 mg QD was not. TEAEs were similar in profile, with additional tremor and abdominal pain at higher doses. At ADP target doses, only isolated vomiting and constipation occurred, and anticholinergic TEAEs were rare.

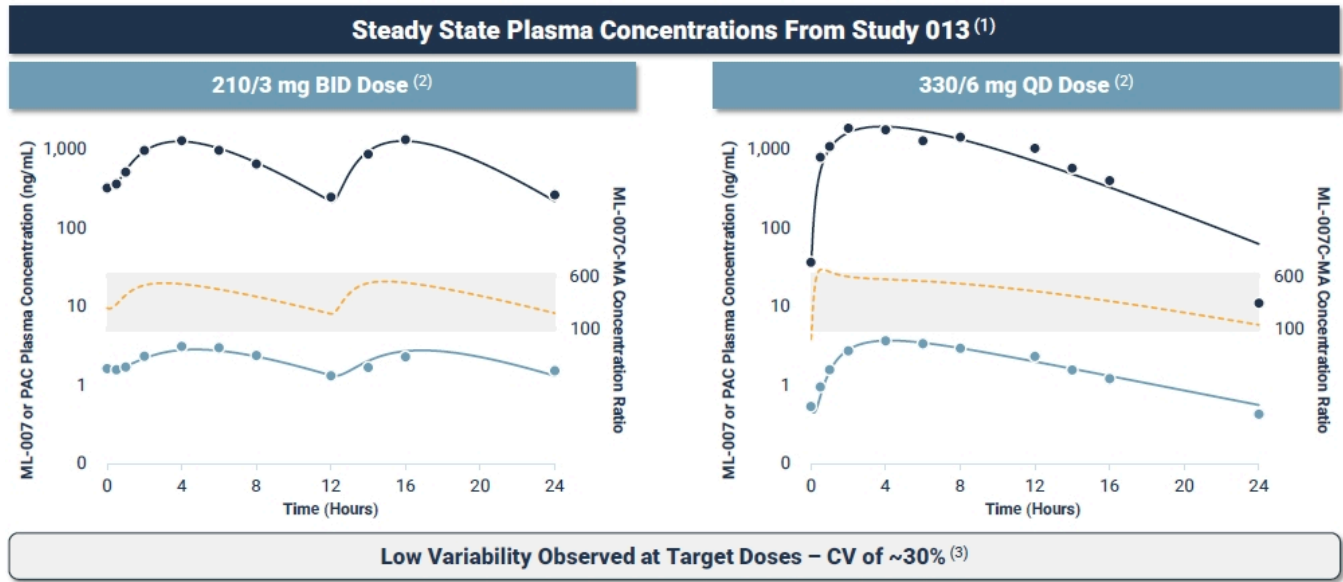
PK Profile: ML-007 and PAC exposures were approximately linear, unaffected by dose combinations, and maintained ML-007:PAC ratios within the desired 100:1-600:1 range. CSF concentrations met or exceeded predicted efficacious levels. PK variability was low (CV ~30% vs. Cobenfy's xanomeline ~100% and trospium ~94%), highlighting MPLT's ability to synchronize ML-007 and PAC exposure. Transient, dose-dependent heart rate increases were observed but were smaller in elderly participants. There were no clinically meaningful changes occurred in labs, vitals, or physical exams.

Key insights for Phase II Studies:

- **No fasting requirement:** Fed dosing improved tolerability; Phase II trials will dose proximal to meals.
- **Minimal titration:** Phase I showed similar AE profiles between single-dose and multi-day titration. ZEPHYR uses a short single-dose titration, VISTA employs a one-week titration.
- **Robust CSF exposure:** Bioanalysis of plasma and CSF samples collected throughout Phase I development show that modeled CSF exposures of ML-007 (based on plasma/PK exposures) are within the targeted efficacious range of 14-27ng/mL based on AIH/PCP ED50's in preclinical models. In contrast, MPLT mgmt asserts that KRTX's Cobenfy's average exposure at the approved clinical dose reaches the ED30.

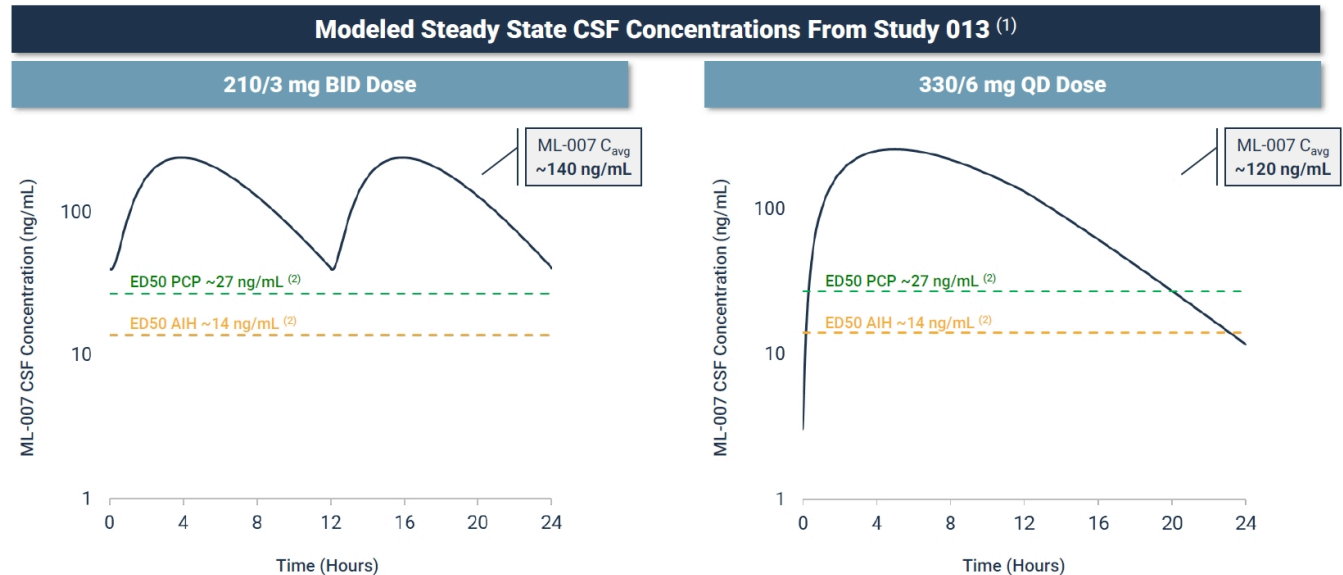
Net-net: ML-007C-MA demonstrated a competitive safety and tolerability profile, supporting advancement to Phase II at 210/3 mg BID and 330/6 mg QD in non-elderly adults (i.e., in schizophrenia), and up to 210/3 mg BID in elderly participants (i.e., in ADP). These findings validate the dosing strategy for ongoing trials, with robust PK, manageable TEAEs, and preserved CNS exposure.

Exhibit 29 - Both QD and BID Dosing Profiles for ML-007C-MA Show ML-007:PAC Ratios Within Target Ranges for the Duration of Exposure



Source: Company reports

Exhibit 30 - Modeled CSF Concentrations From Study 013 Suggest ML-007C-MA Exposure Levels in BID and QD Dosing Remain Above Exposure Levels Necessary for Preclinical Efficacy



Source: Company reports

Comparing the Phase I Safety Profile of ML-007C-MA to Muscarinics

The Phase I study 013 evaluated the safety, tolerability and PK of ML-007C-MA under fasted versus fed conditions, with single and multiple doses of ML-007C-MA in healthy adult participants and multiple doses of ML-007C-MA in healthy elderly participants across 4 cohorts. Here we focus on Cohort 3, which tested maintenance doses of 165/3 mg BID, 210/3 mg BID, 270/6 mg QD, and 330/6 mg QD in healthy adult (non-elderly) participants. Titration regimens of one dose to four days of dosing were assessed. All four dosing regimens of ML-007C-MA were well tolerated during titration and for the full 7-day maintenance dosing duration.

Importantly, the Phase II target doses selected for the ongoing ZEPHYR trial in schizophrenia of 210/3mg BID (N=6) and 330/6mg QD (N=6) were considered safe and well tolerated (Exhibit 31-32). Tolerability of 210/3 mg was improved when a single lower titration dose was administered 12 hours prior. At these target doses during the maintenance dosing period, TEAEs were observed in 50% (33% mild & 17% moderate) and 67% (33% mild & 33% moderate) of patients on 210/3mg BID and 330/6mg QD, respectively, which were mostly mild and transient. No severe or serious TEAEs were observed and no episodes of vomiting or constipation were reported. Additionally, there were no episodes of anticholinergic TEAEs at target doses. However, there was one discontinuation in the 330/6mg QD dosing cohort due to adverse events. The Phase II ZEPHYR study for schizophrenia is designed to employ a single titration dose to reach the target maintenance dose for both BID and QD dosing regimens.

ML-007C-MA was well tolerated in fed and fasted conditions, but tolerability was improved when administered in the fed condition therefore it is recommended to take the medication with food. Additionally, between low-calorie/low-fat and high-calorie/high-fat conditions, the PK exposures for both ML-007 and PAC were comparable.

Exhibit 31 - Study 013: Safety and Tolerability in Healthy Adults (Non-Elderly). Phase 2 Target Doses are 210/3mg and 330/6mg

Safety and Tolerability	Placebo Combined (N=8)	165/3 mg BID (N=6)	210/3 mg BID (N=6)	270/6 mg QD (N=6)	330/6 mg QD (N=6)
Subjects with any TEAE	2 (25%)	6 (100%)	3 (50%)	3 (50%)	4 (67%)
Cholinergic Classification - Subjects (%)					
Procholinergic	0 (0%)	4 (67%)	3 (50%)	2 (33%)	4 (67%)
Anticholinergic	1 (13%)	3 (50%)	0 (0%)	1 (17%)	0 (0%)
Other	1 (13%)	5 (83%)	3 (50%)	3 (50%)	3 (50%)
% of TEAEs by Severity					
Mild	100%	100%	95%	93%	86%
Moderate	0%	0%	5%	7%	14%
Severe	0%	0%	0%	0%	0%

Source: Company Reports

Exhibit 32 - Study 013: Severity of TEAEs by Subject and Most Common TEAEs at Phase 2 Target Doses

Safety and Tolerability	Placebo Combined (N=8)	210/3 mg BID (N=6)	330/6 mg QD (N=6)
Subjects with any TEAE	2 (25%)	3 (50%)	4 (67%)
Mild	2 (25%)	3 (50%)	4 (67%)
Moderate	0 (0%)	1 (17%)	2 (33%)
Severe	0 (0%)	0 (0%)	0 (0%)
Most Common TEAEs (>1 Subject in any Dose Group)			
Chills	0%	0%	2 (33%)
Hyperhidrosis	0%	1 (17%)	2 (33%)
Nausea	0%	3 (50%)	3 (50%)
Dizziness	0%	3 (50%)	2 (33%)
Dyspepsia	0%	0%	2 (33%)

Source: Company Reports

Exhibit 33 - Comparing MPL-007 Phase I Healthy Adult Data vs KRXT's Phase I Healthy Adult with KarXT (Cobenfy) - and KRTX's Data Evolution Into Pivotal Phase II/III SCZ Studies

** Only 100/20mg BID is approved and on Cobenfy Label

Drug Phase	ML-007C-MA Phase 1		KarXT Phase 1		KarXT Phase 2 Schizophrenia Patients	KarXT Phase 3 Schizophrenia Patients	KarXT Phase 3 Schizophrenia Patients
Safety Population	Healthy Volunteers		Healthy Volunteers		EMERGENT-1 (N=89)	EMERGENT-2 (N=126)	EMERGENT-3 (N=125)
Dose	210/3 mg BID (N=6)	330/6 mg QD (N=6)	100/20 mg BID (N=18)	125/40mg BID (N=18)			
Subjects with any TEAE	3 (50%)	4 (67%)	12 (67%)	16 (89%)	52%	74%	70%
Mild	2 (33%)	2 (33%)	9 (50%)	10 (56%)	35%	38%	36%
Moderate	1 (17%)	2 (33%)	3 (17%)	6 (33%)	16%	29%	26%
Severe	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1%	7%	8%
Procholinergic TEAEs							
Nausea	3 (50%)	3 (50%)	4 (22%)	4 (22%)	17%	19%	19%
Vomiting	0%	0%	5 (28%)	1 (6%)	9%	14%	16%
Hyperhidrosis	1 (17%)	2 (33%)	1 (6%)	2 (11%)	2%	-	-
Salivary Hypersecretion	0%	-	1 (6%)	0%	-	-	-
Anticholinergic TEAEs							
Dry mouth	0%	-	2 (11%)	5 (28%)	9%	5%	0%
Dysuria	0%	-	0 (0%)	4 (22%)	-	-	-
Constipation	0%	-	1 (6%)	1 (6%)	17%	21%	13%
Urinary Retention	0%	-	0 (0%)	2 (11%)	-	-	-
Eye disorders	0%	-	0 (0%)	5 (28%)	-	5%	-
Other TEAEs							
Dizziness	3 (50%)	2 (33%)	4 (22%)	7 (39%)	3%	9%	0%
Postural dizziness	0%	-	2 (11%)	2 (11%)	0%	0%	0%
Headache	0%	-	3 (17%)	4 (22%)	7%	13%	11%

Source: Jefferies; Company Reports

(1) MPLT's Phase II Target Doses. We present MPLT's Data Based on the Highest Severity AE Experienced by Patients For Comparability Purposes.

(2) KRTX's Phase II/III EMERGENT Trial Dosing: Xanomeline 50 mg/trospium 20 mg BID on days 1-2 followed by xanomeline 100mg/ trospium 20mg BID on days 3-7. The dose is increased to xanomeline 125mg/ trospium 30mg BID on days 8-34 unless the subject is experiencing adverse events from the xanomeline 100 mg/trospium 20 mg dose.

Comparing Safety of ML-007 to Muscarinic KarXT Phase I Healthy Volunteer Data

We contextualize these safety data by comparing ML-007 to BMJ (Karuna's) Cobenfy Phase I safety data. Specifically, we identified KAR-003 as the most relevant comparator, given this the only other Phase I study of a muscarinic that evaluated the pivotal target dose of 100/20mg BID in healthy adults i.e. this dose is actually on Cobenfy's label. This study also evaluated the 125/40mg BID dose and while this was not evaluated in pivotal studies/not on label, we think it is close enough to the 125/30mg BID that is on label, for the purpose of this exercise. As such, we will use this study and both doses when making our comparisons below with a focus on the frequency and severity of any TEAEs, as well as nausea, and vomiting ([Exhibit 33](#)) That said, we also include data from BMJ's other Cobenfy Phase I, KAR-001, for completeness in our bar charts below.

- **KAR-003:** Phase I randomized, double-blind, placebo-controlled, multiple ascending-dose, clinical trial, the tolerability of different dose combinations of xanomeline and trospium was assessed in 69 healthy volunteers. The fixed ratio combinations of xanomeline and trospium were **100/20mg**, 125/40mg, 150/20mg, and 150/40mg, administered twice daily.
- **KAR-001** was a randomized, double-blind, multiple-dose, inpatient study in 70 healthy adult subject. Subjects were randomized into 1 of 2 treatment arms: 1) xanomeline (75 mg TID/3x daily) and placebo, or 2) xanomeline (75 mg TID) + trospium (20 mg BID/2x daily).

(1) Any TEAEs: ML-007C Phase I showed 50% (33% mild, 17% moderate) and 67% (33% mild, 33% moderate) TEAEs on the 210/3mg BID (N=6) and 330/6mg QD (N=6) doses, respectively. This compares to Cobenfy's 67% (50% mild, 17% moderate) on the 100/20mg BID dose (N=18) and 89% (56% mild, 33% moderate) on the 125/40mg BID dose. To us, the frequency and severity of TEAEs for ML-007C look broadly comparable to Cobenfy at 50-67% TEAEs for ML-007C vs 67%-89% for Cobenfy, with severity of TEAEs being mild in 50%+ of cases for both drugs, though we do flag the smaller N for ML-007C as a caveat to consider when comparing these data.

(2) Nausea and Vomiting: Our review of KRTX's Cobenfy's pivotal EMERGENT studies and OLE studies revealed that nausea and vomiting were the two biggest drivers behind study discontinuations; See [Exhibit 34](#) and [Exhibit 35](#). As such, we see both of these TEAEs as the most important to pay attention to when thinking about the tolerability and safety differentiation of ML-007, as we enter the larger Phase II ZEPHYR study. ML-007C Phase I showed 50% (3/6) nausea rates for both the 210/3mg BID and 330/6mg QD doses, which is higher than the 22% (4/18) nausea rates observed for Cobenfy doses (100/20mg BID and 125/40mg BID). That said, MPLT did share that the majority of the nausea cases observed for ML-007C were mild in severity. Importantly, **no cases of vomiting were reported for ML-007C** on both doses, while Cobenfy showed 28% (5/28) vomiting AEs on the 100/20mg BID dose and 6% (1/6) on 125/40mg BID. Indeed, 1 person discontinued in the Cobenfy study for 6% discontinuation rate (1/18), which we assume could have been related to either nausea or vomiting (specific reason not disclosed). 1 person also discontinued on the 330/6mg QD ML-007C dose for 17% discontinuation rate (1/6), which we believe could have been due to nausea (specific reason not disclosed). Net-Net, while nausea rates for ML-007C were higher than those reported for Cobenfy, we do note that this was offset by zero rates of vomiting observed vs almost 1/3 of patients experiencing vomiting on Cobenfy's 100/20mg BID dose.

(3) Cardiovascular: Cardiovascular AEs are typically seen with second-generation atypical antipsychotics (SGAs) such as QT interval prolongation which can lead to torsades de pointes, ventricular arrhythmias and sudden cardiac death. Tachycardia is also common among SGAs due to the anticholinergic and adrenergic effects, while myocarditis and cardiomyopathy has also been found to be associated with clozapine. As it relates to muscarinics, cardiovascular effects typically emerge through activation of M2 and M3 receptors. However, BMJ's Cobenfy, MPLT's, ML-007C, and NBIX's

direclidine largely overcome CV side effects by directly binding to M1 and M4 receptors, which are not highly expressed in cardiac tissue. This does not mean there is complete elimination of CV side effects as imperfect selectivity could lead to off-target binding to other muscarinic (M2 or M3) subtypes. Said another way, we do see cardiovascular side effects with muscarinics, but to a lesser degree of severity and frequency than what we see with anti-psychotics. As such, we review CV side effects for muscarinics for completeness and to showcase how this could serve as a safety differentiator vs SGAs but note that nausea/vomiting remain the key safety AEs that could ultimately limit uptake.

In Cobenfy's and ML-007C Phase I data, there have been no safety disclosures on any of the cardiovascular AEs, nor do we see any for direclidine's Phase II study. MPLT have only provided qualitative feedback that **'no clinically meaningful changes in mean blood pressure or ECG parameters observed'**. That said, looking at pivotal studies across muscarinics, we do note cases of tachycardia (increased heart rate) and hypertension (raised blood pressure). Specifically, for Cobenfy, in the EMERGENT-2 and EMERGENT-3 studies we see placebo-adjusted hypertension of 8.7% and 4.8%, respectively. We do also note placebo-adjusted supine (lying down) tachycardia of 1.2% in EMERGENT-1. For emraclidine, in the EMPOWER-1 and EMPOWER-2 studies, we see placebo-adjusted sinus tachycardia of 0.8-3.9% and 3.1%, respectively, as well as placebo-adjusted supine (lying) hypertension of 5.4% and 2.3-7.6%, respectively

Furthermore, pooled analyses of Cobenfy's EMERGENT pivotal studies (N=340) shows us that of the 17 discontinuations recorded due to AEs, two of these were due to cardiovascular effects; one due to tachycardia (heart rate increase) and one related to palpitations ([Exhibit 34](#)). Additionally, Cobenfy's OLE studies in N=718 revealed that of the 68 patients that had to dose reduce due to a TEAE, 5 of these cases were due to hypertension. As such, while we are yet to see any notable CV side effects for ML-007c from the Phase I study, our review of the safety data for Cobenfy and emraclidine suggest that **tachycardia and hypertension** could be CV AEs to look out for in MPLT's Phase II ZEPHYR study.

Translatability into Phase 2: Cobenfy AE Rates Generally Decline From Phase I into Phase II

Here we examine the evolution of KRTX's Cobenfy's TEAEs from the Phase I study in healthy volunteers to the larger pivotal Phase II/III studies in schizophrenia patients (EMERGENT-1, EMERGENT-2, EMERGENT-3) to help us understand what we could expect for ML-007C into its Phase II study. In general, TEAEs either stayed the same or declined into the EMERGENT pivotal studies (N=89-126) vs data reported at the 100/20mg and 125/40mg BID doses in the Phase I study ; (See [Exhibit 36](#)) e.g. 67%-89% (78% blended) any TEAEs from Phase I were shown to decline to 52%-74% in the pivotal EMERGENT studies. We also note the severity of TEAEs as a whole was broadly consistent too with the majority still mild/moderate, albeit with a slight shift in mix towards moderate and some severe. This suggests to us that diseased patients with chronic schizophrenia and extensive lifetime exposures to psychotropic medication may be better able to tolerate side effects.

We see nausea and vomiting as the most important AEs to focus on given both were primarily responsible for discontinuations in Cobenfy's pivotal and OLE studies. Specifically, nausea rates for Cobenfy declined from 22% in Phase I to 17-19% in the Phase II/III studies, which equates to an 18% step-down at the mid-point, while vomiting declined from 6%-28% to 9%-16%. The only AE that was shown to increase was constipation which rose from 6% to 13%-21%.

We think the favorable evolution in TEAEs for Cobenfy, particularly on nausea and vomiting, read across positively to ML-007C and suggests, to us, that we could see improved safety into the Phase II ZEPHYR study; we would not expect to see vomiting AEs and would expect nausea AEs to decline.

Exhibit 34 - Pooled Data from Cobenfy EMERGENT Pivotal Studies Reveals Nausea and Vomiting Were Two Main Drivers for Discontinuations

Table 35. Subjects With Adverse Events Leading to Treatment Discontinuation by System Organ Class and Preferred Term, Safety Population, On-Treatment Period (Pooled Studies KAR-007, KAR-009, and KAR-004)

System Organ Class Preferred Term	KarXT (N=340) n (%)	Placebo (N=343) n (%)	Risk Difference % (95% CI)
Any AE leading to discontinuation	17 (5.0)	12 (3.5)	1.5 (-1.6, 4.8)
Cardiac disorders (SOC)	1 (0.3)	0	0.3 (-0.8, 1.6)
Palpitations	1 (0.3)	0	0.3 (-0.8, 1.6)
Gastrointestinal disorders (SOC)	8 (2.4)	0	2.4 (1.2, 4.6)*
Nausea	5 (1.5)	0	1.5 (0.4, 3.4)*
Vomiting	3 (0.9)	0	0.9 (-0.2, 2.6)
Abdominal pain	1 (0.3)	0	0.3 (-0.8, 1.6)
Diarrhoea	1 (0.3)	0	0.3 (-0.8, 1.6)
Dyspepsia	1 (0.3)	0	0.3 (-0.8, 1.6)
Gastroesophageal reflux disease	1 (0.3)	0	0.3 (-0.8, 1.6)
Salivary hypersecretion	1 (0.3)	0	0.3 (-0.8, 1.6)
General disorders and administration site conditions (SOC)	3 (0.9)	0	0.9 (-0.2, 2.6)
Asthenia	1 (0.3)	0	0.3 (-0.8, 1.6)
Chest discomfort	1 (0.3)	0	0.3 (-0.8, 1.6)
Noncardiac chest pain	1 (0.3)	0	0.3 (-0.8, 1.6)
Investigations (SOC)	4 (1.2)	0	1.2 (0.1, 3.0)*
Electrocardiogram abnormal	1 (0.3)	0	0.3 (-0.8, 1.6)
Gamma-glutamyl transferase increased	1 (0.3)	0	0.3 (-0.8, 1.6)
Heart rate increased	1 (0.3)	0	0.3 (-0.8, 1.6)
Hepatic enzyme increased	1 (0.3)	0	0.3 (-0.8, 1.6)
Metabolism and nutrition disorders (SOC)	1 (0.3)	0	0.3 (-0.8, 1.6)
Decreased appetite	1 (0.3)	0	0.3 (-0.8, 1.6)
Musculoskeletal and connective tissue disorders (SOC)	1 (0.3)	0	0.3 (-0.8, 1.6)
Back pain	1 (0.3)	0	0.3 (-0.8, 1.6)
Flank pain	1 (0.3)	0	0.3 (-0.8, 1.6)
Pain in extremity	1 (0.3)	0	0.3 (-0.8, 1.6)
Nervous system disorders (SOC)	2 (0.6)	0	0.6 (-0.5, 2.1)
Dyskinesia	1 (0.3)	0	0.3 (-0.8, 1.6)
Headache	1 (0.3)	0	0.3 (-0.8, 1.6)
Psychiatric disorders (SOC)	4 (1.2)	12 (3.5)	-2.3 (-5.0, -0.1)*
Catatonia	1 (0.3)	0	0.3 (-0.8, 1.6)
Insomnia	1 (0.3)	1 (0.3)	0.0 (-1.4, 1.4)
Psychotic disorder	2 (0.6)	3 (0.9)	-0.3 (-2.0, 1.3)
Mental disorder	0	1 (0.3)	-0.3 (-1.6, 0.8)
Agitation	0	2 (0.6)	-0.6 (-2.1, 0.5)
Schizophrenia	0	6 (1.7)	-1.7 (-3.8, -0.6)*
Renal and urinary disorders (SOC)	1 (0.3)	0	0.3 (-0.8, 1.6)
Dysuria	1 (0.3)	0	0.3 (-0.8, 1.6)

Source: FDA

Exhibit 35 - Data from Cobenfy Open-label Extension Studies Revealed that Nausea was the No.1 Reason for Discontinuations Followed by Vomiting

Table 36. Subject Disposition (Studies KAR-008 and KAR-011)

Disposition Outcome	KAR-008			KAR-011	Pooled KAR-008 and KAR-011
	KarXT to KarXT (N=68) n (%)	Placebo to KarXT (N=84) n (%)	Total KarXT (N=152) n (%)	KarXT (N=566) n (%)	(N=718) n (%)
Subjects randomized	68	84	152	566	718
Safety population	68	84	152	566	718
Completed treatment	68 (100)	84 (100)	152 (100)	104 (18.4)	256 (35.7)
Discontinued study drug	49 (72.1)	68 (81)	117 (77.0)	264 (46.6)	381 (53.1)
Withdrawal of consent	17 (25.0)	33 (39.3)	50 (32.9)	84 (14.8)	134 (18.7)
Adverse event	8 (11.8)	5 (6.0)	13 (8.6)	94 (16.6)	107 (14.9)
Lost to follow-up	9 (13.2)	11 (13.1)	20 (13.2)	39 (6.9)	59 (8.2)
Subject fails to adhere to the protocol requirements	14 (20.6)	9 (10.7)	23 (15.1)	30 (5.3)	53 (7.4)
Other	0	2 (2.4)	2 (1.3)	8 (1.4)	10 (1.4)
Alcohol or illegal drug use	0	3 (3.6)	3 (2.0)	6 (1.1)	9 (1.3)
Death	0	1 (1.2)	1 (0.7)	2 (0.4)	3 (0.4)
Progressive disease	0	2 (2.4)	2 (1.3)	0	2 (0.3)
Applicant decision to discontinue study	0	2 (2.4)	2 (1.3)	0	2 (0.3)
Failure to meet randomization criteria	1 (1.5)	0	1 (0.7)	0	1 (0.1)
General or specific changes in the subject's condition that renders him/her ineligible for further treatment according to the inclusion/exclusion criteria	0	0	0	1 (0.2)	1 (0.1)

Source: Clinical Reviewer/CDS created table from ds.xpt and adsl.xpt; Software: R; Interim data submitted with 120-day update. Abbreviations: CDS, clinical data scientist; KarXT, capsules containing xanomeline and trospium chloride mixed in various ratios; N, number of subjects in treatment arm; n, number of subjects in specified population or group

Table 45. Adverse Events Leading to Dose Reduction in ≥0.5% KarXT-Treated Subjects by Preferred Term, Long-Term Safety (Studies KAR-008 and KAR-011)

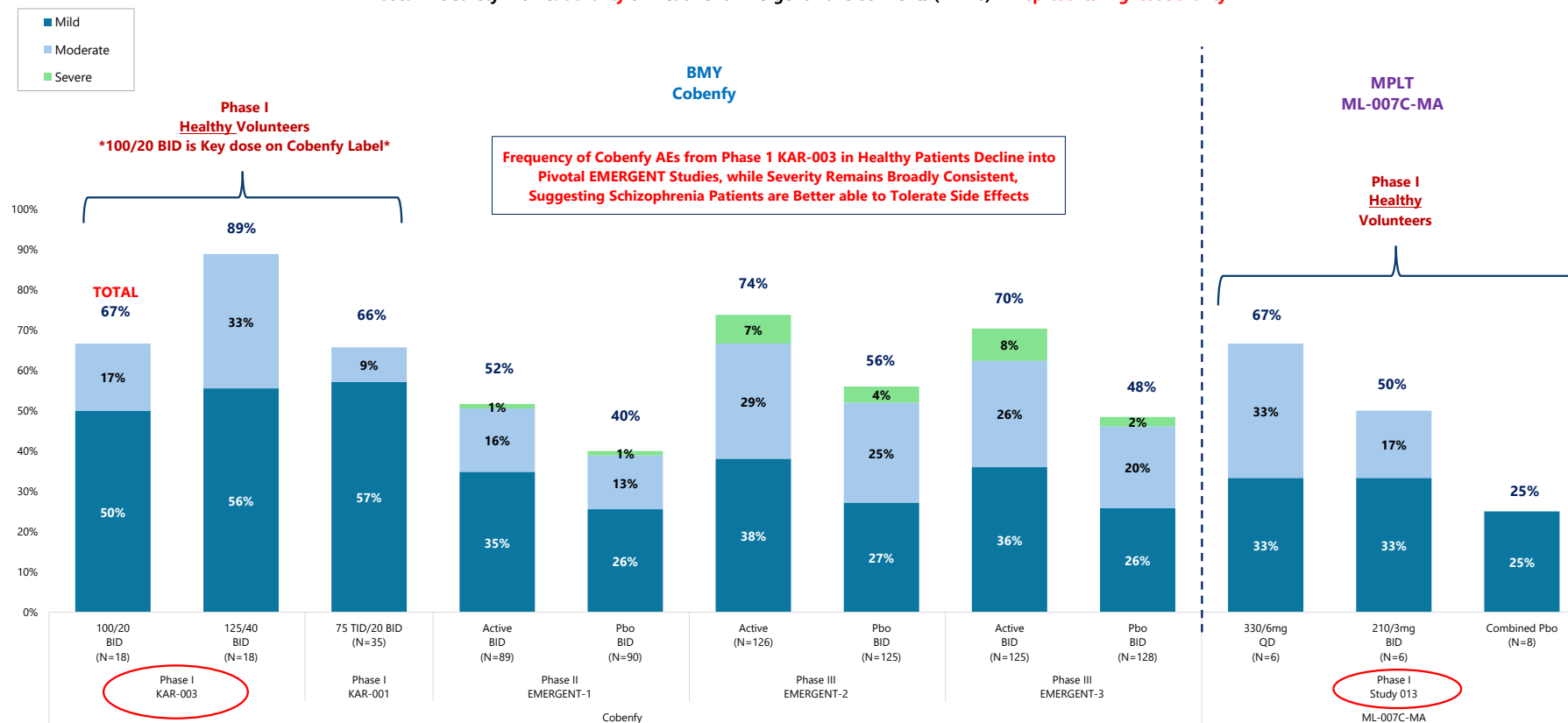
Preferred Term	All KarXT (N=718) n (%)
Subjects with ≥1 TEAE leading to dose reduction	68 (9.5)
Nausea	24 (3.3)
Vomiting	22 (3.1)
Dizziness	13 (1.8)
Constipation	8 (1.1)
Dyspepsia	6 (0.8)
Hypertension	5 (0.7)
Hyperhidrosis	4 (0.6)
Urinary retention	4 (0.6)
Vision blurred	4 (0.6)

Source: 120-day Safety Update, Table 15. Abbreviations: KarXT, capsules containing xanomeline and trospium chloride mixed in various ratios; N, number of subjects in treatment arm; n, number of subjects with adverse event; TEAE, treatment-emergent adverse event

Source: FDA

Exhibit 36 - The severity of TEAEs in ML-007C Phase I are Only Mild-Moderate, on Par with Cobenfy with Mild AEs Accounting for 50%+ of Cases for Both Drugs

Muscarinic Safety Profile: **Severity of Treatment-Emergent Adverse Events (TEAEs) - *Represents Highest Severity***

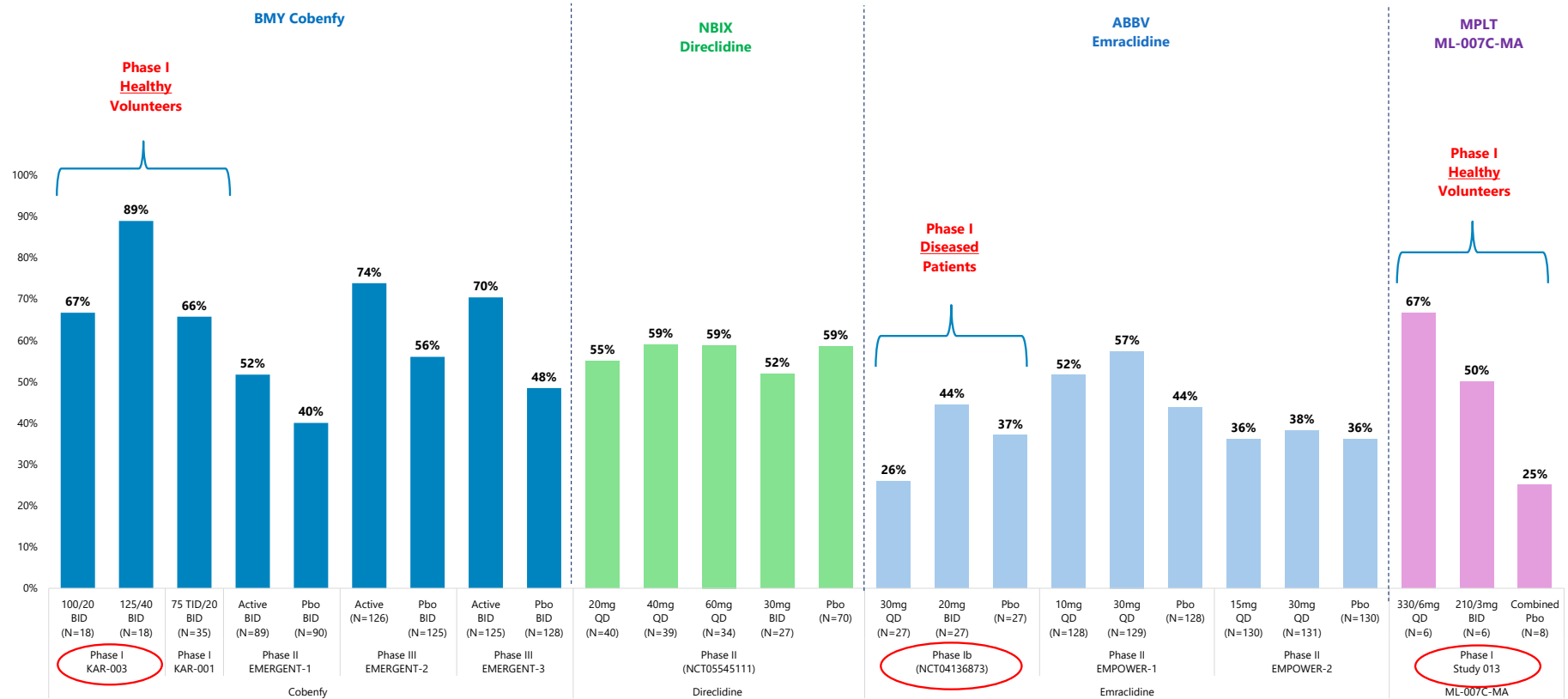


Doses Shown for MPLT are the Phase II Target Doses; We present MPLT's Data Based on the Highest Severity AE Experienced by Patients For Comparability Purposes

100/20mg BID dose in Cobenfy's Phase I is the only dose on the Cobenfy label

Exhibit 37 - ML-007C Phase I Showed Lower % TEAEs (50-67%) vs Cobenfy's Phase I Healthy Volunteer studies (67%-89%)

Muscarinic Safety Profile: Any Treatment-Emergent Adverse Events (TEAEs)

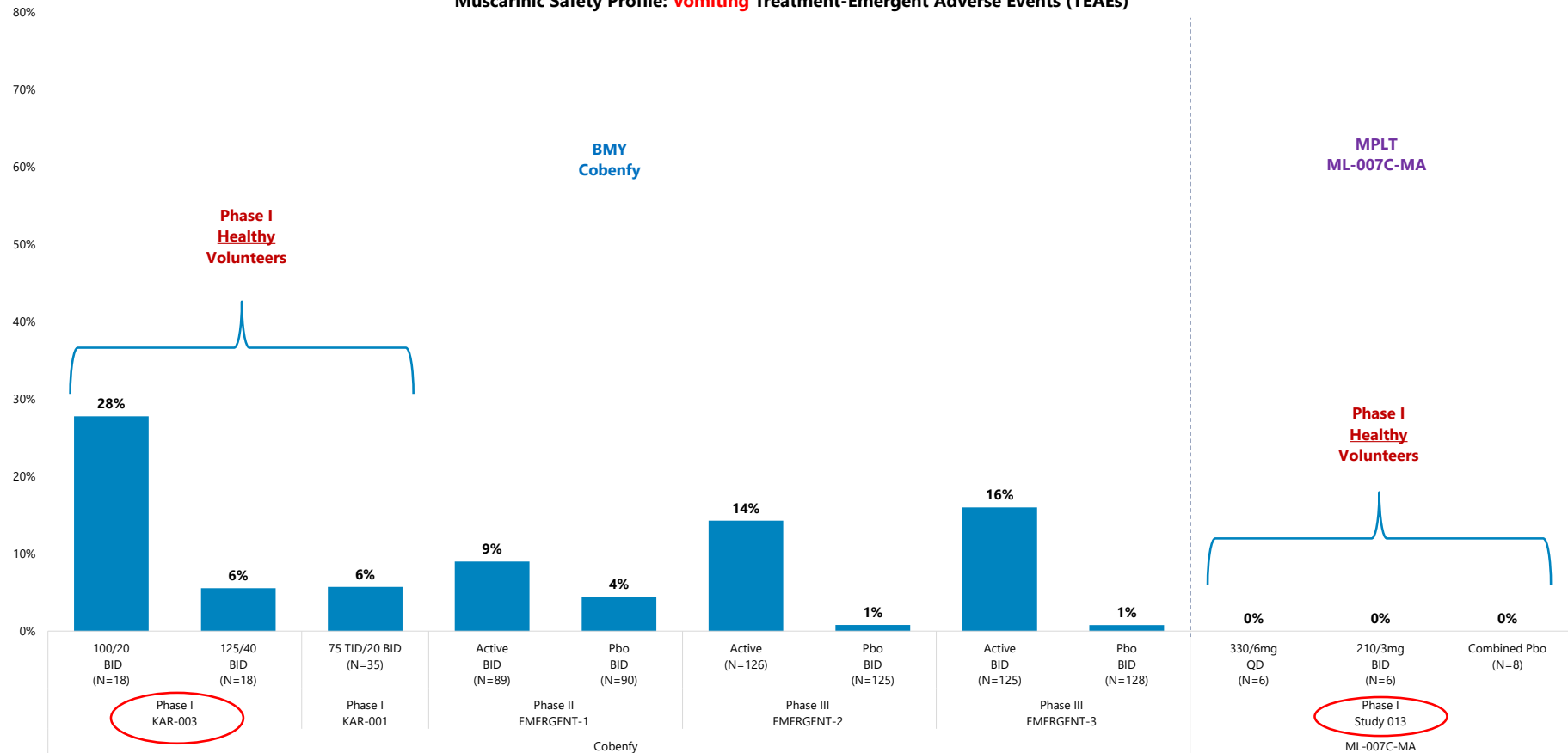


Source: Jefferies

100/20mg BID dose in Cobenfy's Phase I is the only dose on the Cobenfy label

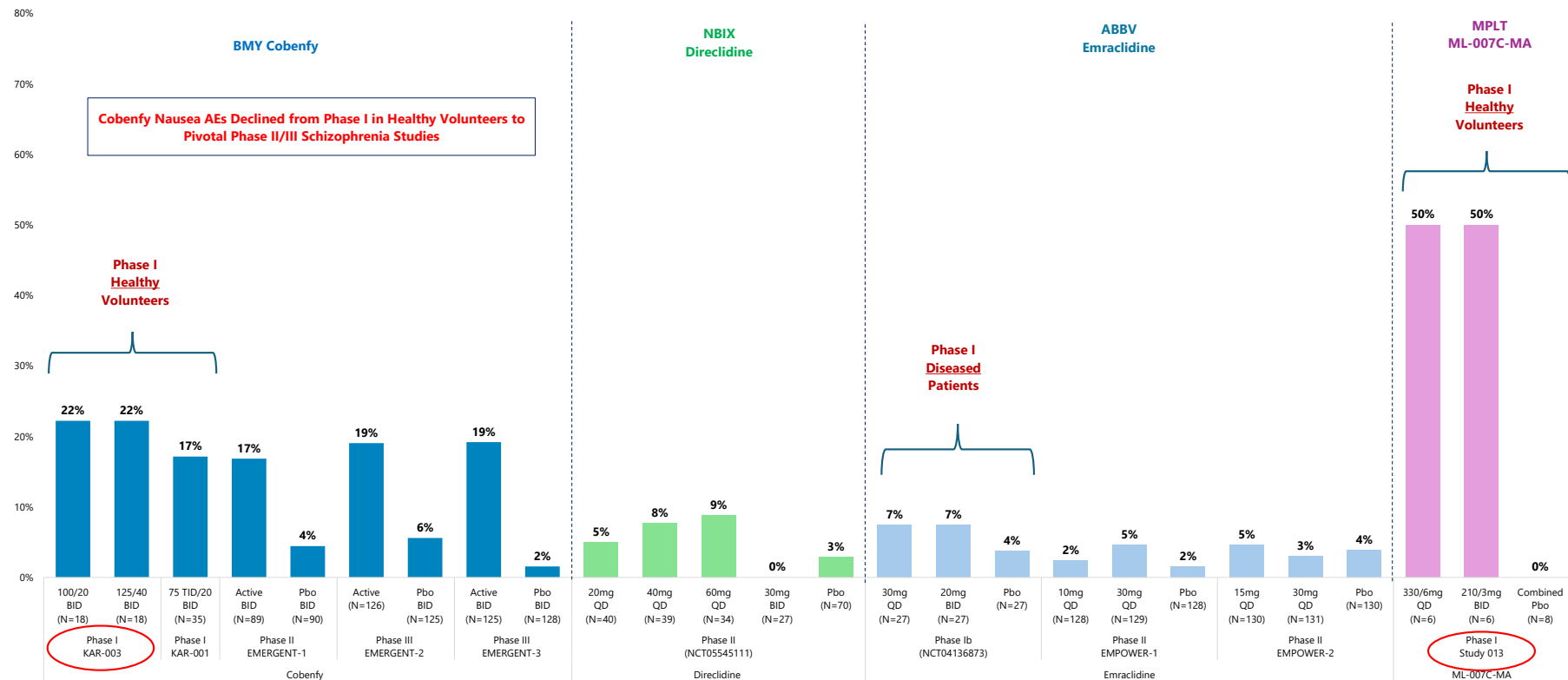
Exhibit 38 - Key Differentiating Factor for ML-007C is the absence of Vomitting AEs in the Phase I study vs 6%-28% Vomitting AEs Observed for Cobenfy's Phase I

Muscarinic Safety Profile: Vomiting Treatment-Emergent Adverse Events (TEAEs)



Source: Jefferies

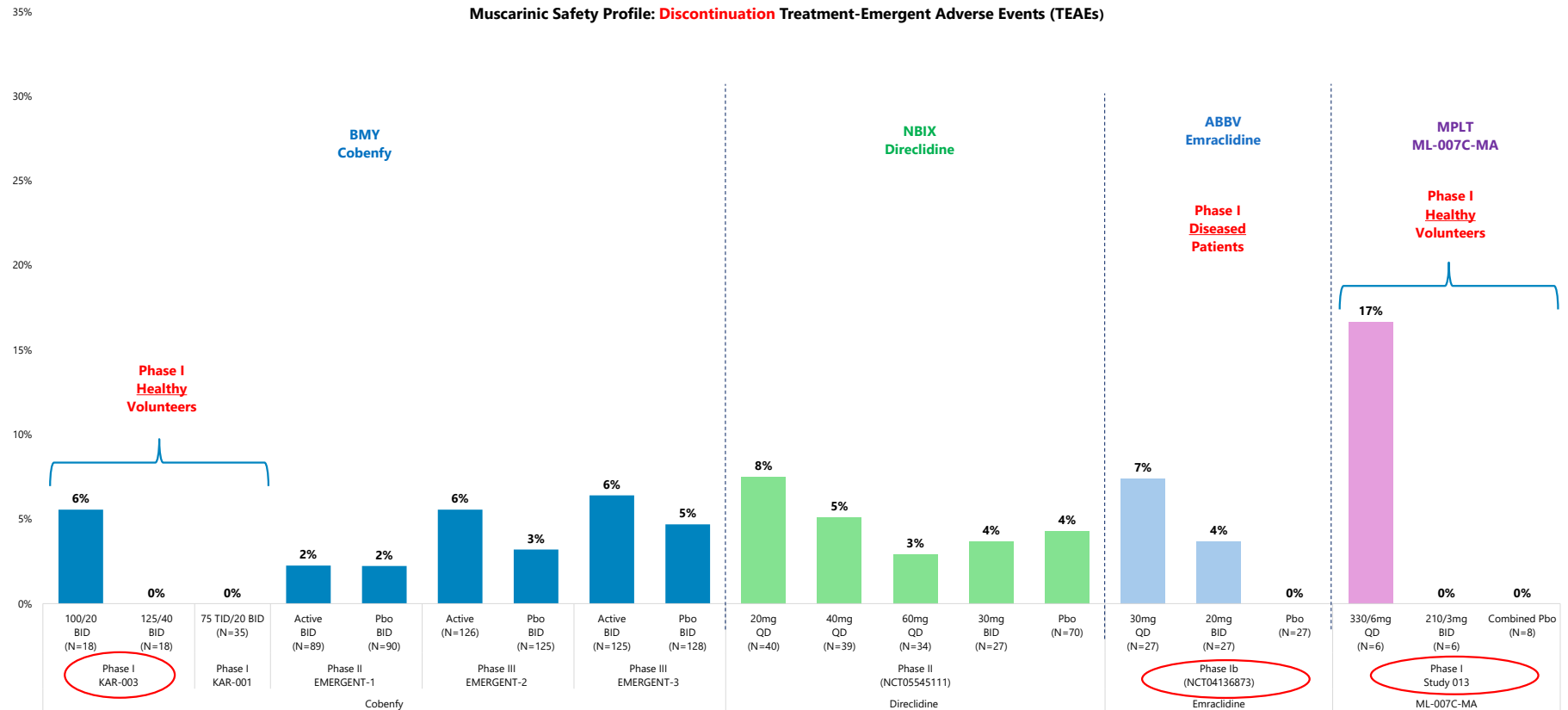
Exhibit 39 - MLC-007C Did Show 50% Nausea AEs in the Phase I, Higher than 17-22% Observed in Cobenfys' Phase I, Though MPLT have shared Nausea for MLC-007C were Mostly Mild in Severity
Muscarinic Safety Profile: Nausea Treatment-Emergent Adverse Events (TEAEs)



Source: Jefferies

Exhibit 40 - 17% Discontinuation (1/6) for ML-007C Does Look High vs 6% for Cobenfy. However this was Attributed to Only 1 person on the 330/6mg QD, Which We Believe Could Have Been Due to Nausea (specific reason not disclosed).

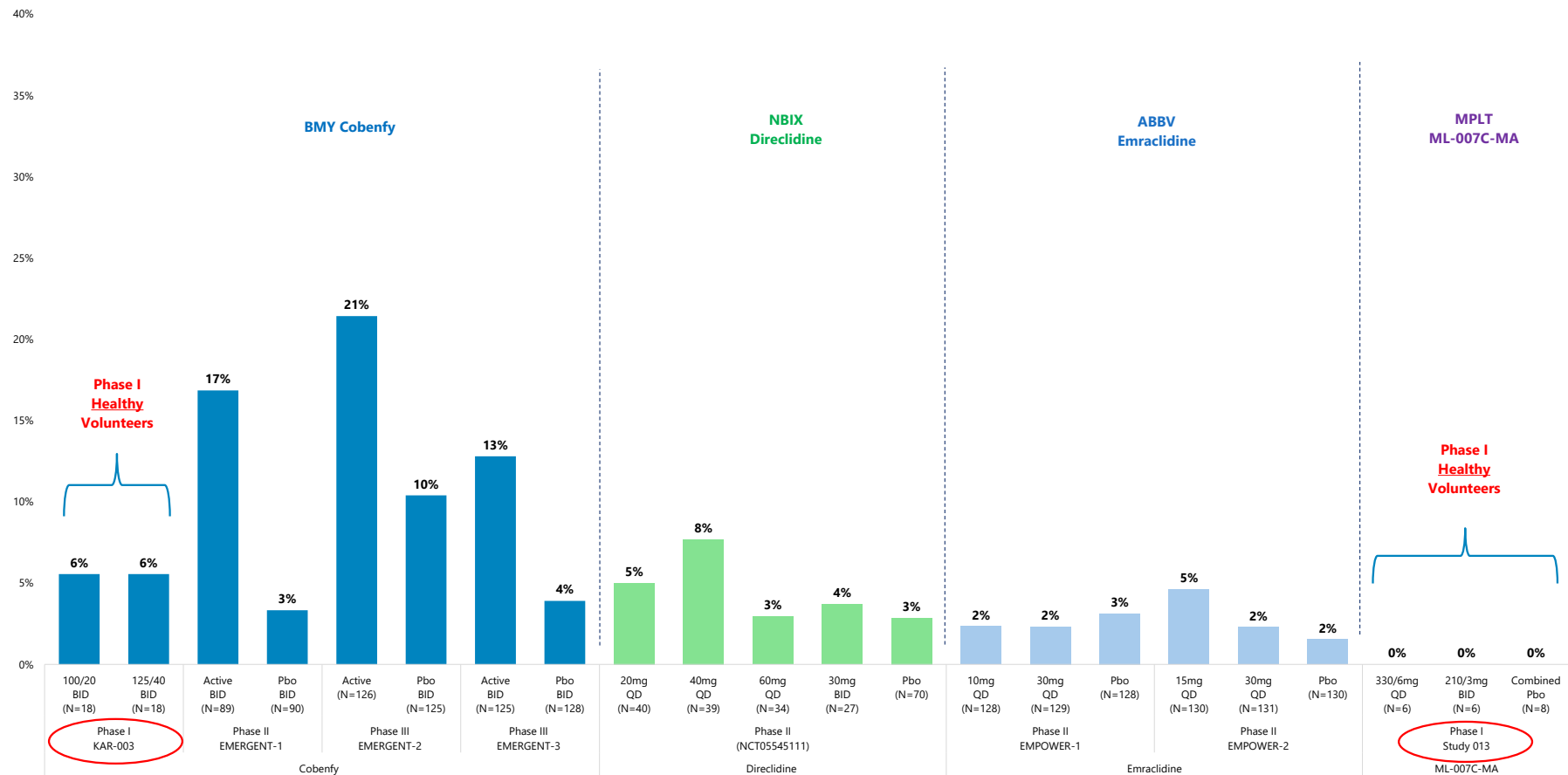
Muscarinic Safety Profile: **Discontinuation** Treatment-Emergent Adverse Events (TEAEs)



Source: Jefferies

Exhibit 41 - Zero Rates of Anticholinergic TEAE Constipation was Observed in ML-007C's Phase I study, which is Better vs 6% for Cobenfy

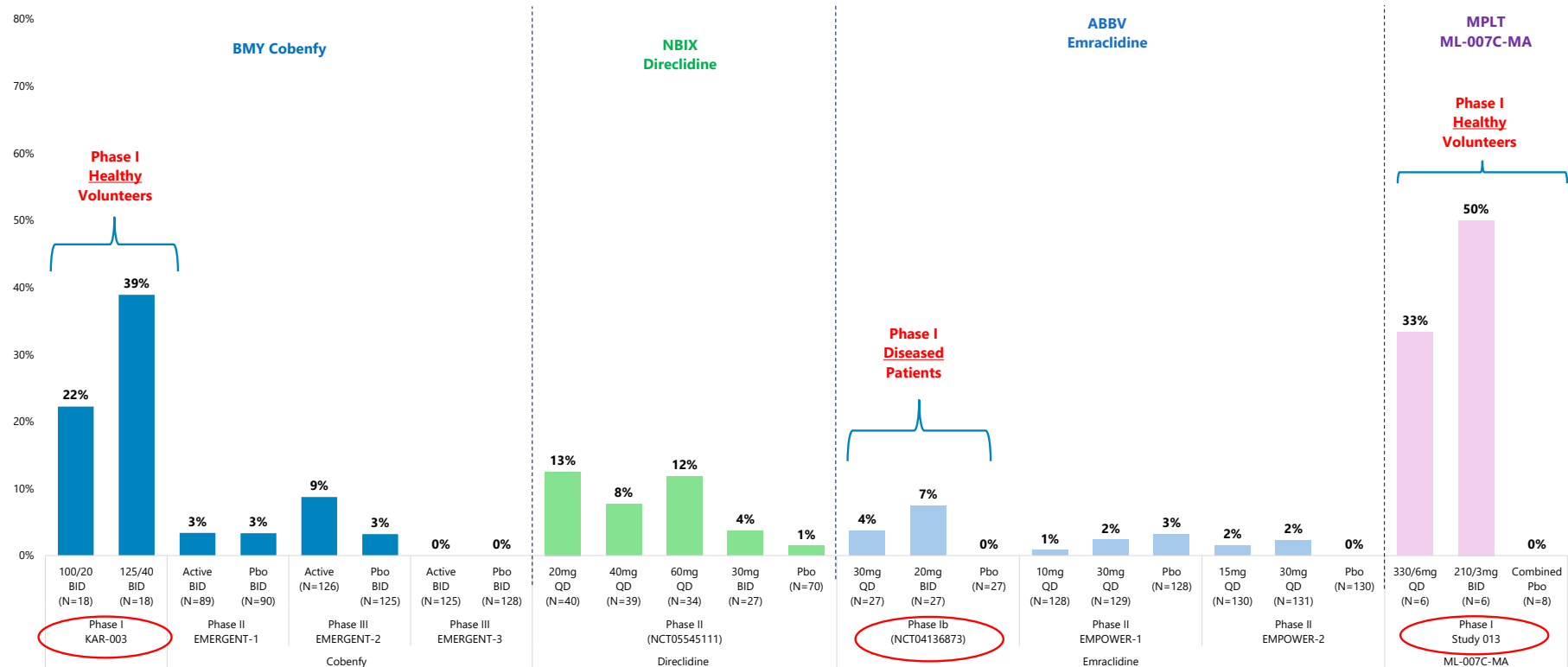
Muscarinic Safety Profile: Constipation Treatment-Emergent Adverse Events (TEAEs)



Source: Jefferies

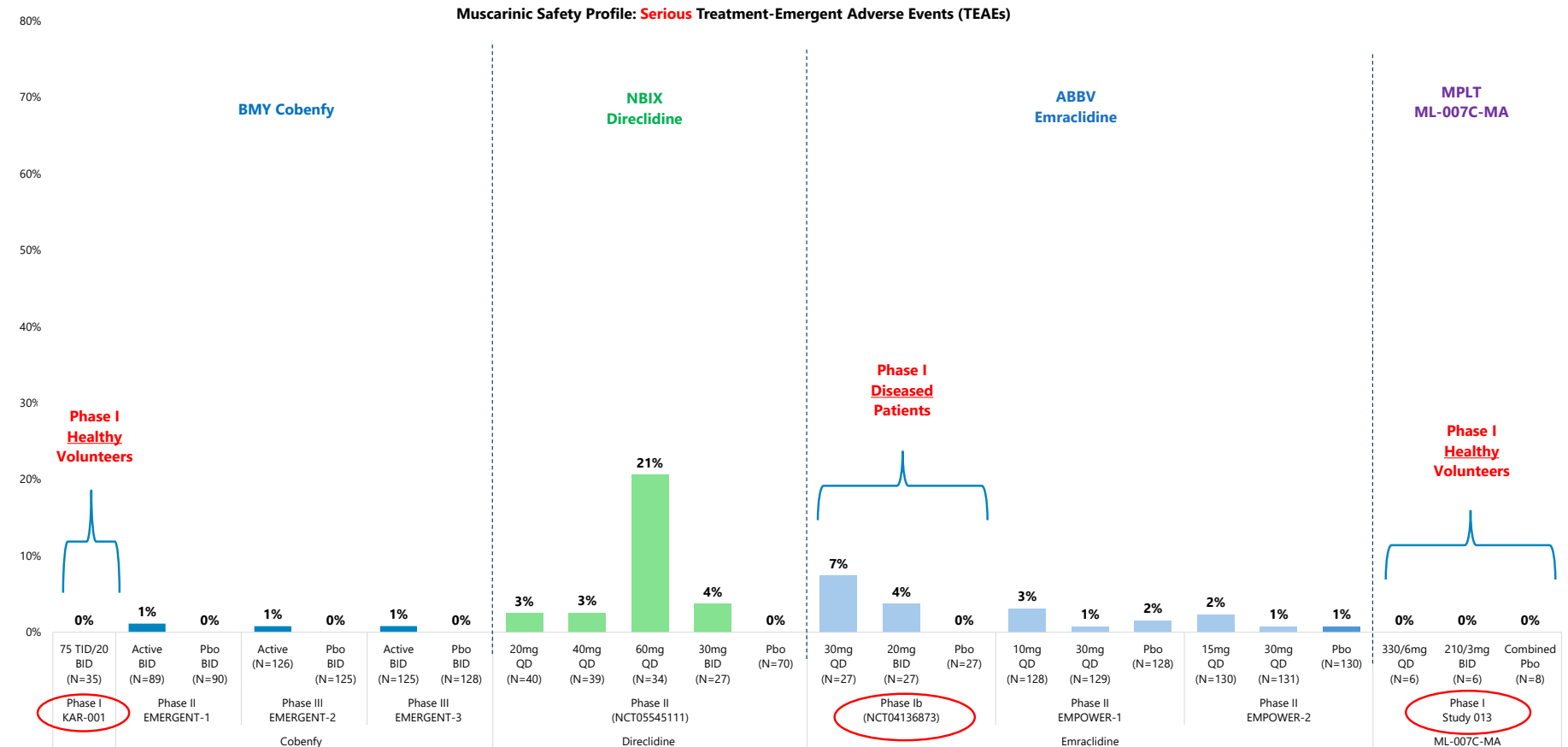
Exhibit 42 - Dizziness for ML-007C at 33-50% Do Look Marginally Higher vs Cobenfy's 22%-39% Though we Note Significant Degradation of These AEs into Cobenfy's Pivotal Studies Giving us Comfort into MPLT's Phase II ZEPHYR Study

Muscarinic Safety Profile: **Dizziness** Treatment-Emergent Adverse Events (TEAEs)



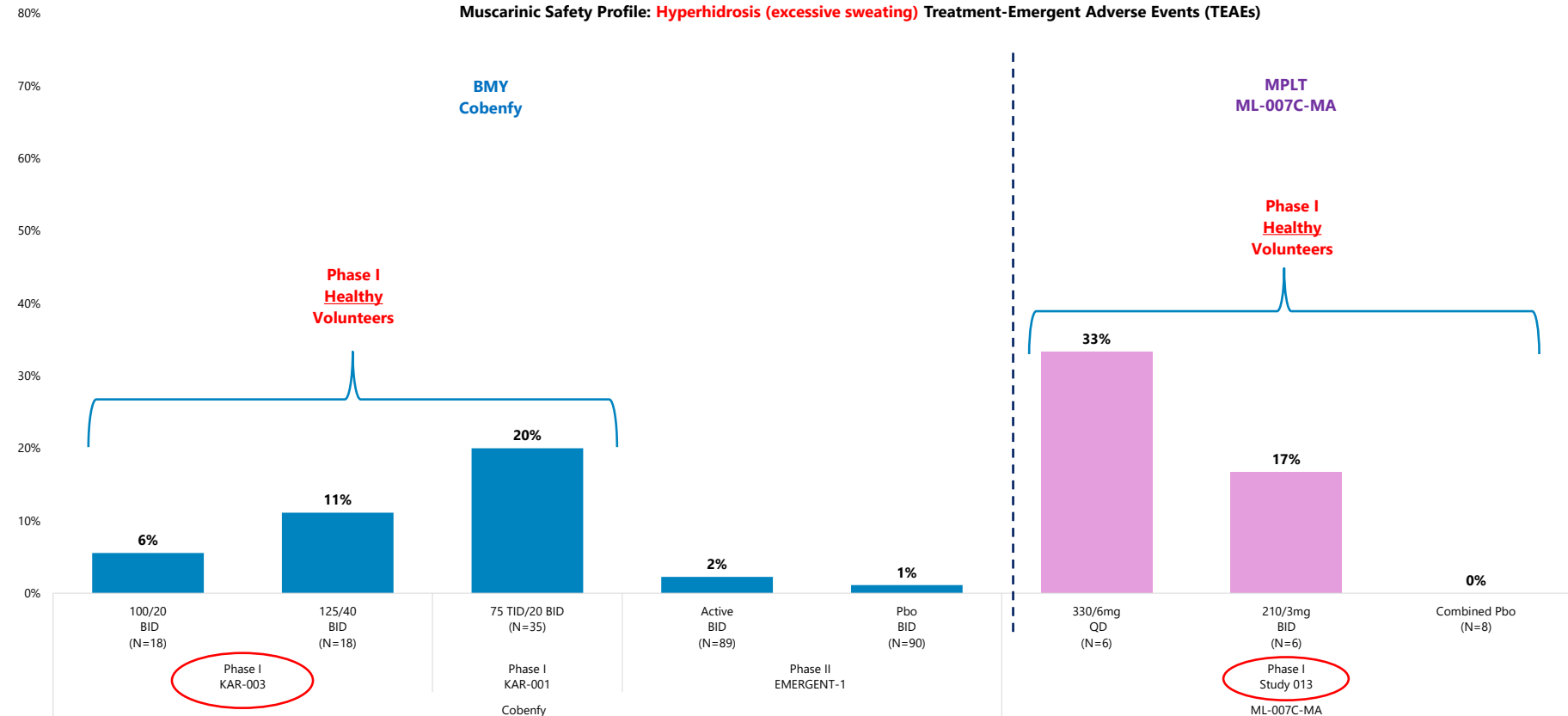
Source: Jefferies

Exhibit 43 - Zero Serious TEAEs were Observed for ML-007C, Similar to Cobenfy's Phase I study



Source: Jefferies

Exhibit 44 - Hyperhidrosis (Excessive Sweating) Rates Looked Higher for ML-007C in Phase I vs Cobenfy Though We See a Significant Step-down in Rates in Cobenfy's pivotal study EMERGENT-1, which Gives us Comfort into MPLT's Phase II ZEPHYR Study



Source: Jefferies

MPLT’s ML-007C-MA Could be Highly Competitive Compared to Other Muscarinic Programs

Unpacking CERE Emraclidine Failure in Schizophrenia - Little/No Readacross to MPLT

We are cautious to assign any readacross to ML-007C-MA from emraclidine’s (M4 PAM) two Phase III failures reported in Nov 2024, which resulted in ABBV taking an impairment charge of approximately \$4.5B (compared to the ~\$9B acquisition value). A prevailing hypothesis in the fallout of the (-) results is that dual M1/M4 agonism might be necessary for muscarinics in schizophrenia, which could indirectly be supported by MPLT’s internal preclinical data (described below), which suggests that M1KO mice treated with ML-007C-MA are most impacted by M1 agonism. Additionally, M1 agonism might have a larger role in cognitive/negative symptoms of schizophrenia, while M4 primarily modulates dopaminergic tone, reducing hyperdopaminergic signaling linked to positive symptoms (e.g., hallucinations/delusions). As such, dual M1/M4 can arguably do a better job of encompassing the entire spectrum of schizophrenia symptoms on PANSS. Week 6 PANSS placebo responses were also somewhat higher in the Phase III studies (e.g., -13.5-16.1) relative to Phase II (-6.8) and our calculated Week 6 PANSS placebo average of -10.2 across 50+ schizophrenia trials. In general though, dose escalation was somewhat limited for emraclidine.

Following the two Phase III failures, ABBV plans to run another multiple-ascending-dose (MAD) study to re-assess doses. They are currently conducting the study to test if they can move the dose higher than 30mg. We wouldn’t be surprised if data from the MAD study could be available as early as 2026-27. In the future, ABBV hopes to apply Phase II learnings to position emraclidine in Phase III trials of schizophrenia as a monotherapy or adjunctive setting. As such, it seems like true Phase II efficacy data for emraclidine could be available in 2028-29 at the earliest. We’d emphasize that ML-007C-MA has been rigorously evaluated across four separate Phase I campaigns, optimizing the dose ahead of Phase II study. Therefore, we believe a (+) Phase II readout for QD dosing would allow us to have greater confidence in Phase II/III translation.

Exhibit 45 - Emraclidine (M4 PAM) Week 6 PANSS Placebo Effects in the Failed Phase II are Higher than Typical Phase II/III Clinical Trials

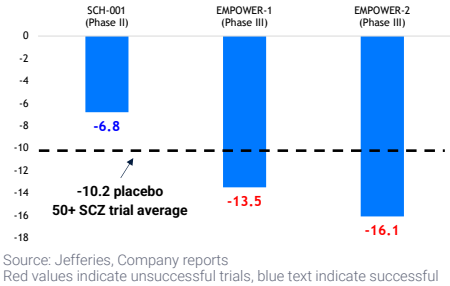


Exhibit 46 - Emraclidine Failed EMPOWER-1/2 Study Results Suggest Additional Dose Optimization is Required

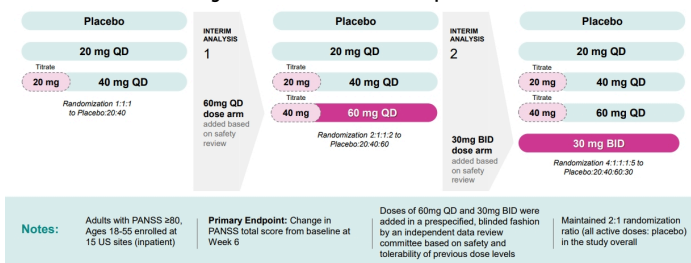
	EMPOWER-1			EMPOWER-2		
	Placebo (N= 127)	Emraclidine 10mg QD (N = 125)	Emraclidine 30mg QD (N = 127)	Placebo (N = 128)	Emraclidine 15mg QD (N = 122)	Emraclidine 30mg QD (N = 123)
Baseline (SD)	98.3 (8.16)	97.6 (7.65)	97.9 (7.89)	97.4 (8.22)	98.0 (8.49)	97.2 (7.75)
LS Mean (95% CI)	-13.5 (-17.0, -10.0)	-14.7 (-18.1, -11.2)	-16.5 (-20.0, -13.1)	-16.1 (-19.4, -12.8)	-18.5 (-22.0, -15.0)	-14.2 (-17.6, -10.8)

Source: Company reports

Pivotal Phase III Data for NBIX's Direclidine M4 Agonist in 2027-28

NBIX has Phase III data for NBI-117568 ('568 or direclidine), which is a >500-fold selective M4-preferring agonist eligible for once-daily (QD) dosing. Direclidine may work directly and selectively at M4, without dependence on endogenous acetylcholine, and without add-back therapy to blunt peripheral cholinergic effects. Indeed, direclidine showed decent rates of constipation (5%) and nausea (5%) at the 20mg dose, albeit higher 13% somnolence/dizziness and 8% discontinuation rates due to TEAEs in the most recent Phase II data the read out in Aug 2024. We'd note direclidine can also be taken with or without food. The data package was somewhat difficult to interpret, given the lack of a dose response. For example, the placebo showed a -10.8 week 6 PANSS, with direclidine treatment resulting in a -18.2 for 20mg QD (e.g., -7.5 pbo-adj, p=0.011), -12.6 40mg QD, -13.7 60mg QD, and -15.8 30mg BID. That said, an inverse dose response is common in CNS/Neuro. On the margin, the QD 20mg dose didn't have a titration requirement in the Phase II, which could provide a convenience advantage, should the Phase III programs read out successfully. NBIX is initiating three Phase III programs for direclidine, with data expected in 2027-28. Meanwhile, NBIX does have a pipeline of other muscarinics, including '567 (M1 agonist) for CNS indications, '569 (M4 agonist) for schizophrenia, '570 (M1/M4 agonist) for schizophrenia, and '986 (M1/M4 antagonist) for movement disorders.

Exhibit 47 - Phase II Design for Direclidine in Schizophrenia



Source: Company reports

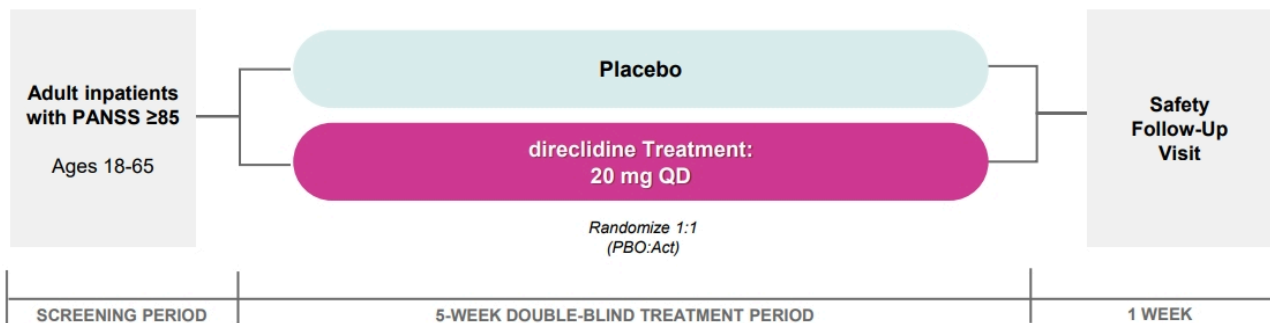
Exhibit 48 - Phase II Direclidine PANSS Efficacy at Week 6 Across All Doses

Week 6 (Day 42)	Placebo (N=68)	20 mg QD (N=35)	40 mg QD (N=38)	60 mg QD (N=34)	30 mg BID (N=26)
PANSS Total Score LS Mean Change from Baseline*	-10.8	-18.2	-12.6	-13.7	-15.8
LS Mean Difference vs. Placebo*	-	-7.5 (p=0.011)	-1.9 (p=0.282)	-2.9 (p=0.189)	-5.0 (p=0.090)
Effect Size**	-	0.61	0.27	0.39	0.23

*Least-squares (LS) means are from a MMRM which includes treatment group, visit, and study period as fixed effects; treatment group-by-visit interaction; baseline PANSS total score as a covariate; and subject as a random effect.
**Effect size (Cohen's D) is based on observed data.

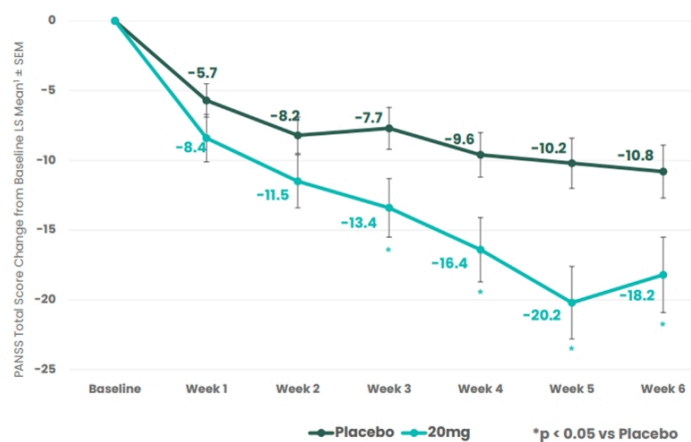
Source: Company reports

Exhibit 50 - Direclidine Phase III Programs are Testing the 20mg QD Dose vs Placebo with Data Expected in 2027-28



Source: Company reports

Exhibit 49 - Direclidine Phase II Efficacy Kinetics for the 20mg Dose Compared to Placebo Showed a -18.2 PANSS Reduction at Week 6 vs -10.8 Placebo

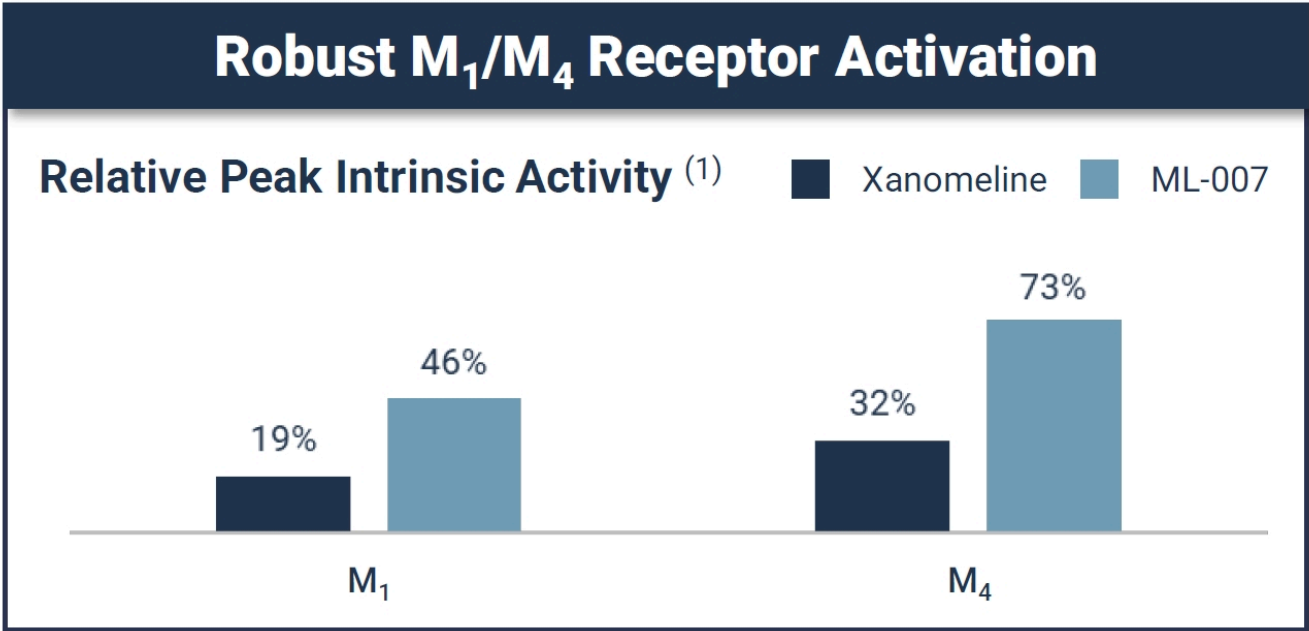


Source: Company reports

Preclinical and *in vitro* Data Supports a Potential Best in Class M1/M4 Muscarinic Agonist in ML-007

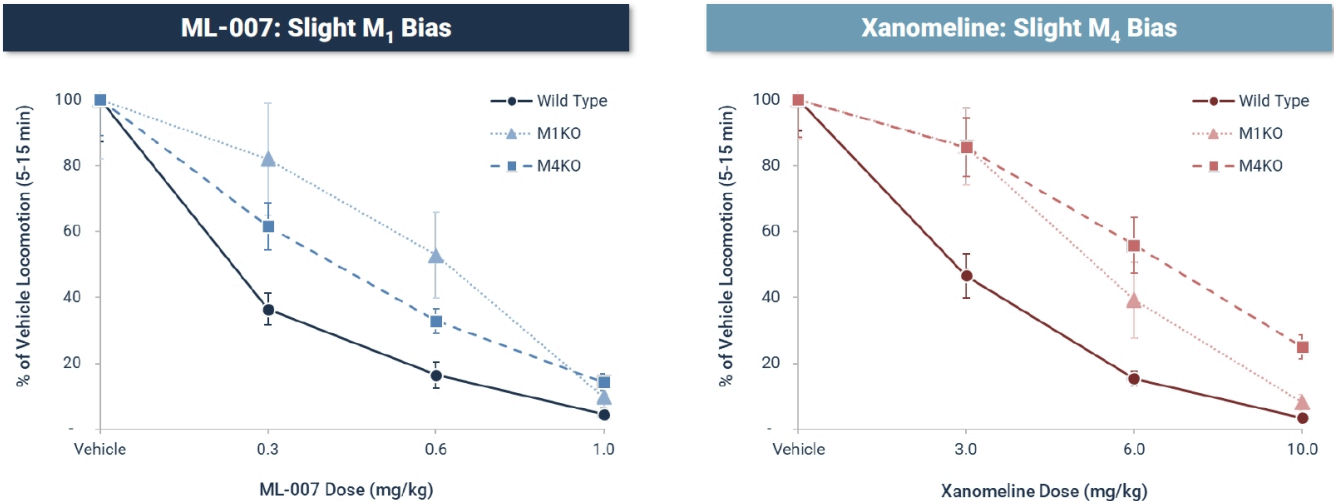
ML-007 has demonstrated promising biological activity in preclinical models and *in vitro* potency testing. (1) In *in vitro* potency testing using a GTPγS assay, ML-007 showed >2x higher peak intrinsic activity relative to xanomeline (e.g., active drug in KRTX's Cobenfy) across both M1 and M4 receptors. Stronger agonism of M1 and M4 for ML-007 (vs xanomeline) in some ways support pot'l biological activity ahead of Phase II schizophrenia data in H2:26. (2) MPLT also shows (+) preclinical data in two separate M1/M4 knockout in AIH mouse models of behavioral/cognitive deficits, supporting a differentiated agonism profile. At high doses of ML-007, M1/M4 activation of ML-007 was sufficient to reverse amphetamine-induced hyperlocomotion in both M1KO and M4KO mice. In contrast, xanomeline showed a slight M4 bias, with M1KO mice showing rescued hyperlocomotion, while M4KO mice did not. (3) The differentiated profile is further supported by hard-to-treat spatial and social memory mouse models of Alzheimer's disease. In a Y-maze spatial reference memory assay, ML-007-treated mice showed improved discrimination (p<0.05) between familiar/novel spaces compared to both vehicle controls and xanomeline treatment. In social memory, ML-007 treatment showed improved memory of a familiar mouse relative to a novel mouse and the test mouse spent more time with the novel mouse (p<0.05). **Net-net:** ML-007 shows improved cognitive benefits in mouse models of behavior/cognition relative to xanomeline.

Exhibit 51 - Relative Peak Intrinsic Activity of M1 and M4 Receptors for ML-007 Are Improved Over Xanomeline



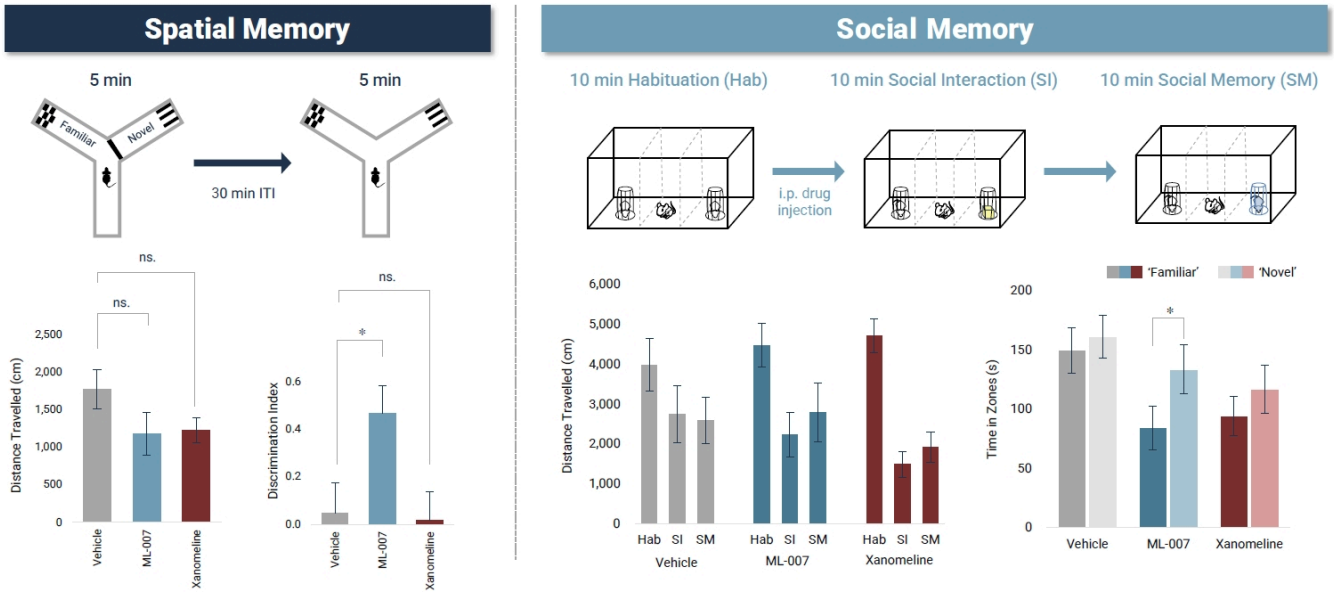
Source: Company reports

Exhibit 52 - Preclinical Pharmacodynamic Effects of ML-007 and Xanomeline in WT, M1KO, and M4KO AIH Mouse Models Underscore Importance of M1 Agonism in Alleviating Schizophrenia-Like Symptoms



Source: Company reports

Exhibit 53 - ML-007 Shows Improved Spatial and Social Memory Over Xanomeline



Source: Company reports

Exhibit 54 - ML-007C-MA Market Model in Schizophrenia

ML-007C-MA Sales Summary (\$M)	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E
US	-	-	-	-	34	116	197	278	357	435	519	605	691	779	869	959	1,051	1,144	1,239	1,335
EU5 (Before royalties)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Sales (\$M)	-	-	-	-	\$34	\$116	\$197	\$278	\$357	\$435	\$519	\$605	\$691	\$779	\$869	\$959	\$1,051	\$1,144	\$1,239	\$1,335
ML-007C-MA Patient Summary	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E
US	-	-	-	-	1,951	6,813	11,749	16,758	21,841	27,000	32,235	37,547	42,937	48,405	53,954	59,582	65,292	71,085	76,960	82,920
EU5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Patients	0	0	0	0	1,951	6,813	11,749	16,758	21,841	27,000	32,235	37,547	42,937	48,405	53,954	59,582	65,292	71,085	76,960	82,920
ML-007C-MA for Schizophrenia - US Market Model	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E
US Population (000s)	347,355	349,960	352,585	355,229	357,894	360,578	363,282	366,007	368,752	371,517	374,304	377,111	379,939	382,789	385,660	388,562	391,467	394,403	397,361	400,341
Annual Growth Rate	0.75%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%
Total SCZ Adult Prevalence (000s)	2,605	2,625	2,644	2,664	2,684	2,704	2,725	2,745	2,766	2,786	2,807	2,828	2,850	2,871	2,892	2,914	2,936	2,958	2,980	3,003
Prevalence	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%
Total Diagnosed SCZ Patients (000s)	2,475	2,493	2,512	2,531	2,550	2,569	2,588	2,608	2,627	2,647	2,667	2,687	2,707	2,727	2,748	2,768	2,789	2,810	2,831	2,852
Diagnosis Rate	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%
Patients Requiring Treatment (000s)	2,104	2,119	2,135	2,151	2,167	2,184	2,200	2,217	2,233	2,250	2,267	2,284	2,301	2,318	2,336	2,353	2,371	2,389	2,407	2,425
% Not requiring treatment at all	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%
Tx Discontinuation Rate	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%
Prevalence	1,262,202	1,271,668	1,281,206	1,290,815	1,300,496	1,310,250	1,320,077	1,329,977	1,339,952	1,350,002	1,360,127	1,370,328	1,380,605	1,390,960	1,401,392	1,411,902	1,422,491	1,433,160	1,443,909	1,454,738
Total ML-007C-MA Patients Treated	-	-	-	-	1,951	6,813	11,749	16,758	21,841	27,000	32,235	37,547	42,937	48,405	53,954	59,582	65,292	71,085	76,960	82,920
% penetration	-	-	-	0%	0.2%	0.5%	0.9%	1.3%	1.6%	2.0%	2.4%	2.7%	3.1%	3.5%	3.9%	4.2%	4.6%	5.0%	5.3%	5.7%
Annual Gross Price	\$23,000	-	-	-	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000
% Price increase	-	-	-	-	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Annual Net Price	-	-	-	-	\$17,250	\$17,020	\$16,790	\$16,560	\$16,330	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100
Gross to net	-	-	-	-	25%	26%	27%	28%	29%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
US Revenue (\$M)	-	-	-	-	\$34	\$116	\$197	\$278	\$357	\$435	\$519	\$605	\$691	\$779	\$869	\$959	\$1,051	\$1,144	\$1,239	\$1,335

Source: Jefferies estimates, Company reports

ML-007C-MA in Alzheimer's Disease Psychosis

Rationale for Muscarinic Agonists in Alzheimer's Disease Psychosis (ADP)

A 6-month landmark study (Bodick et al., 1997) in the 1990's demonstrated promising benefits on cognition in a study of xanomeline in Alzheimer's disease. The study randomized N=343 patients across 17 centers in the US and Canada to receive 75mg, 150mg, or 225mg of xanomeline per day or placebo for six months. Patients receiving high dose xanomeline (225mg) showed significant improvements on the ADAS-Cog domain ($p<0.05$), as well as the Clinician's Interview-Based Impression of Change (CIBIC+), which were stat-sig ($p<0.02$). There were also dose-dependent improvements in outbursts, suspiciousness, delusions, agitation, and hallucinations. However, the study was plagued with high 52% treatment AE-related discontinuations, predominantly due to gastrointestinal side effects (e.g., pot'l cholinergic effects). Ultimately, the behavioral/cognitive efficacy signals for xanomeline (and by extension M1/M4 agonism) in Alzheimer's should technically bode well for incoming datasets for both Cobenfy (data expected YE:25), as well as ML-007C-MA (H2:27), notwithstanding (+) trial execution.

Phase II Expectations for ML-007C-MA in ADP - Topline Data in H2:27

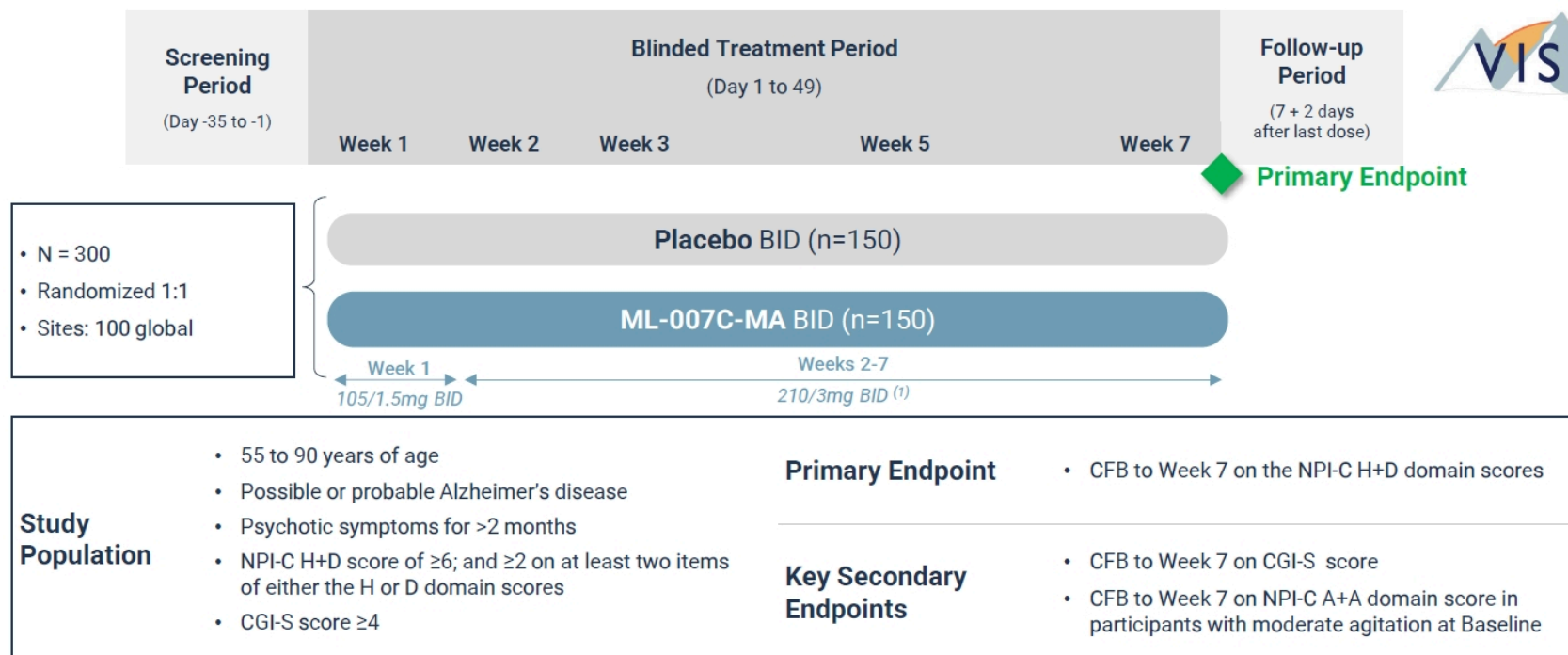
The VISTA Phase II trial is a double-blind, placebo-controlled study evaluating ML-007C-MA for the treatment of hallucinations and delusions associated with Alzheimer's Disease Psychosis (ADP), with topline data expected in H2:27. The trial aims to enroll approximately N=300 participants globally, who will be randomized in a 1:1 ratio to receive either ML-007C-MA at a dose of 210/3mg twice daily or placebo. Topline results from the VISTA trial are anticipated in H2:27.

Patients will begin treatment with a titration dose of ML-007C-MA 105/1.5 mg twice daily (or placebo) for one week before transitioning to the maintenance dose (e.g., 210/3mg). To address tolerability concerns, participants will be permitted a one-time dose reduction; those requiring this adjustment will remain on the reduced dose for the remainder of the study.

- **Primary endpoint:** Change from baseline to Week 7 in the Neuropsychiatric Inventory-Clinician Hallucinations and Delusions (NPI-C H+D) score.
- **Secondary endpoints:** Changes in Clinical Global Impression (e.g., CGI) scores and NPI-C agitation and aggression scores.

There are technically no approved drugs for ADP, making determining meaningful efficacy difficult to handicap. That said, KRTX's Cobenfy has an expected ADEPT-2 readout by YE:25, which could set a bar for MPLT because both trials are evaluating NPI-C H+D, albeit MPLT at 7 weeks and BMY at 14 weeks (based on 1-week and 4-5-week titrations, respectively). We remain constructive on ADP, considering the high unmet need, as well as early clinical trial results for psychosis symptoms associated with Alzheimer's. In theory, an approved safety profile (for both Cobenfy and ML-007C-MA) compared to xanomeline alone could improve trial execution, considering the earlier 6-month trial saw a high 52% discontinuation rate.

Exhibit 55 - Phase II Design of ML-007C-MA in Alzheimer's Disease Psychosis (ADP)



BID = twice daily, CFB = Change from baseline, CGI-S = Clinical Global Impression of Severity, NPI-C A+A = Neuropsychiatric Inventory Clinician, Agitation and Aggression, NPI-C H+D = Neuropsychiatric Inventory Clinician, Hallucinations and Delusions.

1. Increase to the target dose of 210/3 mg BID at Week 1, unless the participant is experiencing tolerability issues that, in the investigator's opinion, would preclude a higher dose. Every effort should be made to increase to the target dose as soon as possible up until Week 3. One-time dose reduction to 105/1.5 mg allowed for tolerability.

Source: Company reports

Cobenfy Data in ADP Expected by YE:25 Could Have Technical
Readacross to MPLT Stock, But We Remain Constructive

We expect N=400 Phase III ADEPT-2 data from Cobenfy in Alzheimer's disease psychosis (ADP) by YE:25, which could have readthrough to MPLT. In Q2:25, BMY reported that they are reviewing key studies to ensure the highest probability of success for their programs, which includes clinical trial site reviews for ADEPT-2. In Q3, BMY mgmt reiterated their confidence in a pot'l (+) readout. Given prior support from early xanomeline trials, there remains precedent for a (+) efficacy readout for Cobenfy. We'd also emphasize that timelines were not altered per se (data still by YE:25), despite mixed perception around mgmt's comments. In a positive outcome for the trial, it could allow us to better handicap (+) results for MPLT's study by creating an efficacy bar because both trials are evaluating NPI-C H+D as a primary endpoint (BMY at 14 weeks and MPLT at 7 weeks). The difference in trial duration endpoints likely stems from ML-007C-MA's 1-week titration requirement vs Cobenfy's 4-5-week titration requirement. Additionally, we'd argue that ML-007C-MA is a more potent M1/M4 agonist compared to Cobenfy (but the same mechanism), de-risking MPLT's own ADP data in H2:27. In a negative/mixed scenario, we could make a similar argument around ML-007C-MA's potency, so it could be important to consider the specifics of the trial readout (e.g., whether there is numerical separation). An improved safety profile from ML-007C-MA could also help to curb discontinuations, which could have (+) implications for the trial's execution and fidelity. Meanwhile, BMY has a second ADEPT-4 readout (an essential duplicate of ADEPT-2), in addition to a randomized withdrawal study (ADEPT-1), both with expected data readouts in 2026.

We're also aware of other select programs pursuing ADP that remain worthwhile to monitor: (1) ITCI/JNJ will have Phase III ADP data for Caplyta (lumateperone, atypical antipsychotic) in 2026-27. (2) ACAD also has Phase II ADP data for ACP-204 (5-HT2A inverse agonist) in mid-2026.

Exhibit 56 - BMY Phase III ADEPT Programs in Alzheimer's Disease Psychosis (ADP) Expected to Readout in 2025-26

Indication		Psychosis in Alzheimer's Disease (ADP)	
Phase/Study	Phase III - ADEPT-1	Phase III - ADEPT-2	Phase III - ADEPT-4
# of Patients	N = 380	N = 400	N = 406
Design	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 66.7/6.67 mg TID* - Placebo	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 66.7/6.67 mg TID* - Placebo	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 66.7/6.67 mg TID* - Placebo
Endpoints	- Primary: Time from randomization to relapse during the 26-week double blind randomized withdrawal period - Key secondary: Time from randomization to discontinuation for any reason during the 26-week Double-Blind Randomized Withdrawal treatment Period	- Primary: Change from baseline in Neuropsychiatric Inventory-Clinician: Hallucinations and Delusions (NPI-C: H+D) score to end of Week 14 - Key secondary: Change from baseline in Clinical Global Impressions-Severity (CGI-S) scale up to week 14.	- Primary: Change from baseline in Neuropsychiatric Inventory-Clinician: Hallucinations and Delusions (NPI-C: H+D) score up to Week 14 - Key secondary: Change from baseline in Clinical Global Impressions-Severity (CGI-S) scale up to week 14.
Status	- Recruiting - Projected data readout 2026	Projected data readout 2H 2025	- Recruiting - Projected data readout 2026
CT Identifier	NCT05511363	NCT06126224	NCT06585787

*Based-on tolerability
Source: Company reports

eharrison@maplightr.com - Maplight Therapeutics - 27/03/2026 06:21 AM UTC

Exhibit 57 - ML-007C-MA Market Model in Alzheimer's Disease Psychosis (ADP)

	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E			
ML-007C-MA Sales Summary (\$M)																							
ML-007C-MA (US)	0	0	0	0	0	47	188	327	462	596	737	879	1,023	1,168	1,315	1,463	1,613	1,764	1,916	2,071			
EU5 (Before royalties)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Total Sales (\$M)	-	-	-	-	-	\$47	\$188	\$327	\$462	\$596	\$737	\$879	#####	###	#####	###	#####	###	\$1,463	\$1,613	\$1,764	\$1,916	\$2,071
ML-007C-MA Patient Summary																							
US	0	0	0	0	0	2,789	11,212	19,719	28,311	36,989	45,752	54,603	63,540	72,566	81,680	90,884	100,177	109,561	119,037	128,605			
EU5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Total Patients	0	0	0	0	0	2,789	11,212	19,719	28,311	36,989	45,752	54,603	63,540	72,566	81,680	90,884	100,177	109,561	119,037	128,605			
ML-007C-MA for ADP - US Market Model																							
US Population (000s)	345,122	346,848	348,582	350,325	352,076	353,837	355,606	357,384	359,171	360,967	362,772	364,585	366,408	368,240	370,082	371,932	373,792	375,661	377,539	379,427			
Annual Growth Rate	0.50%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%			
Alzheimer's Disease Population (000s)	6,801	6,835	6,869	6,904	6,938	6,973	7,008	7,043	7,078	7,113	7,149	7,185	7,220	7,257	7,293	7,329	7,366	7,403	7,440	7,477			
Prevalence	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%			
(1) Alzheimer's with psychosis	2,720,401	2,734,003	2,747,673	2,761,412	2,775,219	2,789,095	2,803,040	2,817,056	2,831,141	2,845,297	2,859,523	2,873,821	2,888,190	2,902,631	2,917,144	2,931,730	2,946,388	2,961,120	2,975,926	2,990,805			
Prevalence	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%			
(2) Alzheimer's without psychosis	4,080,602	4,101,005	4,121,510	4,142,118	4,162,828	4,183,642	4,204,561	4,225,583	4,246,711	4,267,945	4,289,285	4,310,731	4,332,285	4,353,946	4,375,716	4,397,594	4,419,582	4,441,680	4,463,889	4,486,208			
Prevalence	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%			
Total ML-007C-MA Patients Treated	-	-	-	-	-	2,789	11,212	19,719	28,311	36,989	45,752	54,603	63,540	72,566	81,680	90,884	100,177	109,561	119,037	128,605			
Alzheimer's with psychosis	-	-	-	-	-	2,789	11,212	19,719	28,311	36,989	45,752	54,603	63,540	72,566	81,680	90,884	100,177	109,561	119,037	128,605			
% penetration (overall subgroup)	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.4%	0.7%	1.0%	1.3%	1.6%	1.9%	2.2%	2.5%	2.8%	3.1%	3.4%	3.7%	4.0%	4.3%			
Alzheimer's without psychosis	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
% penetration (overall subgroup)	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%			
Annual Gross Price	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000			
% Price increase	-	-	-	-	-	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%			
Annual Net Price	-	-	-	-	-	\$17,250	\$17,020	\$16,790	\$16,560	\$16,330	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100			
Gross to net	-	-	-	-	-	25%	26%	27%	28%	29%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%			
US Revenue (\$M)	-	-	-	-	-	\$47	\$188	\$327	\$462	\$596	\$737	\$879	\$1,023	\$1,168	\$1,315	\$1,463	\$1,613	\$1,764	\$1,916	\$2,071			
Revenue - Alzheimer's with psychosis	-	-	-	-	-	\$47	\$188	\$327	\$462	\$596	\$737	\$879	\$1,023	\$1,168	\$1,315	\$1,463	\$1,613	\$1,764	\$1,916	\$2,071			
Revenue - Alzheimer's without psychosis	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			

Source: Jefferies estimates, Company reports

Earlier Stage Pipeline Products Could Represent Sources of Valuation Upside

(1) ML-004 for Autism Spectrum Disorder (ASD)

MPLT is advancing ML-004, an oral immediate-release/extended-release (IR/ER) formulation of zolmitriptan, a well-characterized 5-HT_{1B/1D} receptor agonist originally approved for migraine treatment. ASD affects ~1 in 36 children in the U.S., with no approved therapies for core symptoms. ML-004 offers a first-in-class mechanism targeting sociability, positioning it as a welcome alternative in a market dominated by symptomatic treatments. The program targets social communication deficits and irritability in ASD-core symptoms for which no FDA-approved therapies currently exist. Existing treatments for ASD-associated irritability are limited to atypical antipsychotics, which carry significant safety liabilities. **Net-net:** If successful, ML-004 could capture significant share by addressing both social deficits and irritability while avoiding antipsychotic-related adverse effects.

The therapeutic hypothesis is rooted in early stage evidence linking serotonergic dysregulation to ASD pathology. Preclinical and optogenetic studies demonstrate that activation of the dorsal raphe nucleus (DRN) to nucleus accumbens (NAc) serotonin circuit enhances sociability, an effect mediated by 5-HT_{1B} receptor activation. Blocking these receptors disrupts social reward, while agonists restore sociability and reduce aggression in ASD models. Additional validation comes from mechanistic insights into MDMA's pro-social effects, which are abolished by 5-HT_{1B} antagonism. ML-004 aims to replicate these benefits without MDMA's addictive liability, leveraging zolmitriptan's established brain penetration and safety profile. MPLT conducted 5 *in vivo* studies assessing locomotion, sociability, and aggression in ASD models, consistently showing statistically significant improvements in social behavior and reductions in aggression. These findings support ML-004's potential to address unmet needs in ASD.

ML-004 is currently in the Phase II IRIS trial, evaluating efficacy on social communication deficits (primary endpoint) and irritability (secondary endpoint). Topline results are expected in H2:26. Chronic dosing at higher exposures than migraine indications necessitated new IR/ER formulations and additional safety/PK work prior to efficacy trials. The IRIS trial is a multicenter, randomized, double-blind, placebo-controlled Phase II study evaluating ML-004 in N=150 adolescents (12-17) and adults (18-45) with Autism Spectrum Disorder (ASD) across Australia, Canada and the US. The primary goal is to assess improvement in core social communication deficits, a domain with no approved therapies. Patients will enter a 12-week maintenance phase following dose titration to flexible maintenance doses of 24, 48, or 72mg.

- **Primary Endpoint:** Change from baseline on Autism Behavior Inventory Social Communication (ABI-SC) Domain Score
- **Secondary Endpoint:** Change from baseline on Aberrant Behavior Checklist - Irritability (ABC-I) for patients with moderate or greater irritability at baseline

Patients completing IRIS may enter a 52-week open-label extension to evaluate long-term safety of ML-004. Past trials of Abilify and Risperdal (atypical antipsychotics) have instead used the ABC-I, making ABI-SC a more differentiated approach to ASD treatment, focused on cognition/communication (vs irritability). That said, we may still expect some benefits on ABC-I secondary endpoints, which will be worthwhile to compare to atypical antipsychotics. For context, Abilify/Risperdal have shown **-6-11 pbo-adj** changes on ABC-I, corresponding to **-44-57%** changes.

Exhibit 58 - ABC-I Subscale - Percent Reduction (or Improvement) Achieved by Drugs for Irritability in Autism Spectrum Disorder (ASD)

Company	Drug	Status	Study	Design	Duration	Age	N	Dose	Baseline ABC-I	Efficacy Change	Pbo Δ Change	%	Pbo Δ Change
Otsuka/ BMY	Abilify	Approved	Study 1	Pbo- Controlled	8 Weeks	6-17	98	2-15mg QD	29.6	-12.9	-7.9	-44%	-27%
								Placebo	30.2	-5.0	-	-17%	-
			Study 2	Pbo- Controlled	8 Weeks	6-17	218	15mg QD	28.9	-14.4	-6.0	-50%	-20%
								10mg QD	28.2	-13.2	-4.8	-47%	-17%
								5mg QD	28.6	-12.4	-4.0	-43%	-13%
								Placebo	28.0	-8.4	-	-30%	-
JNJ	Risperdal	Approved	Study 1	Pbo- Controlled	8 Weeks	5-16	101	0.5-3.5mg BID	26.2	-14.9	-11.3	-57%	-43%
								Placebo	25.5	-3.6	-	-14%	-
			Study 2	Pbo- Controlled	8 Weeks	5-12	55	0.02-0.06mg/kg QD	18.9	-12.1	-5.6	-64%	-33%
								Placebo	21.2	-6.5	-	-31%	-

Source: Jefferies, Company reports

Exhibit 59 - ML-004 Market Model in Autism Spectrum Disorder (ASD)

ML-004 Sales Summary (\$M)	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E
US	-	-	-	-	-	-	39	119	200	282	365	450	535	622	711	800	891
EUS (Before royalties)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Sales (\$M)	-	-	-	-	-	-	\$39	\$119	\$200	\$282	\$365	\$450	\$535	\$622	\$711	\$800	\$891
ML-004 Patient Summary	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E
US	-	-	-	-	-	-	2,763	8,350	14,022	19,778	25,619	31,547	37,563	43,667	49,860	56,144	62,519
EUS	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Patients	0	0	0	0	0	0	2,763	8,350	14,022	19,778	25,619	31,547	37,563	43,667	49,860	56,144	62,519
ML-004 for Autism Spectrum Disorder - US Market Model	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E
US Population (000s)	347,355	349,960	352,585	355,229	357,894	360,578	363,282	366,007	368,752	371,517	374,304	377,111	379,939	382,789	385,660	388,552	391,467
Annual Growth Rate	0.75%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%
Total ASD Prevalence (000s)	8,416	8,479	8,543	8,607	8,672	8,737	8,802	8,868	8,935	9,002	9,069	9,137	9,206	9,275	9,344	9,414	9,485
Prevalence	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%
Total Diagnosed ASD Patients (000s)	7,575	7,631	7,689	7,746	7,804	7,863	7,922	7,981	8,041	8,101	8,162	8,223	8,285	8,347	8,410	8,473	8,536
Diagnosis Rate	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%
Moderate to Severe (000s)	3,522	3,549	3,575	3,602	3,629	3,656	3,684	3,711	3,739	3,767	3,795	3,824	3,853	3,881	3,911	3,940	3,969
% Not requiring treatment at all	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%
Tx Discontinuation Rate	1,761,091	1,774,299	1,787,606	1,801,013	1,814,521	1,828,130	1,841,841	1,855,654	1,869,572	1,883,594	1,897,720	1,911,953	1,926,293	1,940,740	1,955,296	1,969,960	1,984,735
Prevalence	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%
Total ML-004 Patients Treated	-	-	-	-	-	-	2,763	8,350	14,022	19,778	25,619	31,547	37,563	43,667	49,860	56,144	62,519
% penetration	0%	0%	0%	0%	0%	0%	0.2%	0.5%	0.8%	1.1%	1.4%	1.7%	2.0%	2.3%	2.6%	2.9%	3.2%
Annual Gross Price	\$19,000	-	-	-	-	-	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000
% Price increase	-	-	-	-	-	-	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Annual Net Price	-	-	-	-	-	-	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250
Gross to net	25%	-	-	-	-	-	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%
US Revenue (\$M)	-	-	-	-	-	-	\$39	\$119	\$200	\$282	\$365	\$450	\$535	\$622	\$711	\$800	\$891

Source: Jefferies estimates, Company reports

Clinical Development History of ML-004 - Summary of Phase I Studies

For context, MPLT developed ML-004 for chronic use at higher doses than approved doses for zolmitriptan (e.g., Zomig is 2.5-5mg) and has completed two Phase I trials to evaluate safety, tolerability and PK.

MAP-ZOL-HV-001 (MAD PK Study):

- **Design:** Randomized, double-blind, placebo-controlled, N=40 healthy adults across five ascending-dose cohorts (5-40mg TID/3x daily).
- **Findings:** Most adverse events (headache, dizziness, nausea, hiccups) were mild, dose-related, and more frequent at 40mg TID. No serious adverse events (SAEs), no clinically significant QTc changes.
- **Implication:** Confirms safety margin for chronic high-dose exposure.

ML-004-ER-001 (Bioavailability Study):

- **Design:** Open-label, N=12 healthy adults under fasted/fed conditions, comparing IR zolmitriptan (20mg) vs ML-004 (24mg bi-layer IR/ER).
- **Findings:** ML-004 achieved prolonged absorption and reduced Cmax vs IR zolmitriptan, supporting a smoother PK profile. Modeling predicts 48-72 mg doses sustain target plasma exposure for 12-15 hours.
- **Safety:** Well tolerated, fewer TEAEs vs IR zolmitriptan, which were all mild/moderate, and resulted in no discontinuations.

(2) ML-021 in Parkinson's Disease

ML-021 is a selective antagonist of the muscarinic M4 receptor, a target highly expressed in the direct pathway neurons of the striatum. In Parkinson's disease, the loss of striatal dopamine leads to increased acetylcholine signaling, which activates M4 receptors and suppresses direct pathway activity, ultimately contributing to motor deficits. While anticholinergic agents have been used for decades (e.g., benztropine and trihexyphenidyl) to treat Parkinson's symptoms, their clinical utility has been limited by off-target antagonism at M1 receptors, which is associated with cognitive impairment, psychosis, and gastrointestinal side effects. By selectively blocking M4 receptors, ML-021 is designed to restore direct pathway activity and improve motor symptoms without the liabilities of M1 antagonism.

The therapeutic rationale for ML-021 is supported by robust preclinical evidence. Optogenetic experiments have demonstrated that activation of the direct pathway enhances movement and reverses motor deficits in Parkinson's models. Using both tool compounds and ML-021 itself, MPLT has shown improvements in motor function across multiple *in vitro* and *in vivo* studies. These findings underscore the potential for ML-021 to deliver clinically meaningful benefit through a non-dopaminergic mechanism.

Development is progressing toward IND-enabling studies, which are expected to be completed in H2:26. This positions ML-021 for pot'l first-in-human (FIH) trials in 2027. If successful, ML-021 could redefine the role of cholinergic modulation in Parkinson's disease, offering a differentiated adjunctive therapy with a more favorable safety profile. Given the global prevalence of Parkinson's disease and the limitations of current treatment options, even modest penetration could translate into a \$1B+ commercial opportunity in theory.

(3) ML-009 in Impulsivity, Agitation, and Irritability

ML-009 is a positive allosteric modulator (PAM) of G-protein-coupled receptor 52 (GPR52), an orphan receptor selectively expressed in the indirect pathway neurons of the striatum. GPR52 plays a critical role in regulating neuronal function through cAMP-dependent signaling and is colocalized with dopamine D2 receptors, the primary target of antipsychotics. Activation of GPR52 counteracts D2 receptor activity, enhancing indirect pathway signaling and offering a novel mechanism to address symptoms such as impulsivity, hyperactivity, and agitation across a range of central nervous system (CNS) disorders.

Preclinical studies have demonstrated that GPR52 activation reduces stimulant-induced hyperactivity, impulsivity, and aggression without inducing catalepsy (e.g., extrapyramidal side effects), a common liability of dopaminergic modulation. Leveraging internal medicinal chemistry capabilities, MPLT has identified multiple selective, orally bioavailable PAMs of GPR52. These candidates have shown promising pharmacological profiles in both *in vitro* and *in vivo* models, validating the therapeutic potential of this approach.

The program is currently in advanced preclinical development, with multiple compounds under evaluation. MPLT expects to nominate a lead candidate for IND-enabling studies in 2026, marking a significant milestone toward clinical translation. If successful, ML-009 could emerge as a first-in-class therapy targeting behavioral dysregulation in a variety of CNS/Neuro disorders.

Intellectual Property

ML-007

MPLT holds a robust patent portfolio protecting ML-007 and its therapeutic applications. The company owns three issued US patents covering the composition of matter (CoM) for ML-007, which are expected to expire between 2031 and 2032, excluding potential patent term adjustments or extensions. Additionally, one issued US patent covers the use of ML-007 for treating schizophrenia and agitation-related disorders (ADP), with an expected expiration in 2031 under similar conditions.

Beyond issued patents, MPLT has actively pursued global protection for combination therapies involving ML-007 and a positive allosteric modulator (PAC). The portfolio includes one pending US application and 16 pending foreign applications across major jurisdictions including Australia, Canada, China, the European Patent Office, Japan, and others, with any resulting patents anticipated to expire in 2042. Further, MPLT has filed a Patent Cooperation Treaty (PCT) application directed to pharmaceutical compositions containing ML-007 and PAC, which could extend exclusivity through 2044.

MPLT also owns a pending U.S. provisional application covering salts and crystalline polymorphs of ML-007. Patents issuing from this priority application **could extend protection until 2045**, subject to adjustments or extensions. As such, MPLT could have a therapy in ML-007C-MA with significant tail value.

ML-004

MPLT owns one pending US application and 13 pending foreign applications covering the use of ML-004 to treat ASD symptoms across key jurisdictions, including Australia, Canada, China, the European Patent Office, Japan, Korea, and others. If granted, these patents would provide exclusivity through 2040, subject to potential term adjustments or extensions.

In addition, MPLT has filed one pending US application and 21 pending foreign applications directed to its IR/ER oral ML-004 compositions and their use in ASD treatment. These filings span major markets such as the U.S., Europe, China, Japan, India, and the Middle East, **with any resulting patents expected to expire in 2042**, excluding possible extensions.

ML-009

PLT owns one pending PCT application directed to small molecule GPR52 agonists. Any issued patents from this PT application would be expected to **expire in 2044**, excluding extension.

Licensing and Other Collaboration Agreements

NeuroSolis Agreement

In June 2020, the MPLT entered into an asset purchase agreement with NeuroSolis, Inc., acquiring proprietary M1/M4 agonist molecules and associated IP, including programs for identifying modulators of muscarinic receptors. Under this agreement, MPLT assumed no liabilities and committed to using commercially reasonable efforts to achieve specified development and regulatory milestones. In May 2024, NeuroSolis granted a waiver extending one milestone deadline, which MPLT subsequently met in June 2025. To date, MPLT has paid \$150,000 in upfront and milestone payments and issued 447,006 shares of common stock, with up to 1,043,014 shares remaining contingent on future milestones.

Michael J. Fox Foundation Grant Agreement

MPLT maintains grant agreements with the Michael J. Fox Foundation, which could require future payments up to twice the grant amount based on net product sales. The timing and likelihood of these obligations remain uncertain.

University Agreements - Stanford and Vanderbilt

MPLT also holds license and collaboration agreements with Stanford University and other academic institutions, which include nominal annual maintenance fees and potential milestone or royalty payments tied to regulatory and commercial achievements.

In November 2024, MPLT entered into an exclusive license agreement with Vanderbilt University. This agreement grants worldwide rights to develop and commercialize certain compounds and licensed products, along with non-exclusive rights to use Vanderbilt's know-how and tool compounds. Initial consideration included a \$0.3 million upfront payment and \$0.3 million reimbursement for past patent costs, totaling \$0.6 million recorded as R&D expense. The agreement also provides for up to \$52.4 million in development and commercial milestone payments and low single-digit royalties on net sales, subject to reductions if market exclusivity lapses. As of YE:24, no milestone or royalty payments have been incurred under this agreement.

Key Risk Factors

Limited Operating History: MPLT is a clinical-stage biopharmaceutical company formed in 2018 with no history of commercializing products, making it difficult to assess its long-term viability.

Unproven Development Approach: MPLT's platform and methods for discovering and developing product candidates are untested, and the company has not demonstrated the ability to obtain regulatory approvals or commercialize products.

Persistent Losses and No Profitability: The company has incurred substantial losses since inception and expects significant, increasing losses for the foreseeable future, with no assurance of achieving profitability.

Dependence on Future Financing: MPLT requires additional capital beyond current resources. Failure to secure funding on acceptable terms could halt development and commercialization efforts.

Regulatory Approval Uncertainty: FDA and foreign approval processes are lengthy, costly, and unpredictable. Failure to obtain timely approvals could impact MPLT's ability to market products.

Clinical Development Risks: MPLT's trials may face delays, enrollment challenges, or failure to demonstrate safety and efficacy, which could derail development timelines and increase costs.

Commercialization Challenges: Even if approved, MPLT's products may fail to gain market acceptance due to entrenched standards of care, pricing pressures, or inadequate reimbursement.

Competitive Pressures: Rapidly evolving technologies and well-funded competitors could render MPLT's products less effective or obsolete, limiting commercial opportunity.

Intellectual Property Limitations: MPLT's ability to protect proprietary technology is uncertain; competitors could develop similar products, undermining its competitive position.

Operational and Third-Party Reliance: MPLT depends on CROs, manufacturers, and strategic partners, introducing risks of delays, compliance failures, and integration challenges.

Management Biographies

Christopher Kroeger, M.D., MBA (CEO): Chris brings over two decades of leadership experience in development-stage therapeutic and medical device companies. Prior to joining MPLT, Chris served as CEO of Cardioxyl Pharmaceuticals, where he guided the development of three investigational new drug candidates and oversaw six successful Phase 1 and Phase 2 clinical trials, culminating in the company's acquisition by Bristol Myers Squibb for \$2.1B. Chris also led OvaScience, a publicly traded fertility company, through a strategic turnaround that resulted in its merger with Millendo Therapeutics. Earlier in his career, Chris was a Partner at The Aurora Funds, a venture capital firm focused on early-stage biotech and medical device investments, and held roles at Genzyme and Decision Resources. He earned a bachelor's degree in Biology from Harvard University, an M.D. from Stanford University School of Medicine, and an MBA from Harvard Business School. Chris completed his General Surgery residency at Brigham and Women's Hospital, an affiliate of Harvard Medical School. He currently serves on the boards of Stellaromics and Antlia Biosciences.

Vishwas Setia (CFO): Vishwas serves as MPLT's Chief Financial Officer, bringing extensive experience in healthcare investment banking and strategic advisory. Before joining, he was Managing Director in the healthcare investment banking group at Bank of America Securities, where he advised biopharmaceutical companies on capital markets and strategic transactions. Over his career, Vishwas executed more than \$15 billion in equity and equity-linked financings and over \$100 billion in M&A and other strategic deals. He holds an MBA from the Kellogg School of Management at Northwestern University and a Bachelor of Engineering in Mechanical Engineering from Delhi College of Engineering, India.

Erin Pennock Foff, M.D., Ph.D. (CMO): Erin serves as MPLT's Chief Medical Officer, bringing over 13 years of experience in clinical trial design and execution, alongside a strong foundation in molecular biology and genetics. Prior to joining MPLT, Erin was Executive Director of Clinical Development at ACAD, where she led programs focused on dementia and neurodegenerative disorders. Before transitioning to industry, Erin held an academic physician-scientist role at the University of Virginia as a cognitive neurologist specializing in early-onset dementias and directed a translational research program investigating RNA toxicity and neurodegeneration. She earned a doctorate in Molecular and Human Genetics from Baylor College of Medicine, a medical degree from Jefferson Medical College, and completed her neurology residency at the University of Virginia. Erin is board-certified by the American Board of Psychiatry and Neurology.

Anatol Kreitzer, Ph.D. (CDO): Anatol serves as MPLT's Chief Discovery Officer, overseeing platform development and target discovery initiatives. He brings more than 20 years of experience in academic neuroscience research, specializing in neuromodulation and the use of optogenetics, electrophysiology, and imaging to study neural circuit function and dysfunction. Prior to joining MPLT, Anatol was a Professor of Physiology and Neurology at the University of California, San Francisco, where his laboratory focused on the basal ganglia, a key brain region governing motivation, decision-making, and behavioral automation. His team made significant contributions to understanding the cellular, synaptic, and circuit mechanisms underlying motor deficits in Parkinson's disease and dyskinesia. Anatol earned a bachelor's degree from the University of California, Berkeley, and a Ph.D. in Neurobiology from Harvard University. He completed postdoctoral training at Stanford University and advanced to Full Professor at UCSF, where he also served as Director of the Neuroscience Graduate Program from 2016 to 2019.

Exhibit 60 - ML-007C-MA DCF in Schizophrenia

Source: Jefferies estimates, Company reports

Peak Sales Calculator	Downside	Base Case	Upside
Peak sales (\$M)	\$500	\$2,071	\$2,000
Multiple	2.0x	2.0x	2.5x
Valuation	\$1,000	\$4,141	\$5,000
Assumed PoS	0%	15%	30%
\$/Share (adjusted)	\$0	\$12	\$29

76

Exhibit 62 - ML-004 DCF in Autism Spectrum Disorder (ASD)

DCF	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	
Revenue (in \$M)	-	-	-	-	-	-	\$39	\$119	\$200	\$282	\$365	\$450	\$535	\$622	\$711	\$800	\$891	
US	-	-	-	-	-	-	39	119	200	282	365	450	535	622	711	800	891	
EU5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Milestone Payments	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Royalties	0%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Cost of Goods sold	-	-	-	-	-	-	2	6	10	14	18	22	27	31	36	40	45	
% of Revenue	0%	0%	0%	0%	0%	0%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	
R&D	27	41	46	51	35	35	20	10	10	10	10	10	10	10	10	10	10	
% of Revenue	-	-	-	-	-	-	51%	8%	5%	4%	3%	2%	2%	2%	1%	1%	1%	
SG&A	4	5	5	6	20	50	100	100	100	100	100	100	100	100	100	100	100	
% of Revenue	-	-	-	-	-	-	264%	84%	50%	35%	27%	22%	19%	16%	14%	12%	11%	
Total EBIT	(\$31)	(\$45)	(\$51)	(\$58)	(\$55)	(\$85)	(\$83)	\$3	\$80	\$158	\$237	\$317	\$399	\$481	\$565	\$650	\$736	
Tax rate	0%	0%	0%	0%	0%	0%	0%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	
Profit	(\$31)	(\$45)	(\$51)	(\$58)	(\$55)	(\$85)	(\$83)	\$3	\$68	\$134	\$201	\$270	\$339	\$409	\$480	\$553	\$626	
D&A	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Stock Option Expense	\$1	\$1	\$1	\$1	\$4	\$10	\$20	\$20	\$20	\$20	\$20	\$20	\$20	\$20	\$20	\$20	\$20	
Capex	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Working Capital	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
FCF (in \$M)	(\$30)	(\$44)	(\$50)	(\$57)	(\$51)	(\$75)	(\$63)	\$23	\$88	\$154	\$221	\$290	\$359	\$429	\$500	\$573	\$646	\$1,425
Discount period	2026	-	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
NPV		(\$30)	(\$41)	(\$42)	(\$44)	(\$36)	(\$49)	(\$37)	\$12	\$44	\$71	\$93	\$112	\$128	\$140	\$150	\$157	\$163 \$359
NPV Sum (in \$M)	\$1,150	\$/share		Downside	Base Case	Upside												
Terminal Growth (%)	-25%	Probability adjusted NPV		\$0	\$230	\$230												
Discount Rate	9%	Assumed PoS		0%	20%	20%												
Share count (in M)	58.3	ML-004		\$0	\$4	\$4												
Probability of Success	20%																	
Probability adjusted NPV	\$230																	
Price Per Share	\$4																	
Peak Sales Calculator		Downside	Base Case	Upside														
Peak sales (\$M)		\$500	\$891	\$1,500														
Multiple		2.0x	2.0x	2.5x														
Valuation		\$1,000	\$1,782	\$3,750														
Assumed PoS		0%	20%	20%														
\$/Share		\$0	\$6	\$13														

Source: Jefferies estimates, Company reports

Market Models

Exhibit 63 - ML-007C-MA Market Model in Schizophrenia

ML-007C-MA Sales Summary (\$M)	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E
US	-	-	-	-	34	116	197	278	357	435	519	605	691	779	869	959	1,051	1,144	1,239	1,335
EU5 (Before royalties)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Sales (\$M)	-	-	-	-	\$34	\$116	\$197	\$278	\$357	\$435	\$519	\$605	\$691	\$779	\$869	\$959	\$1,051	\$1,144	\$1,239	\$1,335
ML-007C-MA Patient Summary	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E
US	-	-	-	-	1,951	6,813	11,749	16,758	21,841	27,000	32,235	37,547	42,937	48,405	53,954	59,582	65,292	71,085	76,960	82,920
EU5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Patients	0	0	0	0	1,951	6,813	11,749	16,758	21,841	27,000	32,235	37,547	42,937	48,405	53,954	59,582	65,292	71,085	76,960	82,920
ML-007C-MA for Schizophrenia - US Market Model	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E
US Population (000s)	347,355	349,960	352,585	355,229	357,894	360,578	363,282	366,007	368,752	371,517	374,304	377,111	379,939	382,789	385,660	388,552	391,467	394,403	397,361	400,341
Annual Growth Rate	0.75%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%
Total SCZ Adult Prevalence (000s)	2,605	2,625	2,644	2,664	2,684	2,704	2,725	2,745	2,766	2,786	2,807	2,828	2,850	2,871	2,892	2,914	2,936	2,958	2,980	3,003
Prevalence	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%
Total Diagnosed SCZ Patients (000s)	2,475	2,493	2,512	2,531	2,550	2,569	2,588	2,608	2,627	2,647	2,667	2,687	2,707	2,727	2,748	2,768	2,789	2,810	2,831	2,852
Diagnosis Rate	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%	95.0%
Patients Requiring Treatment (000s)	2,104	2,119	2,135	2,151	2,167	2,184	2,200	2,217	2,233	2,250	2,267	2,284	2,301	2,318	2,336	2,353	2,371	2,389	2,407	2,425
% Not requiring treatment at all	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%
Tx Discontinuation Rate	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%
Prevalence	1,262,202	1,271,668	1,281,206	1,290,815	1,300,496	1,310,250	1,320,077	1,329,977	1,339,952	1,350,002	1,360,127	1,370,328	1,380,605	1,390,960	1,401,392	1,411,902	1,422,491	1,433,160	1,443,909	1,454,738
Total ML-007C-MA Patients Treated	-	-	-	-	1,951	6,813	11,749	16,758	21,841	27,000	32,235	37,547	42,937	48,405	53,954	59,582	65,292	71,085	76,960	82,920
% penetration	-	-	-	0%	0.2%	0.5%	0.9%	1.3%	1.6%	2.0%	2.4%	2.7%	3.1%	3.5%	3.9%	4.2%	4.6%	5.0%	5.3%	5.7%
Annual Gross Price	\$23,000	-	-	-	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000
% Price Increase	-	-	-	-	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Annual Net Price	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250	\$17,250
Gross to net	-	-	-	-	25%	26%	27%	28%	29%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
US Revenue (\$M)	-	-	-	-	\$34	\$116	\$197	\$278	\$357	\$435	\$519	\$605	\$691	\$779	\$869	\$959	\$1,051	\$1,144	\$1,239	\$1,335

Source: Jefferies estimates, Company reports

Exhibit 64 - ML-007C-MA Market Model in Alzheimer's Disease Psychosis (ADP)

ML-007C-MA Sales Summary (\$M)																							
	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E			
ML-007C-MA (US)	0	0	0	0	0	47	188	327	462	596	737	879	1,023	1,168	1,315	1,463	1,613	1,764	1,916	2,071			
EU5 (Before royalties)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Total Sales (\$M)	-	-	-	-	-	\$47	\$188	\$327	\$462	\$596	\$737	\$879	#####	###	#####	###	#####	###	\$1,463	\$1,613	\$1,764	\$1,916	\$2,071
ML-007C-MA Patient Summary																							
	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E			
US	0	0	0	0	0	2,789	11,212	19,719	28,311	36,989	45,752	54,603	63,540	72,566	81,680	90,884	100,177	109,561	119,037	128,605			
EU5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Total Patients	0	0	0	0	0	2,789	11,212	19,719	28,311	36,989	45,752	54,603	63,540	72,566	81,680	90,884	100,177	109,561	119,037	128,605			
ML-007C-MA for ADP - US Market Model																							
	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E			
US Population (000s)	345,122	346,848	348,582	350,325	352,076	353,837	355,606	357,384	359,171	360,967	362,772	364,585	366,408	368,240	370,082	371,932	373,792	375,661	377,539	379,427			
Annual Growth Rate	0.50%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%			
Alzheimer's Disease Population (000s)	6,801	6,835	6,869	6,904	6,938	6,973	7,008	7,043	7,078	7,113	7,149	7,185	7,220	7,257	7,293	7,329	7,366	7,403	7,440	7,477			
Prevalence	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%			
(1) Alzheimer's with psychosis	2,720,401	2,734,003	2,747,673	2,761,412	2,775,219	2,789,095	2,803,040	2,817,056	2,831,141	2,845,297	2,859,523	2,873,821	2,888,190	2,902,631	2,917,144	2,931,730	2,946,388	2,961,120	2,975,926	2,990,805			
Prevalence	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%			
(2) Alzheimer's without psychosis	4,080,602	4,101,005	4,121,510	4,142,118	4,162,828	4,183,642	4,204,561	4,225,583	4,246,711	4,267,945	4,289,285	4,310,731	4,332,285	4,353,946	4,375,716	4,397,594	4,419,582	4,441,680	4,463,889	4,486,208			
Prevalence	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%	60.0%			
Total ML-007C-MA Patients Treated	-	-	-	-	-	2,789	11,212	19,719	28,311	36,989	45,752	54,603	63,540	72,566	81,680	90,884	100,177	109,561	119,037	128,605			
Alzheimer's with psychosis	-	-	-	-	-	2,789	11,212	19,719	28,311	36,989	45,752	54,603	63,540	72,566	81,680	90,884	100,177	109,561	119,037	128,605			
% penetration (overall subgroup)	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.4%	0.7%	1.0%	1.3%	1.6%	1.9%	2.2%	2.5%	2.8%	3.1%	3.4%	3.7%	4.0%	4.3%			
Alzheimer's without psychosis	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
% penetration (overall subgroup)	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%			
Annual Gross Price	\$23,000	-	-	-	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000	\$23,000			
% Price increase	-	-	-	-	-	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%			
Annual Net Price	-	-	-	-	\$17,250	\$17,020	\$16,790	\$16,560	\$16,330	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100	\$16,100			
Gross to net	-	-	-	-	-	25%	26%	27%	28%	29%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%			
US Revenue (\$M)																							
Revenue - Alzheimer's with psychosis	-	-	-	-	-	\$47	\$188	\$327	\$462	\$596	\$737	\$879	\$1,023	\$1,168	\$1,315	\$1,463	\$1,613	\$1,764	\$1,916	\$2,071			
Revenue - Alzheimer's without psychosis	-	-	-	-	-	-	-	-	-	-	-	-	\$879	\$1,023	\$1,168	\$1,315	\$1,463	\$1,613	\$1,764	\$1,916			

Source: Jefferies estimates, Company reports

Exhibit 65 - ML-004 Market Model in Autism Spectrum Disorder (ASD)

ML-004 Sales Summary (\$M)	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E
US	-	-	-	-	-	-	39	119	200	282	365	450	535	622	711	800	891
EUS (Before royalties)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Sales (\$M)	-	-	-	-	-	-	\$39	\$119	\$200	\$282	\$365	\$450	\$535	\$622	\$711	\$800	\$891
ML-004 Patient Summary	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E
US	-	-	-	-	-	-	2,763	8,350	14,022	19,778	25,619	31,547	37,563	43,667	49,860	56,144	62,519
EUS	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Patients	0	0	0	0	0	0	2,763	8,350	14,022	19,778	25,619	31,547	37,563	43,667	49,860	56,144	62,519
ML-004 for Autism Spectrum Disorder - US Market Model	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E
US Population (000s)	347,355	349,960	352,585	355,229	357,894	360,578	363,282	366,007	368,752	371,517	374,304	377,111	379,939	382,789	385,660	388,552	391,467
Annual Growth Rate	0.75%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%
Total ASD Prevalence (000s)	8,416	8,479	8,543	8,607	8,672	8,737	8,802	8,868	8,935	9,002	9,069	9,137	9,206	9,275	9,344	9,414	9,485
Prevalence	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%	2.4%
Total Diagnosed ASD Patients (000s)	7,575	7,631	7,689	7,746	7,804	7,863	7,922	7,981	8,041	8,101	8,162	8,223	8,285	8,347	8,410	8,473	8,536
Diagnosis Rate	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%	90.0%
Moderate to Severe (000s)	3,522	3,549	3,575	3,602	3,629	3,656	3,684	3,711	3,739	3,767	3,795	3,824	3,853	3,881	3,911	3,940	3,969
% Not requiring treatment at all	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%
Tx Discontinuation Rate	1,761,091	1,774,299	1,787,606	1,801,013	1,814,521	1,828,130	1,841,841	1,855,654	1,869,572	1,883,594	1,897,720	1,911,953	1,926,293	1,940,740	1,955,296	1,969,960	1,984,735
Prevalence	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%
Total ML-004 Patients Treated	-	-	-	-	-	-	2,763	8,350	14,022	19,778	25,619	31,547	37,563	43,667	49,860	56,144	62,519
% penetration	0%	0%	0%	0%	0%	0%	0.2%	0.5%	0.8%	1.1%	1.4%	1.7%	2.0%	2.3%	2.6%	2.9%	3.2%
Annual Gross Price	\$19,000	-	-	-	-	-	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000	\$19,000
% Price increase	-	-	-	-	-	-	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Annual Net Price	-	-	-	-	-	-	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250	\$14,250
Gross to net	25%	-	-	-	-	-	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%
US Revenue (\$M)	-	-	-	-	-	-	\$39	\$119	\$200	\$282	\$365	\$450	\$535	\$622	\$711	\$800	\$891

Source: Jefferies estimates, Company reports

Income Statement

Exhibit 66 - MPLT Income Statement

Fiscal Year (Ending in December)	FY23	1Q24	2Q24	3Q24	4Q24	FY24	1Q25	2Q25	3Q25E	4Q25E	FY25E	1Q26E	2Q26E	3Q26E	4Q26E	FY26E	FY27E	FY28E	FY29E	FY30E
(\$ figures in MM, except per share)	Mar-24	Jun-24	Sep-24	Dec-24		Mar-25	Jun-25	Sep-25	Dec-25		Mar-26	Jun-26	Sep-26	Dec-26						
GAAP Income Statement																				
Revenue																				
Net Product Revenue	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	\$35.0
ML-007C-MA - Total	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	35.0
Schizophrenia	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	35.0
Alzheimer's Disease Psychosis	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ML-004 - Autism Spectrum Disorder	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ML-021 - Parkinson's Disease	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ML-009 - Hyperactivity/Impulsivity	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other Revenue	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Licensing revenue	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other revenue	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Revenues	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	\$35.0
Cost of Sales	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.8
% of Product Revenue	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	5.0%
Operating Expenses	\$57.3	\$19.3	\$20.0	\$21.5	\$22.2	\$82.9	\$23.5	\$30.7	\$34.2	\$38.8	\$127.2	\$43.4	\$48.0	\$53.6	\$60.2	\$205.2	\$302.0	\$341.2	\$385.8	\$436.6
Research and Development (R&D)	49.7	15.3	15.6	18.0	19.5	68.5	19.8	26.8	30.0	34.0	110.6	38.0	42.0	47.0	53.0	180.0	271.9	305.2	342.7	385.0
Sales, General, and Administrative (SG&A)	7.6	3.9	4.3	3.5	2.6	14.4	3.8	3.8	4.2	4.8	16.6	5.4	6.0	6.6	7.2	25.2	30.1	36.0	43.1	51.5
Other Expenses	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Expenses	\$57.3	\$19.3	\$20.0	\$21.5	\$22.2	\$82.9	\$23.5	\$30.7	\$34.2	\$38.8	\$127.2	\$43.4	\$48.0	\$53.6	\$60.2	\$205.2	\$302.0	\$341.2	\$385.8	\$438.3
Operating Income (EBIT)	(\$57.3)	(\$19.3)	(\$20.0)	(\$21.5)	(\$22.2)	(\$82.9)	(\$23.5)	(\$30.7)	(\$34.2)	(\$38.8)	(\$127.2)	(\$43.4)	(\$48.0)	(\$53.6)	(\$60.2)	(\$205.2)	(\$302.0)	(\$341.2)	(\$385.8)	(\$403.3)
Total Other Income	1.6	0.4	1.6	1.4	2.0	5.4	1.2	0.8	1.0	1.7	4.7	4.2	5.5	5.0	5.6	20.3	23.2	24.5	15.8	22.5
Interest income	1.1	0.9	1.6	0.9	1.1	4.5	0.8	0.7	1.0	1.7	4.2	4.2	5.5	5.0	5.6	20.3	23.2	24.5	15.8	22.5
Interest expense	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other income, net	0.5	(0.5)	(0.0)	0.5	0.9	0.9	0.4	0.1	-	-	0.5	-	-	-	-	-	-	-	-	-
Pre-Tax Income	(55.7)	(18.9)	(18.4)	(20.1)	(20.1)	(77.6)	(22.3)	(29.8)	(33.2)	(37.1)	(122.5)	(39.2)	(42.5)	(48.6)	(54.6)	(184.9)	(278.8)	(316.7)	(370.0)	(380.8)
Provision for taxes (benefit)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tax rate	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Net Income	(55.7)	(18.9)	(18.4)	(20.1)	(20.1)	(77.6)	(22.3)	(29.8)	(33.2)	(37.1)	(122.5)	(39.2)	(42.5)	(48.6)	(54.6)	(184.9)	(278.8)	(316.7)	(370.0)	(380.8)
Other	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Net Income attributable to common stockholders	(55.7)	(18.9)	(18.4)	(20.1)	(20.1)	(77.6)	(22.3)	(29.8)	(33.2)	(37.1)	(122.5)	(39.2)	(42.5)	(48.6)	(54.6)	(184.9)	(278.8)	(316.7)	(370.0)	(380.8)
EPS (Basic)	(\$5.03)	(\$1.68)	(\$1.54)	(\$1.68)	(\$1.63)	(\$6.52)	(\$1.75)	(\$2.33)	(\$1.31)	(\$0.87)	(\$5.23)	(\$0.92)	(\$0.99)	(\$1.02)	(\$1.14)	(\$4.08)	(\$5.27)	(\$5.98)	(\$6.37)	(\$6.55)
EPS (Diluted, as reported)	(\$5.03)	(\$1.68)	(\$1.54)	(\$1.68)	(\$1.63)	(\$6.52)	(\$1.75)	(\$2.33)	(\$1.31)	(\$0.87)	(\$5.23)	(\$0.92)	(\$0.99)	(\$1.02)	(\$1.14)	(\$4.08)	(\$5.27)	(\$5.98)	(\$6.37)	(\$6.55)
Shares Outstanding - Basic	11.1	11.2	12.0	12.0	12.4	11.9	12.8	12.8	25.3	42.7	23.4	42.8	42.8	47.8	47.9	45.3	52.9	53.0	58.0	58.1
Shares Outstanding - Diluted (as reported)	11.1	11.2	12.0	12.0	12.4	11.9	12.8	12.8	25.3	42.7	23.4	42.8	42.8	47.8	47.9	45.3	52.9	53.0	58.0	58.1
Shares Outstanding - Fully Diluted + Preferred/Warrants	11.1	13.2	28.3	28.4	28.7	24.7	29.2	29.2	41.6	49.6	37.4	49.7	49.7	54.7	54.8	52.2	59.8	59.9	64.9	65.0
Shares Outstanding - Fully Diluted + Options/Warrants, etc	11.1	14.4	29.3	29.4	29.7	25.7	30.2	30.1	42.6	53.5	39.1	53.6	53.6	58.6	58.7	56.1	63.7	63.8	68.8	68.9

Source: Jefferies estimates, Company reports

Balance Sheet

Exhibit 67 - MPLT Balance Sheet

Fiscal Year (Ending in December) (\$ figures in MM, except per share)	FY23	1Q24 Mar-24	2Q24 Jun-24	3Q24 Sep-24	4Q24 Dec-24	FY24	1Q25 Mar-25	2Q25 Jun-25	3Q25E Sep-25	4Q25E Dec-25	FY25E	1Q26E Mar-26	2Q26E Jun-26	3Q26E Sep-26	4Q26E Dec-26	FY26E	FY27E	FY28E	FY29E	FY30E
Current assets:																				
Cash and cash equivalents	79.8	179.8	38.7	18.7	38.3	38.3	50.2	33.5	189.1	431.8	431.8	393.7	352.5	499.4	446.5	446.5	539.6	232.3	358.0	9.6
Marketable securities	-	-	121.5	121.5	70.5	70.5	42.1	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0	7.0
Prepaid expenses & other current assets	3.8	3.8	3.8	3.8	5.8	5.8	6.0	12.7	12.7	12.7	12.7	12.7	12.7	12.7	12.7	12.7	12.7	12.7	12.7	12.7
Restricted cash	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Accounts receivable	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Inventory	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other current assets	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total current assets	83.6	183.6	164.1	144.0	114.6	114.6	98.3	73.2	228.9	471.5	471.5	433.4	392.2	539.1	486.2	486.2	579.3	272.0	397.7	29.4
Marketable securities - long-term	-	0.0	0.3	0.3	11.4	11.4	2.5	-	-	-	-	-	-	-	-	-	-	-	-	-
Restricted cash - long-term	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Other intangible assets (Right-of-use)	5.6	5.6	5.6	5.6	6.4	6.4	6.1	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9
Property and equipment, net	1.0	1.0	1.0	1.0	1.2	1.2	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Other long-term assets	1.0	1.0	1.0	1.0	3.2	3.2	3.4	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8
Total assets	91.4	191.4	172.2	152.1	136.9	136.9	111.6	84.1	239.8	482.4	482.4	444.4	403.2	550.1	497.1	497.1	590.2	282.9	408.7	40.3
Current liabilities:																				
Accounts payable	3.3	3.3	3.3	3.3	1.9	1.9	0.9	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Accrued expenses	6.8	6.8	6.8	6.8	10.0	10.0	8.1	9.6	9.6	9.6	9.6	9.6	9.6	9.6	9.6	9.6	9.6	9.6	9.6	9.6
Lease liability - current	0.5	0.5	0.5	0.5	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Deferred grant earnings	4.2	4.2	4.2	4.2	3.2	3.2	3.2	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Current portion of long-term debt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total current liabilities	14.7	14.7	14.7	14.7	15.9	15.9	13.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Deferred revenue, net of current portion	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Long-term debt	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Preferred stock warrant liability	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Deferred tax liabilities	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other long-term liabilities	5.4	5.4	5.4	5.4	5.8	5.8	5.6	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4
Total liabilities	20.1	20.1	20.1	20.1	21.7	21.7	18.6	20.4	20.4	20.4	20.4	20.4	20.4	20.4	20.4	20.4	20.4	20.4	20.4	20.4
Stockholders' Equity:																				
Convertible preferred stock	188.9	188.9	188.9	188.9	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8	308.8
Common stock	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Additional paid-in capital	4.2	4.2	4.2	4.2	5.6	5.6	5.8	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4
Accumulated deficit and other	(121.8)	(21.8)	(41.0)	(61.1)	(199.2)	(199.2)	(221.6)	(251.5)	(95.8)	146.8	146.8	108.7	67.5	214.4	161.5	161.5	254.6	(52.7)	73.0	(295.3)
Treasury stock	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total stockholders' equity	71.3	171.3	152.1	132.0	115.2	115.2	93.0	63.8	219.4	462.1	462.1	424.0	382.8	529.7	476.7	476.7	569.9	262.6	388.3	19.9
Total liabilities and stockholders' equity	91.4	191.4	172.2	152.1	136.9	136.9	111.6	84.1	239.8	482.4	482.4	444.4	403.2	550.1	497.1	497.1	590.2	282.9	408.7	40.3

Source: Jefferies estimates, Company reports

Cash Flow Statement

Exhibit 68 - MPLT Cash Flow Statement

Fiscal Year (Ending in December)	FY23	1Q24	2Q24	3Q24	4Q24	FY24	1Q25	2Q25	3Q25E	4Q25E	FY25E	1Q26E	2Q26E	3Q26E	4Q26E	FY26E	FY27E	FY28E	FY29E	FY30E
(\$ figures in MM, except per share)	Mar-24	Jun-24	Sep-24	Dec-24	Mar-25	Jun-25	Sep-25	Dec-25	Mar-26	Jun-26	Sep-26	Dec-26	Mar-26	Jun-26	Sep-26	Dec-26	Mar-26	Jun-26	Sep-26	Dec-26
Cash flows from operating activities:																				
Net income	(55.7)	(18.9)	(18.4)	(20.1)	(20.1)	(77.6)	(22.3)	(29.8)	(33.2)	(37.1)	(122.5)	(39.2)	(42.5)	(48.6)	(54.6)	(184.9)	(278.8)	(316.7)	(370.0)	(380.8)
Adjustments to reconcile net income to net cash provided -																				
Depreciation and amortization	0.6	0.1	(0.2)	(0.2)	(0.2)	(1.0)	(0.1)	0.2	0.2	0.2	0.5	0.2	0.2	0.2	0.2	0.8	0.8	0.8	0.8	0.8
Non-cash lease expense	0.5	0.1	0.2	-	-	0.7	0.2	0.2	0.2	0.2	0.9	0.2	0.2	0.2	0.2	0.9	0.9	0.9	0.9	0.9
Loss (gain) from equity method investment	(1.0)	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Stock-based compensation	1.0	0.3	0.2	0.3	0.3	1.1	0.2	0.1	0.4	0.5	1.3	0.7	0.9	1.1	1.2	3.9	6.5	7.7	9.1	10.8
Loss from revaluation of preferred stock purchase rig	0.5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Loss on disposal of property and equipment	-	-	-	-	-	0.1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Non-cash interest expense (income)	-	-	-	-	-	-	-	0.5	-	-	0.5	-	-	-	-	-	-	-	-	-
Change in deferred tax, other	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Changes in operating assets and liabilities																				
Increase in inventory	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
(Increase) decrease in prepaid expenses	0.5	0.0	0.2	-	(2.0)	(2.3)	(0.2)	(6.7)	-	-	(6.9)	-	-	-	-	-	-	-	-	-
(Increase) decrease in receivables	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
(Increase) decrease in other assets	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Deferred revenue	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Increase (decrease) in accounts payable	(0.4)	0.8	(2.0)	-	(1.3)	(1.3)	(1.1)	1.2	-	-	0.2	-	-	-	-	-	-	-	-	-
Accrued expenses	3.9	(2.6)	2.4	-	3.2	2.2	(2.1)	1.3	-	-	(0.8)	-	-	-	-	-	-	-	-	-
Operating lease liability	(0.3)	(0.1)	(0.2)	-	0.3	(0.7)	(0.2)	(0.2)	-	-	(0.4)	-	-	-	-	-	-	-	-	-
Increase (decrease) in other liabilities	(1.7)	(0.2)	(0.2)	-	(1.0)	(1.0)	-	(0.8)	-	-	(0.8)	-	-	-	-	-	-	-	-	-
Net cash provided by operating activities	(52.0)	(19.9)	(17.5)	(20.1)	(22.8)	(78.8)	(25.6)	(34.0)	(32.4)	(36.2)	(128.1)	(38.1)	(41.2)	(47.1)	(52.9)	(179.3)	(270.6)	(307.3)	(359.3)	(368.4)
Cash flows from investing activities:																				
Maturities/sale of marketable securities	-	-	-	-	-	45.0	37.5	17.5	-	-	55.0	-	-	-	-	-	-	-	-	20.0
Purchases of marketable securities	-	-	(121.5)	-	-	(125.0)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Capital expenditures	(0.5)	(0.0)	(0.3)	-	-	(0.8)	-	(0.1)	-	-	(0.1)	-	-	-	-	-	-	-	-	-
Purchases of other securities	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Net cash used in investing activities	(0.5)	(0.0)	(121.8)	-	-	(80.8)	37.5	17.4	-	-	54.9	-	-	-	-	-	-	-	-	20.0
Cash flows from financing activities:																				
Proceeds from issuance of common stock/series D, n	104.5	120.0	-	-	-	120.0	-	-	188.1	278.8	466.9	-	-	194.0	-	194.0	363.8	-	485.0	-
Proceeds from the exercise of stock options	0.0	0.0	0.3	-	-	0.3	0.0	0.0	-	-	0.0	-	-	-	-	-	-	-	-	-
Proceeds from issuance of convertible preferred stoc	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Proceeds from debt borrowings	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Repayment of debt borrowings	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other financing activities	-	-	(2.0)	-	-	(2.2)	(0.1)	(0.2)	-	-	(0.2)	-	-	-	-	-	-	-	-	-
Net cash provided by (used in) financing activities	104.6	120.0	(1.7)	-	-	118.1	(0.0)	(0.2)	188.1	278.8	466.7	-	-	194.0	-	194.0	363.8	-	485.0	-
Effect of foreign currency exchange rate changes on cash :	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Net increase in cash and cash equivalents	52.1	100.0	(141.1)	(20.1)	(22.8)	(41.5)	11.9	(16.7)	155.7	242.6	393.5	(38.1)	(41.2)	146.9	(52.9)	14.7	93.1	(307.3)	125.7	(348.4)
Cash and cash equivalents at beginning of period	27.9	80.0	180.0	39.0	18.9	80.0	38.5	50.4	33.7	189.3	38.5	432.0	393.9	352.7	499.6	432.0	446.7	539.8	232.5	358.2
Cash and cash equivalents at end of period	80.0	180.0	39.0	18.9	(3.9)	38.5	50.4	33.7	189.3	432.0	432.0	393.9	352.7	499.6	446.7	446.7	539.8	232.5	358.2	9.8

Source: Jefferies estimates, Company reports

Company Description

Maplight Therapeutics

MPLT is a clinical-stage biopharmaceutical company focused on addressing central nervous system disorders, including schizophrenia, alzheimer's disease psychosis (ADP), autism spectrum disorder (ASD), and Parkinson's disease.

Company Valuation/Risks

Maplight Therapeutics

Valuation is a probability of success (PoS)-weighted SOTP-based DCF analysis of pipeline products. Key risks include regulatory, clinical development, and commercialization.

Bristol-Myers Squibb Co

Valuation: Our DCF-based PT is supported by PE and PEG-relative valuations. Risks: R&D, LOEs, pricing, competition, legal/regulatory, M&A.

Neurocrine Biosciences

Our PT is DCF-based. Risks include commercial, regulatory, and clinical.

Analyst Certification:

I, Andrew Tsai, certify that all of the views expressed in this research report accurately reflect my personal views about the subject security(ies) and subject company(ies). I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed in this research report.

I, John Cox, Ph.D., certify that all of the views expressed in this research report accurately reflect my personal views about the subject security(ies) and subject company(ies). I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed in this research report.

I, Brian Balchin, ACA, certify that all of the views expressed in this research report accurately reflect my personal views about the subject security(ies) and subject company(ies). I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed in this research report.

I, Matthew Barcus, Ph.D., certify that all of the views expressed in this research report accurately reflect my personal views about the subject security(ies) and subject company(ies). I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed in this research report.

As is the case with all Jefferies employees, the analyst(s) responsible for the coverage of the financial instruments discussed in this report receives compensation based in part on the overall performance of the firm, including investment banking income. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Aside from certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgement.

Investment Recommendation Record

(Article 3(1)e and Article 7 of MAR)

Recommendation Published	November 20, 2025 18:31 P.M.
Recommendation Distributed	November 21, 2025 0:05 A.M.

Company Specific Disclosures

Within the past twelve months, Maplight Therapeutics has been a client of Jefferies Financial Group Inc., its affiliates or subsidiaries and investment banking services are being or have been provided.

Jefferies Financial Group Inc., its affiliates or subsidiaries expect to receive or intend to seek compensation for investment banking services from Maplight Therapeutics within the next three months.

Within the past 12 months, Jefferies Financial Group Inc., its affiliates or subsidiaries has received compensation from investment banking services from Maplight Therapeutics.

Jefferies Financial Group Inc., its affiliates or subsidiaries has acted as a manager or co-manager in the underwriting or placement of securities for Maplight Therapeutics or one of its affiliates within the past twelve months.

Within the past twelve months, Jefferies Financial Group Inc. and/or its affiliates received compensation for products and services other than investment banking services from non-investment banking, securities related compensation for client services it provided to Neurocrine Biosciences.

Explanation of Jefferies Ratings

Buy - Describes securities that we expect to provide a total return (price appreciation plus yield) of 15% or more within a 12-month period.

Hold - Describes securities that we expect to provide a total return (price appreciation plus yield) of plus 15% or minus 10% within a 12-month period.

Underperform - Describes securities that we expect to provide a total return (price appreciation plus yield) of minus 10% or less within a 12-month period.

The expected total return (price appreciation plus yield) for Buy rated securities with an average security price consistently below \$10 is 20% or more within a 12-month period as these companies are typically more volatile than the overall stock market. For Hold rated securities with an average security price consistently below \$10, the expected total return (price appreciation plus yield) is plus or minus 20% within a 12-month period. For Underperform rated securities with an average security price consistently below \$10, the expected total return (price appreciation plus yield) is minus 20% or less within a 12-month period.

NR - The investment rating and price target have been temporarily suspended. Such suspensions are in compliance with applicable regulations and/or Jefferies policies.

CS - Coverage Suspended. Jefferies has suspended coverage of this company.

NC - Not covered. Jefferies does not cover this company.

Restricted - Describes issuers where, in conjunction with Jefferies engagement in certain transactions, company policy or applicable securities regulations prohibit certain types of communications, including investment recommendations.

Monitor - Describes securities whose company fundamentals and financials are being monitored, and for which no financial projections or opinions on the investment merits of the company are provided.

Valuation Methodology

Jefferies' methodology for assigning ratings may include the following: market capitalization, maturity, growth/value, volatility and expected total return over the next 12 months. The price targets are based on several methodologies, which may include, but are not restricted to, analyses of market risk, growth rate, revenue stream, discounted cash flow (DCF), EBITDA, EPS, cash flow (CF), free cash flow (FCF), EV/EBITDA, P/E, PE/growth, P/CF, P/FCF, premium (discount)/average group EV/EBITDA, premium (discount)/average group P/E, sum of the parts, net asset value, dividend returns, and return on equity (ROE) over the next 12 months.

Jefferies Franchise Picks

Jefferies Franchise Picks include stock selections from among the best stock ideas from our equity analysts over a 12 month period. Stock selection is based on fundamental analysis and may take into account other factors such as analyst conviction, differentiated analysis, a favorable risk/reward ratio and investment themes that Jefferies analysts are recommending. Jefferies Franchise Picks will include only Buy rated stocks and the number can vary depending on analyst recommendations for inclusion. Stocks will be added as new opportunities arise and removed when the reason for inclusion changes, the stock has met its desired return, if it is no longer rated Buy and/or if it triggers a stop loss. Stocks having 120 day volatility in the bottom quartile of S&P stocks will continue to have a 15% stop loss, and the remainder will have a 20% stop. Franchise Picks are not intended to represent a recommended portfolio of stocks and is not sector based, but we may note where we believe a Pick falls within an investment style such as growth or value.

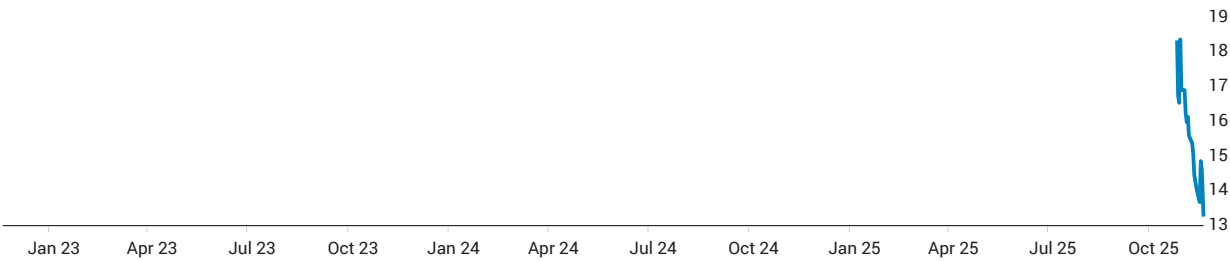
Risks which may impede the achievement of our Price Target

This report was prepared for general circulation and does not provide investment recommendations specific to individual investors. As such, the financial instruments discussed in this report may not be suitable for all investors and investors must make their own investment decisions based upon their specific investment objectives and financial situation utilizing their own financial advisors as they deem necessary. Past performance of the financial instruments recommended in this report should not be taken as an indication or guarantee of future results. The price, value of, and income from, any of the financial instruments mentioned in this report can rise as well as fall and may be affected by changes in economic, financial and political factors. If a financial instrument is denominated in a currency other than the investor's home currency, a change in exchange rates may adversely affect the price of, value of, or income derived from the financial instrument described in this report. To the extent prices are shown in non-US currency, please note that our local currency price targets are based on a currency conversion using an exchange rate as of the prior trading day (unless otherwise noted). Should there be fluctuations in the exchange rate after this date, that will affect the non-US target prices and should no longer be relied upon. In addition, investors in securities such as ADRs, whose values are affected by the currency of the underlying security, effectively assume currency risk.

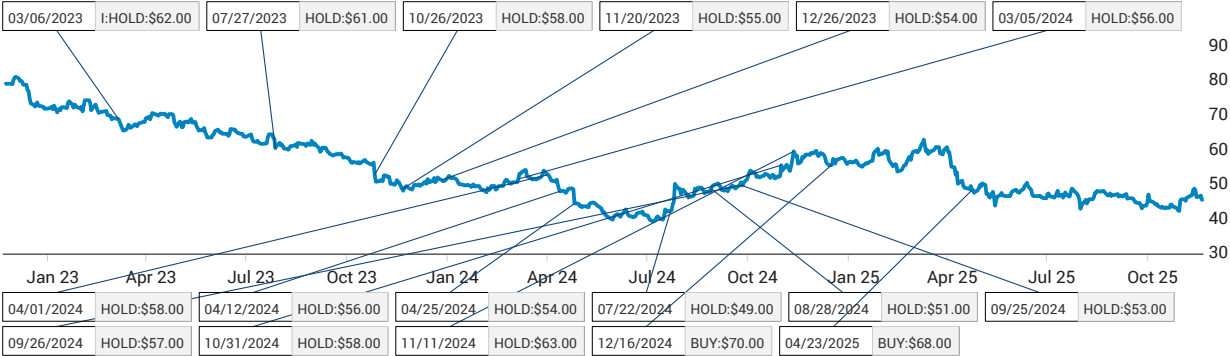
Other Companies Mentioned in This Report

- Bristol-Myers Squibb Co (BMY: \$45.94, BUY)
- Neurocrine Biosciences (NBIX: \$138.98, BUY)

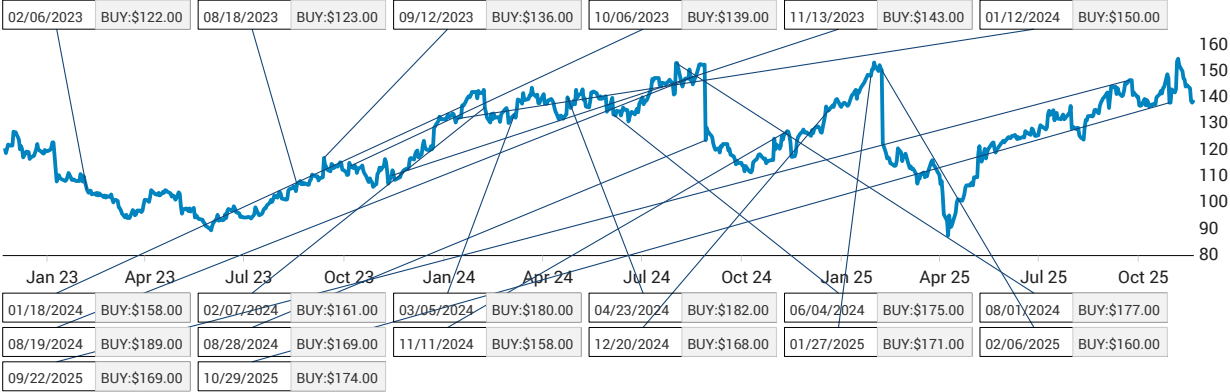
Rating and Price Target History for: Maplight Therapeutics (MPLT) as of 11-20-2025



Rating and Price Target History for: Bristol-Myers Squibb Co (BMY) as of 11-20-2025



Rating and Price Target History for: Neurocrine Biosciences (NBIX) as of 11-20-2025



Notes: Each box in the Rating and Price Target History chart above represents actions over the past three years in which an analyst initiated on a company, made a change to a rating or price target of a company or discontinued coverage of a company.

Legend:

I: Initiating Coverage

D: Dropped Coverage

B: Buy

H: Hold

UP: Underperform

Distribution of Ratings			IB Serv./Past12 Mos.		JIL Mkt Serv./Past12 Mos.	
	Count	Percent	Count	Percent	Count	Percent
BUY	2152	61.07%	390	18.12%	125	5.81%
HOLD	1216	34.51%	108	8.88%	18	1.48%
UNDERPERFORM	156	4.43%	2	1.28%	2	1.28%

Other important disclosures

Other Important Disclosures

Jefferies does business and seeks to do business with companies covered in its research reports, and expects to receive or intends to seek compensation for investment banking services among other activities from such companies. As a result, investors should be aware that Jefferies may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Jefferies Equity Research refers to research reports produced by analysts employed by one of the following Jefferies Financial Group Inc. ("Jefferies") companies:

United States: Jefferies LLC which is an SEC registered broker-dealer and a member of FINRA (and distributed by Jefferies Research Services, LLC, an SEC registered Investment Adviser, to clients paying separately for such research).

Canada: Jefferies Securities Inc., which is an investment dealer registered in each of the thirteen Canadian jurisdictions and a dealer member of the Canadian Investment Regulatory Organization, including research reports produced jointly by Jefferies Securities Inc. and another Jefferies entity (and distributed by Jefferies Securities Inc.).

Where Jefferies Securities Inc. distributes research reports produced by Jefferies LLC, Jefferies International Limited, Jefferies (Japan) Limited, Tokyo Branch or Jefferies India Private Limited, you are advised that each of Jefferies LLC, Jefferies International Limited, Jefferies (Japan) Limited, Tokyo Branch and Jefferies India Private Limited operates as a dealer in your jurisdiction under an exemption from the dealer registration requirements contained in National Instrument 31-103 Registration Requirements, Exemptions and Ongoing Registrant Obligations and, as such, each of Jefferies LLC, Jefferies International Limited, Jefferies (Japan) Limited, Tokyo Branch and Jefferies India Private Limited is not required to be and is not a registered dealer or adviser in your jurisdiction. You are advised that where Jefferies LLC or Jefferies International Limited prepared this research report, it was not prepared in accordance with Canadian disclosure requirements relating to research reports in Canada.

United Kingdom: Jefferies International Limited, which is authorized and regulated by the Financial Conduct Authority; registered in England and Wales No. 1978621; registered office: 100 Bishopsgate, London EC2N 4JL; telephone +44 (0)20 7029 8000; facsimile +44 (0)20 7029 8010.

Germany: Jefferies GmbH, which is authorized and regulated by the Bundesanstalt fuer Finanzdienstleistungsaufsicht, BaFin-ID: 10150151; registered office: Bockenheimer Landstr. 24, 60323 Frankfurt a.M., Germany; telephone: +49 (0) 69 719 1870

Hong Kong: Jefferies Hong Kong Limited, which is licensed by the Securities and Futures Commission of Hong Kong with CE number ATS546; located at Level 26, Two International Finance Center, 8 Finance Street, Central, Hong Kong; telephone: +852 3743 8000.

Singapore: Jefferies Singapore Limited, which is licensed by the Monetary Authority of Singapore; located at 80 Raffles Place #15-20, UOB Plaza 2, Singapore 048624, telephone: +65 6551 3950.

Japan: Jefferies (Japan) Limited, Tokyo Branch, which is a securities company registered by the Financial Services Agency of Japan and is a member of the Japan Securities Dealers Association; located at Tokyo Midtown Hibiya 30F Hibiya Mitsui Tower, 1-1-2 Yurakucho, Chiyoda-ku, Tokyo 100-0006; telephone +813 5251 6100; facsimile +813 5251 6101.

India: Jefferies India Private Limited (CIN - U74140MH2007PTC200509), licensed by the Securities and Exchange Board of India for: Stock Broker (NSE & BSE) INZ000243033, Research Analyst INH000000701 and Merchant Banker INM000011443, located at Level 16, Express Towers, Nariman Point, Mumbai 400 021, India; Tel +91 22 4356 6000. Compliance Officer name: Sanjay Pai, Tel No: +91 22 42246150, Email: spai@jefferies.com, Grievance officer name: Sanjay Pai, Tel no. +91 22 42246150, Email: compliance_india@jefferies.com. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors.

Australia: Jefferies (Australia) Pty Limited (ACN 623 059 898), which holds an Australian financial services license (AFSL 504712) and is located at Level 20, 60 Martin Place, Sydney NSW 2000; telephone +61 2 9364 2800.

Dubai: Jefferies International Limited, Dubai branch, which is licensed by the Dubai Financial Services Authority (DFSA Reference Number F007325); registered office Unit L31-06, L31-07, Level 31, ICD Brookfield Pace, DIFC, PO Box 121208, Dubai, UAE.

This report was prepared by personnel who are associated with Jefferies (Jefferies Securities Inc., Jefferies International Limited, Jefferies GmbH, Jefferies Hong Kong Limited, Jefferies Singapore Limited, Jefferies (Japan) Limited, Tokyo Branch, Jefferies India Private Limited), and Jefferies (Australia) Pty Ltd; or by personnel who are associated with both Jefferies LLC and Jefferies Research Services LLC ("JRS"). Jefferies LLC is a US registered broker-dealer and is affiliated with JRS, which is a US registered investment adviser. JRS does not create tailored or personalized research and all research provided by JRS is impersonal. If you are paying separately for this research, it is being provided to you by JRS. Otherwise, it is being provided by Jefferies LLC. Jefferies LLC, JRS, and their affiliates are collectively referred to below as "Jefferies". Jefferies may seek to do business with companies covered in this research report. As a result, investors should be aware that Jefferies may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only one of many factors in making their investment decisions. Specific conflict of interest and other disclosures that are required by FINRA, the Canadian Investment Regulatory Organization and other rules are set forth in this disclosure section.

If you are receiving this report from a non-US Jefferies entity, please note the following: Unless prohibited by the provisions of Regulation S of the U.S. Securities Act of 1933, as amended, this material is distributed in the United States by Jefferies LLC, which accepts responsibility for its contents in accordance with the provisions of Rule 15a-6 under the US Securities Exchange Act of 1934, as amended. Transactions by or on behalf of any US person may only be effected through Jefferies LLC. In the United Kingdom and European Economic Area this report is issued and/or approved for distribution by Jefferies International Limited ("JIL") and/or Jefferies GmbH and is intended for use only by persons who have, or have been assessed as having, suitable professional experience and expertise, or by persons to whom it can be otherwise lawfully distributed. Jefferies LLC, JIL, Jefferies GmbH and their affiliates, may make a market or provide liquidity in the financial instruments referred to in this report; and where they do make a market, such activity is disclosed specifically in this report under "company specific disclosures".

For Canadian investors, this material is intended for use only by professional or institutional investors. None of the investments or investment services mentioned or described herein is available to other persons or to anyone in Canada who is not a "permitted client" as defined by National Instrument 31-103 Registration Requirements, Exemptions and Ongoing Registrant Obligations, as applicable. This research report is a general discussion of the merits and risks of a security or securities only, and is not in any way meant to be tailored to the needs and circumstances of any recipient. The information contained herein is not, and under no circumstances is to be construed as, an offer to sell securities described herein, or solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators, if applicable, and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. The information contained herein is under no circumstances to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed judgment upon this research report, the information contained herein or the merits of the securities described herein, and any representation to the contrary is an offence.

In Singapore, Jefferies Singapore Limited ("JSL") is regulated by the Monetary Authority of Singapore. For investors in the Republic of Singapore, where this material is prepared and issued by a Jefferies affiliate outside of Singapore, it is distributed by JSL pursuant to Regulation 32C of the Financial Advisers Regulations. The material contained in this document is intended solely for accredited, expert or institutional investors, as defined under the Securities and Futures Act 2001 (Singapore). If there are any matters arising from, or in connection with this material, please contact JSL, located at 80 Raffles Place #15-20, UOB Plaza 2, Singapore 048624, telephone: +65 6551 3950. In Dubai, this material is issued and distributed by Jefferies International Limited, Dubai branch, and is intended solely for Professional Clients and should not be distributed to, or relied upon by, Retail Clients (as defined by DFSA). A distribution of ratings in percentage terms in each sector covered is available upon request from your sales representative. In Japan, this material is issued and distributed by Jefferies (Japan) Limited to institutional investors only. In Hong Kong, this report is issued and approved by Jefferies Hong Kong Limited and is intended for use only by professional investors as defined in the Hong Kong Securities and Futures Ordinance and its subsidiary legislation. In the Republic of China (Taiwan), this report should not be distributed. The research in relation to this report is conducted outside the People's Republic of China ("PRC"). This report does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. In India, this report is made available by Jefferies India Private Limited. In Australia, this report is issued and/or approved for distribution by, or on behalf of, Jefferies (Australia) Securities Pty Ltd (ACN 610 977 074), which holds an Australian financial services license (AFSL 487263). It is directed solely at wholesale clients within the meaning of the Corporations Act 2001 (Cth) of Australia (the "Corporations Act"), in connection with their consideration of any investment or investment service that is the subject of this report. This report may contain general financial product advice. Where this report refers to a particular financial product, you should obtain a copy of the relevant product disclosure statement or offer document before making any decision in relation to the product. Recipients of this document in any other jurisdictions should inform themselves about and observe any applicable legal requirements in relation to the receipt of this document. This report is not an offer or solicitation of an offer to buy or sell any security or derivative instrument, or to make any investment. Any opinion or estimate constitutes the preparer's best judgment as of the date of preparation, and is subject to change without notice. Jefferies assumes no obligation to maintain or update this report based on subsequent information and events. Jefferies, and their respective officers, directors, and employees, may have long or short positions in, or may buy or sell any of the securities, derivative instruments or other investments mentioned or described herein, either as agent or as principal for their own account. This material is provided solely for informational purposes and is not tailored to any recipient, and is not based on, and does not take into account, the particular investment objectives, portfolio holdings, strategy, financial situation, or needs of any recipient. As such, any advice or recommendation in this report may not be suitable for a particular recipient. Jefferies assumes recipients of this report are capable of evaluating the information contained herein and of exercising independent judgment. A recipient of this report should not make any investment decision without first considering whether any advice or recommendation in this report is suitable for the recipient based on the recipient's particular circumstances and, if appropriate or otherwise needed, seeking professional advice, including tax advice. Jefferies does not perform any suitability or other analysis to check whether an investment decision made by the recipient based on this report is consistent with a recipient's investment objectives, portfolio holdings, strategy, financial situation, or needs.

By providing this report, neither JRS nor any other Jefferies entity accepts any authority, discretion, or control over the management of the recipient's assets. Any action taken by the recipient of this report, based on the information in the report, is at the recipient's sole judgment and risk. The recipient must perform his or her own independent review of any prospective investment. If the recipient uses the services of Jefferies LLC (or other affiliated broker-dealers), in connection with a purchase or sale of a security that is a subject of these materials, such broker-dealer may act as principal for its own accounts or as agent for another person. Only JRS is registered with the SEC as an investment adviser; and therefore neither Jefferies LLC nor any other Jefferies affiliate has any fiduciary duty in connection with distribution of these reports.

The price and value of the investments referred to herein and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

This report may contain forward looking statements that may be affected by inaccurate assumptions or by known or unknown risks, uncertainties, and other important factors. As a result, the actual results, events, performance or achievements of the financial product may be materially different from those expressed or implied in such statements.

This report has been prepared independently of any issuer of securities mentioned herein and not as agent of any issuer of securities. No Equity Research personnel have authority whatsoever to make any representations or warranty on behalf of the issuer(s). Any comments or statements made herein are those of the Jefferies entity producing this report and may differ from the views of other Jefferies entities.

This report may contain information obtained from third parties, including ratings from credit ratings agencies such as Standard & Poor's, and information derived from third-party or proprietary generative artificial intelligence (Gen AI) models. Jefferies does not guarantee the accuracy, completeness, timeliness or availability of this information, and is not responsible for any errors or omissions (negligent or otherwise), regardless of the cause, or for the results obtained from the use of such content. Neither Jefferies nor any third-party content providers, including providers of Gen AI models, give any express or implied warranties, including, but not limited to, any warranties of merchantability or fitness for a particular purpose or use. Neither Jefferies nor any third-party content provider shall be liable for any direct, indirect, incidental, exemplary, compensatory, punitive, special or consequential damages, costs, expenses, legal fees, or losses (including lost income or profits and opportunity costs) in connection with any use of their content, including ratings. Credit ratings are statements of opinions and are not statements of fact or recommendations to purchase, hold or sell securities. They do not address the suitability of securities or the suitability of securities for investment purposes, and should not be relied on as investment advice. Reproduction and distribution of third party content in any form is prohibited except with the prior written permission of the related third party.

Jefferies research reports are disseminated and available electronically, and, in some cases, also in printed form. Electronic research is simultaneously made available to all clients. This report or any portion hereof may not be copied, reprinted, sold, or redistributed or disclosed by the recipient or any third party, by content scraping or extraction, automated processing, or any other form or means, without the prior written consent of Jefferies. Any unauthorized use is prohibited. Neither Jefferies nor any of its respective directors, officers or employees, is responsible for guaranteeing the financial success of any investment, or accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this report or its contents. Nothing herein shall be construed to waive any liability Jefferies has under applicable U.S. federal or state securities laws.

For Important Disclosure information relating to JRS, please see https://adviserinfo.sec.gov/IAPD/Content/Common/crd_iapd_Brochure.aspx?BRCHR_VRSN_ID=483878 and <https://adviserinfo.sec.gov/Firm/292142> or visit our website at <https://javatar.bluematrix.com/sellside/Disclosures.action>, or www.jefferies.com, or call 1.888.JEfferies.

© 2025 Jefferies